

Review

Robotic disassembly for end-of-life products focusing on task and motion planning: A comprehensive survey

Mohammed Eesa Asif¹, Alireza Rastegarpanah^{*,1}, Rustam Stolkin

Extreme Robotics Lab, School of Metallurgy & Materials, University of Birmingham, Birmingham, B15 2TT, UK

ARTICLE INFO

Keywords:

Electric vehicles
Lithium-ion batteries
Robotic disassembly
Recycling
Circular Economy
Task and motion planning

ABSTRACT

The rise of mass production and the resulting accumulation of end-of-life (EoL) products present a growing challenge in waste management and highlight the need for efficient resource recovery. In response to this challenge, robotic disassembly has emerged as a vital tool for the circular economy. Combining accuracy, adaptability, and the potential for handling hazardous materials offers a sustainable solution for dismantling complex EoL objects. This comprehensive survey delves into the motivations for robotic disassembly and the pivotal role of task and motion planning (TAMP) in optimising disassembly processes. It analyses the evolution of disassembly strategies, from conventional methods to those driven by cutting-edge artificial intelligence (AI) techniques, for the future of waste management. Additionally, the survey explores several case study applications, focusing on the disassembly of EV lithium-ion batteries. It highlights how TAMP and AI integration can bolster adaptability, safety, and informed decision-making within real-world disassembly challenges. Finally, the review examines promising future research directions in robotics that hold the potential to advance further improvement in robotic disassembly to increase sustainability and the responsible management of EoL products.

Contents

1. Introduction	2
1.1. Search methodologies	3
1.2. Comparison to other surveys and literature reviews	3
2. Background	3
2.1. Destructive vs non-destructive disassembly	10
2.2. Level of autonomy	10
2.3. Robotic disassembly challenges	10
2.3.1. Computational complexity	10
2.3.2. Complex geometries and interlocked parts	11
2.3.3. Simulation-to-reality (Sim2Real) problem	11
2.3.4. Variability and uncertainty	11
3. Disassembly approaches	11
3.1. Finite state machines	11
3.2. Matrix-based	12
3.3. Graph-based	13
3.3.1. Petri Nets	13
3.3.2. AND/OR graph	13
3.4. Tree-based approaches	13
3.5. Machine learning and AI approaches	14
3.5.1. Deep learning and Computer vision	15
3.5.2. Reinforcement learning	16
3.6. Human Robot Collaboration	19

* Corresponding author.

E-mail address: a.rastegarpanah@bham.ac.uk (A. Rastegarpanah).

¹ Mohammed Eesa Asif and Alireza Rastegarpanah should be considered joint first author.

3.7.	Telerobotics	19
3.8.	Optimisation	19
3.8.1.	Single objective and Multi objective	20
3.8.2.	Meta-heuristic algorithm	20
3.8.3.	Fuzzy Logic	20
3.8.4.	Reinforcement learning	21
3.8.5.	Optimal nonlinear control	21
4.	Task and Motion Planning	22
4.1.	Hierarchical-based Task and Motion Planning	22
4.2.	Belief-based Task and Motion Planning	25
4.3.	Optimisation-based Task and Motion Planning	25
4.3.1.	Constraint-based TAMP	27
4.4.	Sampling-based TAMP	27
4.5.	Innovative application and frameworks	28
5.	Expanded applications	29
5.1.	Electronic waste	29
5.2.	Automotive components	30
5.3.	Electric vehicle batteries	30
5.4.	Future application	32
6.	Future directions	32
6.1.	Digital twin technology	32
6.2.	Improving industrial robot communication	33
6.3.	Building on foundation models for robotic control	33
6.4.	Improvement in multi-agent task planning for disassembly	33
6.5.	Domain adaptation	35
7.	Conclusion	35
	CRedit authorship contribution statement	36
	Declaration of competing interest	36
	Acknowledgements	36
	References	36

1. Introduction

Mass production has been a significant driver in raising many people’s living standards in the past decades. It has made several goods more affordable and accessible, making life easier and more comfortable. Unfortunately, this increase in production has also led to more **end-of-life** (EoL) products, also known as spent items, reaching the end of their functional lifespan that need to be disposed of. This has raised concerns about environmental degradation and the depletion of natural resources. Current research shows the presence of approximately 1.3 billion tonnes of metal waste globally, with forecasts predicting a surge to 2.7 billion tonnes by 2050 [1]; the same trend is being observed in Electronic waste (E-waste) with projections of tonnes of electronic equipment that will end up in landfills [2]. This rise has led to governments taking action by creating policies aimed at fostering greener, more competitive markets [3], specifically through initiatives that promote circular economic models and sustainability efforts. As such, the remanufacturing and recycling sectors emerge as pivotal forces in the transition to a circular economy [4], which emphasises the reduction, reuse, and recycling of products integral to tackling the e-waste crisis. In a circular economy, products are designed with the end of their life in mind, ensuring that they can be kept in use for as long as possible and then recycled to minimise waste [2,5]. This economic model also positions the remanufacturing industry to make significant socio-economic impacts. Among these benefits are the creation of job opportunities across various regions, including the UK, USA, and Europe. Notably, the UK’s remanufacturing sector contributes £2.4 billion to the industrial output [6].

In comparison, the automotive remanufacturing market in the USA was valued at \$60.8 billion in 2022, with expectations to double to \$118.2 billion by 2030 [7]. Similarly, the European remanufacturing market, valued at €45 billion, is anticipated to reach €90 billion by 2030 [8]. Additional benefits of remanufacturing include significant reductions in greenhouse gas emissions and lower resource consumption enabled by valuable secondary sources of materials extracted from

spent EoL products. Recent studies have also shown changing consumer behaviours shifting towards demanding more sustainable practices [9].

The rising consumer demand and governmental support have significantly fuelled interest in remanufacturing engineering. Unlike recycling or refurbishing, remanufacturing is a process that revitalises EoL products and champions environmental sustainability. This process involves disassembly, restoration, and reassembly in a controlled and sustainable manner. At the heart of the remanufacturing process is disassembly, specifically robotic disassembly, which tackles several challenges, including the labour-intensive nature of manual disassembly, the need for precise component separation, and the safe handling of hazardous materials [2,5].

The progression of robotic disassembly systems is characterised by the growing incorporation of **artificial intelligence** (AI), machine vision, and sophisticated planning algorithms [10]. This technological progress improves the efficiency of the disassembly process and boosts material recovery, particularly in waste sorting and precise disassembly in dangerous settings. Additionally, optimisation models focus on determining the most cost-effective and environmentally sustainable disassembly methods, moving towards automation and away from manual labour. Emerging techniques are optimising robotic disassembly tasks and scheduling, enhancing the efficiency of disassembly line balancing [11]. These advancements, alongside the development of modular task and motion planning architectures, offer dynamic and flexible disassembly processes. The introduction of hybrid **human-robot collaborative** (HRC) cells addresses the variability of products in remanufacturing, combining robotic precision with human adaptability and dexterity [12]. This shift minimises the risks of handling hazardous materials such as EV batteries, reduces operational expenses, and enhances worker safety [13,14].

Task and motion planning (TAMP) is crucial for robotic disassembly; it integrates high-level decision-making with detailed execution strategies to ensure efficient and safe breakdown of products into their components [15]. This involves sequence planning to determine the order and method of disassembly and motion planning to navigate the robot’s movements around obstacles and through confined spaces.

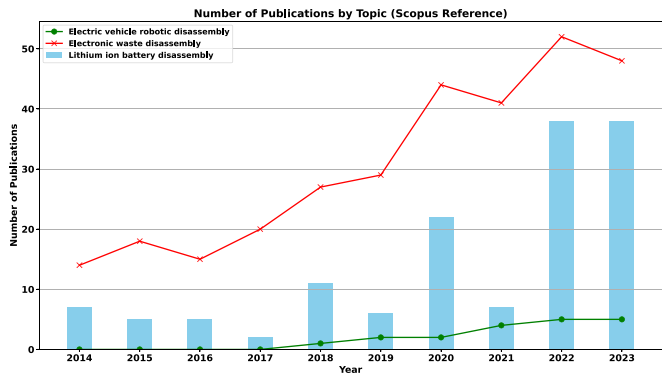


Fig. 1. Related papers in the literature reflecting the trend.

The complexity of this planning is in accommodating the physical constraints of the product and the robot’s capabilities and finding the optimal sequence that minimises time or energy while maintaining safety and feasibility. In parallel, disassembly line balancing is concerned with optimising task distribution across multiple workstations to reduce idle time and maintain a smooth workflow, aiming for efficient task allocation and resource management to maximise throughput rates across the entire production line [16].

1.1. Search methodologies

The importance of robotic disassembly in managing EoL products is evident, as illustrated in Fig. 1, which shows a steady increase in related publications over the last decade. This growth is driven by significant industrial, societal, ecological, and ethical demands for EoL disassembly topics. Consequently, there is a pressing need to consolidate information on emerging technological advancements, current obstacles, and prospective future directions that could benefit industrial practitioners and academic researchers engaged in this domain. This paper aims to fulfil this requirement by exploring the methodologies, identifying technological deficiencies, and highlighting recent progress in robotic disassembly. It concentrates on TAMP within and how it can be adapted for disassembly tasks. The following research databases were used to conduct the survey: Google Scholar, Scopus, ScienceDirect, IEEE Xplore, Web of Science, and Connected Papers. The literature screened was peer-reviewed academic conference papers, journal articles, and books within the research domain of robotics, automation, computer science, sustainability, and manufacturing engineering. (The search criteria overview for the papers is illustrated in Table 1.)

1.2. Comparison to other surveys and literature reviews

In Table 2 we present a comparison of this survey with multiple existing surveys in the field of robotics, mainly focusing on aspects related to TAMP, **Robotic disassembly sequence planning** (RDSP), **Robotic disassembly line balancing** (RDLB) and the use of meta-heuristics and machine learning in these domains. A common issue identified in these reviews is the shallow examination of how the discussed methods apply in real-world situations and their ability to adapt to new EoL applications. Additionally, there is a noticeable lack of studies on integrating TAMP specifically for robotic disassembly dealing with EoL of objects. This highlights a research gap, emphasising the importance of developing TAMP strategies to address the specific challenges of disassembling objects with geometric constraints, which is crucial for advancing sustainable manufacturing. An overview of the entire study can be seen in Fig. 2.

Hence, the objective of the survey paper is to review the integration and implications of advanced planning approaches in robotic disassembly within industrial contexts, focusing on TAMP, and to point out the

Table 1 Overview of the survey criteria applied during the search process for relevant literature.	
Search criteria	
Search terms	“Robotic Disassembly”, “robot-assisted disassembly”, “Human-Robot Collaboration” AND “Disassembly”, “Task and motion planning”, “Hierarchical task planners”, “Optimization” AND “Robotic Disassembly”, “Machine learning” AND “disassembly”, “Simulation-to-reality” AND “Robotics”
Period	January 2014–March 2024
Publication type	Peer-reviewed academic conference papers, journal articles, and books focusing on technical and scientific advancements
Exclusion Criteria	
Language	Non-English
Contextual	Economic evaluation/feasibility studies, profitability scores, disassembly line design, destructive disassembly, tool design, and non-technical aspects
Scope	Papers not directly addressing advancements in robotic disassembly technology, methodologies, or applications

challenges, opportunities, and future directions. The Key Questions to be addressed by this review are:

1. *Prevailing Approaches and Enhancements:* What are the prevailing approaches to robotic disassembly, and how could integrating TAMP enhance these methods to improve adaptability and operational efficiency?
2. *Impact of Recent AI Advancements:* How do recent AI advancements, such as Reinforcement Learning and computer vision, enhance the flexibility and efficacy of robotic disassembly? What new opportunities could emerge from integrating these methodologies with current practices?
3. *Integration of AI Planning Methodologies:* How are advanced AI planning methodologies, particularly TAMP, being integrated into robotic disassembly? What are the gaps between theoretical advancements and practical applications?

The remainder of this article is organised as follows: Section 2 delves into the background of robotic disassembly, highlighting its distinctions from robotic assembly. It further explores the differences between destructive and non-destructive disassembly and the various levels of autonomy used. This section also thoroughly discusses the inherent challenges and strategies employed to overcome them. In Section 3, explores the diverse approaches to disassembly, tracing developments from historical to cutting-edge methods and detailing various optimisation strategies employed in disassembly processes. Section 4 reviews related works in task and motion planning, building upon previous studies and discussing modern approaches applicable to disassembly. In Section 5, we discuss the different robotic disassembly applications and propose additional industries where they can be used. Finally, Section 6 concludes the article, outlining future directions for research in robotic disassembly and presenting a road map for advancing state-of-the-art technologies and methodologies, aiding future research in robotic disassembly.

2. Background

Robotic disassembly research is an area that mainly covers the most difficult challenges and methodologies used in decision-making and control. Examples of such approaches in disassembly include AND/OR graphs, Finite State Machines, Graph-based, Matrix-based, and AI and **Machine Learning** (ML) based approaches, which are contributing to the evolution of disassembly systems. Table 3 presents a compilation of the different disassembly approaches of robotic disassembly over

Table 2
Related surveys to summarise robotic disassembly and task and motion planning.

Survey paper	Topic	Synopsis
[14]	TAMP approaches	This article systematically reviews TAMP approaches for autonomous robots in unstructured human environments, integrating abstract reasoning with geometric feasibility, categorises challenges, and presents solutions via online and machine learning-based TAMP methods to enhance real-world application and generalisation.
[12]	Human–robot collaborative disassembly	This paper reviews the progress and challenges in (HRC) disassembly from 2009 to 2020, focusing on safety, collaboration modes, communication interfaces, and design characteristics to enhance sustainability in recycling processes by improving efficiency in hybrid disassembly robot cells.
[13]	Algorithms for TAMP problems	This article surveys algorithms for TAMP problems, which involve a robot navigating and interacting with many objects, integrating discrete task planning, mathematical programming, and continuous motion planning to address challenges not solvable by these fields alone, focusing on continuous-space solutions and integrating discrete-continuous search components.
[17]	Robotic Disassembly Sequence Planning	This article reviews advancements in RDSP for recycling and sustainable development, covering theory and methods in modelling, mathematical programming, artificial intelligence, and uncertainty handling, guiding optimal strategies for managing obsolete products in the Industry 4.0 context.
[11]	Robotic Disassembly Line Balancing	This paper reviews the use of meta-heuristics to solve RDLB, Planning, and Scheduling Problems. The related literature was summarised and analysed, the current optimisation methodologies employed, and trends were outlined
[18]	Robotic motion optimisation	The paper is an exploratory survey that shows optimisation of motion for robots in industrial production by the use of automation in place of manual computations of the best trajectories, with attention drawn to improvement in efficiency, elimination of errors in kinematic redundancy, etc., including task sequencing, collision avoidance.
This survey	All of the above topics	Outlines methodologies, technological gaps (with a focus on TAMP), and recent developments in RDSP focusing on advancements in Robotic Vision AI planning and reinforcement learning. It also discusses the applications of robotic disassembly and suggests future directions to encourage wider adoption of automation.

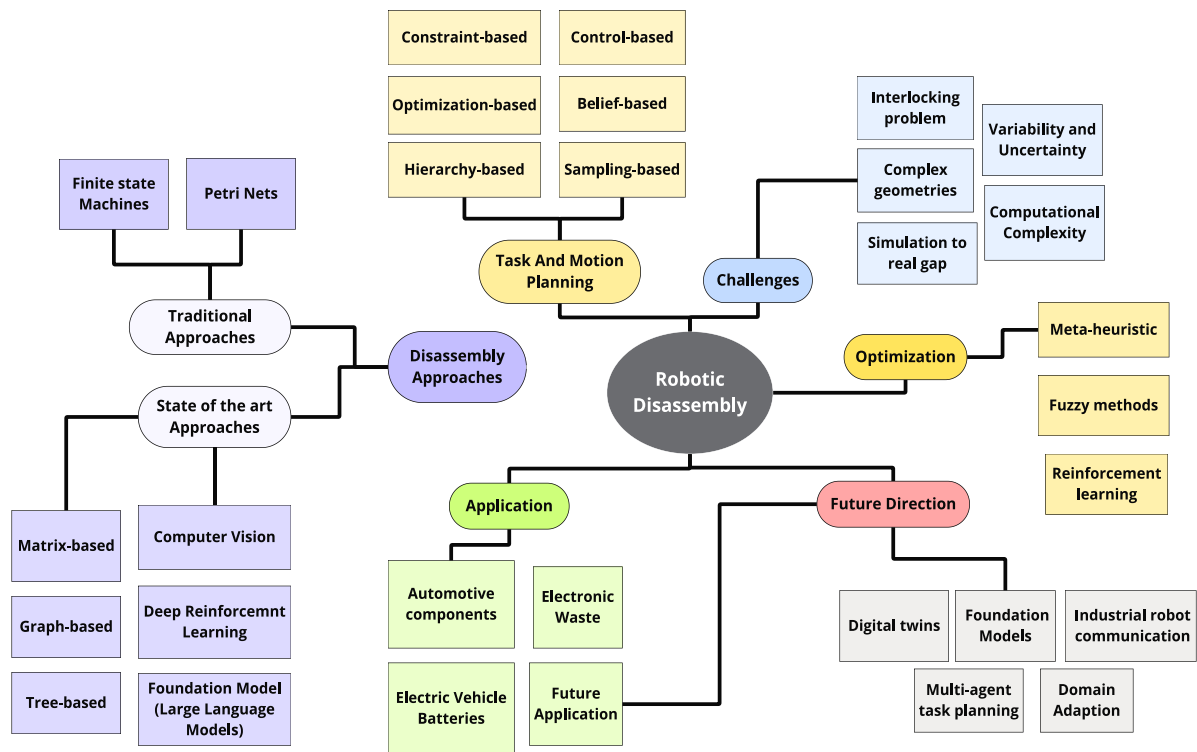


Fig. 2. Graphical representation of the review scope, covering challenges and methodologies in robotic disassembly for end-of-life objects, with an emphasis on task and motion planning.

Table 3

The categorisation of research on robotic disassembly of end-of-life products by outlining study objectives, outputs, evaluations, and applications across various methodologies.

Approach	Author & Year	Objective	Performance evaluation	Output	Application
Matrix-based	Alshibli et al. (2019) [19]	Orthogonal Arrays and Task Allocation in RDSP using disassembly task, constraints, and robotic capabilities	Robustness of disassembly sequence, efficiency of task allocation, and adaptability to variability in disassembly tasks	Optimised disassembly sequence, task allocation among	N/A
	Wang et al. (2018) [20]	Matrix manipulation to improve Efficiency in Disassembly for 4-stroke Engine using Assembly, contact and relation matrix whilst & applying logic gates	Accuracy of sub-assemblies detection	Sub-assemblies identification	Automotive components
	Chen et al. (2017) [21]	Matrix manipulation and Radio frequency identification (RFID) to record and trace the lifecycle information, optimising disassembly process and improving recovery efficiency using disassembly matrix, fastener matrix and probability fastener matrix	Accuracy of sub-assemblies detection	Sub-assemblies identification	Waste management and recycling of products
	Wegner et al. (2014) [22]	Matrix manipulation, for Generating disassembly graph of Hybrid Q5 Audi battery using Priority matrix	N/A	Disassembly sequence	EV battery
	Jin et al. (2014) [23]	Using Matrix manipulation and Dynamic programming for Generating disassembly sequence LCD screens using CAD as models	N/A	Interference matrices for spatial information between components in Cartesian workspace	Electronic waste
Reinforcement Learning-based	Xiao et al. (2023) [24]	Multi-agent RL (MARL) to optimise the disassembly process of retired EVB, aiming to enhance efficiency and safety in recycling through effective HRC using disassembly graph representation of EVB	Average optimal solution	Sequence of disassembly tasks, with task assignments	EV battery
	Qu et al. (2023) [25]	Reinforcement learning with data-driven learning (modified DDPG) to automate precision disassembly on performance of learned skills and explore transfer-ability using state of the environment (Cartesian coordinates, robot angles, force, torque along the x, y, and z-axis), actions defined by movement commands	Success rate, stability Index, performance on transfer ability	Optimal policy based on current state and env (performing disassembly task, with constraints of minimal clearance and avoiding excessive force or torque that could lead to failure)	N/A
	Allagui et al. (2023) [26]	Reinforcement learning for handling disassembly parameters and minimise time and cost using CAD data and collision matrix	Disassembly time	Disassembly sequence	Automotive components
	Aflakian et al. (2023) [27]	Reinforcement learning (TD3), Convex hull workspace bounding to accelerate the training process of RL agents and reducing the search space using human demonstrations on Random Surfaces, 3D Convex Hull Computation, Task and Desired Parameters as Reward Functions	Training Progress, policy performance	Optimal Solution within the Bounded Workspace, Trained Policy's Behavior (resilience to local minima and learning from imperfect demonstrations)	EV battery
	Cai et al. (2022) [28]	Actor-Critic (A2C) algorithm in RDLB to simultaneously find feasible disassembly sequences and allocate tasks to workstations in an optimal manner and minimise the cycle time of workstations	Computation complexity	Optimal sequence	Electronic waste
	Liu et al. (2022) [29]	Q-learning algorithm for optimising HRC in two-sided disassembly line balancing problems for minimising total disassembly time using disassembly tasks AND/OR graph, detailing the relationships, sequences of subassemblies/parts, including precedence and conflict matrices	Benchmarked against the SARSA algorithm across various case studies	Disassembly sequence	N/A
	Zhao et al. (2021) [30]	RL-based selective disassembly using finite Markov Decision Process, and DQN for optimal disassembly sequences, using the structure of products shown by Multi-Level Selective Disassembly Hybrid Graph Model	Benchmarking	Optimised sequences	N/A
	Mao et al. (2021) [31]	Adaptive disassembly planning using Deep Q-Network (DQN) for dynamic, optimal disassembly sequences using GA for Virtual Reality training for Aircraft engine using disassembly Petri Net to model	GA fitness function	Optimised sequences	Aircraft (aerospace)

(continued on next page)

Table 3 (continued).

Approach	Author & Year	Objective	Performance evaluation	Output	Application
	Zakka et al. (2020) [32]	Learning shape priors - RL for generalisation end-to-end pick and place for disassembly	Area under the curve comparing proposed method to baseline methods, success rate	Visual observations of the workspace, represented as grayscale-depth heightmap images, using pick & place location along with angle that defines change in orientation for this location change	N/A
	Liu et al. (2019) [33]	HRC integrated with RL to Improve Sustainability of EoL products like diaphragm coupling: Joining gearbox shafts in machinery using disassembly components	Efficiency, safety	Disassembled component sequence	Automotive components
	Kristensen et al. (2019) [34]	Reinforcement learning, Q-learning algorithm for Robotic disassembly unscrewing operations for e-waste using CAD drawings	N/A	Gazebo simulation	Electronic waste
	Chen et al. (2019) [35]	RL-based Assigning Policy (RLAP) for scheduling multi-projects in cloud manufacturing to Minimise both the total make-span and the logistical distance, using interrelated tasks, each with precedence relation	Schedule quality and distribution of service	Non-dominated solution set for the scheduling of tasks	Cloud Manufacturing
	Tuncel et al. (2014) [36]	Monte-Carlo based Q-learning unique dynamics for balancing the disassembly line minimises idle time of using precedence graph of cell phones and PC component sub-assemblies, the approach also incorporates tasks times	N/A	Optimise disassembly	Electronic waste
Computer vision-based	Biehler et al. (2023) [37]	PLURAL (3D Point Cloud Transfer Learning via contrastive learning With Augmentations) to overcome challenges in the 3D point cloud machine learning model application	Benchmarked against state-of-the-art transfer learning methods	Domain-invariant features	Defect detection
	Chen et al. (2022) [38]	Improved YOLOv5 for Classification and positioning of circuit board components using images of circuit board components	38% higher algorithm effect than original YOLOv5 model	Component detection and identification	Electronic waste
	Zheng et al. (2022) [39]	PointNet Deep Neural Network for Identification of mechanical parts and engine disassembly using 3D point clouds of turbocharger components	Accuracy in recognition	Part identification	Automotive components
	Foo et al. (2022) [40]	Computer vision for Disassembly sequence of LCD screens using disassembly component image	Accuracy in identification of components	N/A	Electronic waste
	Braadland et al. (2021) [41]	3D Perception Methods (Graph-based and Voxel-based Convolution) for Point cloud instance segmentation using 3D point clouds of battery packs	Accuracy and processing time	Instance-based segmentation	EV battery
	Su et al. (2021) [42]	Mask R-CNN for Laptop parts detection using images	Improved accuracy through quantity and space constraints	Parts detection and required rotation angles	Electronic waste
	Bai et al. (2021) [43]	Modified Faster R-CNN and SVDD algorithms for Fastener detection for Screw loosening	Detection rates and accuracy compared to baseline method	Images of railway fasteners using fastener states and position	N/A
	Li et al. (2021) [44]	Faster R-CNN and Rotation Edge Similarity for Screw detection	Accuracy; localisation deviation	Images of mobile phone main-boards using screw location and classification	Electronic waste
	Foo et al. (2021) [45]	Deep learning model for Screw detection	Logic reasoning on common screw location combinations; re-training on missed detection	Images of LCD monitors using coordinates of screws	Electronic waste
	Brogan et al. (2021) [46]	Tiny-YOLOv2 for Fastener detection using images of cross-recessed screws	Highest average precision	Screw detection	Electronic waste
	He et al. (2020) [47]	Vision-based for Dismantling circuit boards from used mobile phones	Visual inspection, template matching	Disassembled and recovered CPUs	Electronic waste
	Zhang et al. (2020) [48]	Deep Learning (Faster R-CNN) for Bolt loosening detection using images of bolts in various states	Average precision; real-time monitoring capability	Bolt state (tight or loosened)	EV battery
	DiFilippo et al. (2018) [49]	Vision-force control system to Automate non-destructive disassembly of Laptops using Vision data	Identification Accuracy, Time Efficiency, Screw Removal Success Rate	Automated unscrewing	Electronic waste

(continued on next page)

Table 3 (continued).

Approach	Author & Year	Objective	Performance evaluation	Output	Application
Human robot collaboration	Vongbunyong et al. (2017) [50]	Vision-based cognitive robotics system to develop a platform capable of transferring disassembly skills from humans to robots by demonstrating the disassembly (LCD screens) process using a vision system and other sensors. The knowledge and skills acquired are then transferred to a robot to automate the disassembly process using Vision tracking data	The precision and accuracy of the system were assessed, with marginal errors noted due to the limitations of the RGB-D sensor. Improvement suggestions include using 3-D markers for more precise tracking and higher-performance sensors.	Disassembly experiments	Electronic waste
	Friedrich et al. (2016) [51]	Vision-based control for disassembly planning and object detection using CAD and vision data	N/A	Disassembly planner	N/A
	Wegener et al. (2015) [52]	Human-robot collaboration for Automating simple tasks (unscrewing) using coordinates of screw bolt type and fastener	Camera based detection -(TPR, FPR ROC) and bit changing mechanism	Disassembly sequence using regions of interest	EV battery
	Chu et al. (2023) [53]	HRC disassembly using a hybrid particle swarm optimisation with the Q-learning algorithm to minimise disassembly completion time using battery characteristics	Number of minimum and average optimal solution (superior to other meta-heuristics in efficiency and robustness.)	Optimal disassembly sequence and task allocation	EV battery
	Yuan et al. (2023) [54]	New heuristic algorithm for multi-criteria resilience assessment to support spent lithium-ion battery recycling through HRC disassembly assessment using spent lithium-ion batteries characteristics, human and robot collaboration parameters	Number of minimum and average optimal solution	Resilience assessment for HRC, optimal disassembly strategies	EV battery
	Kay et al. (2022) [55]	Integrates human skills with robotic precision. Robots perform dangerous and precise tasks (cutting battery tabs). At the same time, humans handle tasks that require more dexterity and decision-making (sorting components and removing fasteners) to perform semi-automated robotic disassembly of LIB modules from EVs to enhance the recycling process, addressing high labour costs, safety risks from manual disassembly, loss of valuable materials	The success is measured by its ability to perform tasks faster and with greater precision than human technicians, particularly in cutting operations, while reducing exposure to hazardous conditions.	Disassembly process that includes precise cutting and gripping with force feedback	EV battery
	Zhou et al. (2022) [56]	HRC disassembly + innovative Particle Swarm Optimisation (PSO)-Pareto algorithm to enhance the cooperation, safety, and disassembly efficiency of human-robot collaboration in the removal of screws from EoL products using dynamic and uncertain characteristics of human operators, rust conditions of screws	Average running time	Pareto optimal solutions	EV battery
	Wu et al. (2022) [57]	Multi-objective optimisation for cell-level disassembly in HRC for disposal of waste power battery modules (WPBMs) from Tesla model S	Number of workstations & workers, workstation idle time, disassembly cost	Disassembly sequence	EV battery
	Huang et al. (2021) [58]	Active compliance, Operator touch (impact) sensing and position control (HRC) to create a disassembly cell for complex geometries (automotive turbocharger) which uses human-robot and robot-robot collaboration	Cycle time vs Idle time	Disassembly sequence	N/A
Evolutionary optimisation-based	Wu et al. (2023) [59]	Sim2real Transfer Learning for Point Cloud Segmentation to Overcome low sample efficiency in the real world using simulated data using synthetic point cloud data	Novel patch-based attention network to tackle imbalanced learning	Segmented result, screw location and orientation	Electric motors
	Rosenberg et al. (2022) [60]	Fuzzy Logic to Calculate the fuzzy and defuzzified disassembling times applying PHEV battery system	Disassembly completion time	Disassembly graph	EV battery
	Xu et al. (2020) [61]	Modified Discrete Bees Algorithm based on Pareto to Minimize disassembly cost and time of PC disassembly using task classification	Generational Distance and Hyper-volume	Optimal Disassembly Sequence	Electronic waste
	Ramirez et al. (2020) [62] Zhang et al. (2019) [63]	Genetic Algorithm, hill climbing, constructive greedy for optimal disassembly of gear pump Two-Pointer Detection Strategy and Super-Fast Bees Algorithm to Robotic Re-Planning for peg hole using geometrical parameters, initial position errors, clearance ratio, compliance centre location	Fitness value, sensitivity analysis Extraction force and path optimisation	Disassembly sequence Optimised path of compliance centre, forces acting, extraction force	Automotive components N/A

(continued on next page)

Table 3 (continued).

Approach	Author & Year	Objective	Performance evaluation	Output	Application
	Laili et al. (2019) [64]	Super-Fast Bees Algorithm, Greedy search to Robotic Disassembly Re-Planning using initial plan	Disassembly time	Optimized Re-Planning Strategy	N/A
	Tao et al. (2019) [65]	Multi-Objective Meta-Heuristic Algorithm to Balance Disassembly Profit with Environmental Impact	Recovery Profit, Energy Consumption	Automated Disassembly Schemes and recovery rate for LCD screens, electrolytic capacitors and batteries	Electronic waste
	Ren et al. (2018) [66]	Genetic Algorithm combined with AND/OR precedence relations to Minimize Disassembly Time of an HG5-20 triaxial five-speed mechanical transmission	Disassembly Time	Optimized Disassembly Sequence	Automotive components
	Feng et al. (2018) [67]	Hybrid Disassembly for Multi-objective Optimization of EOL product data	Profit, Environmental Impact, Pareto	Recovery Plan	Spindle from CNC machine
	Liu et al. (2018) [68]	Genetic Algorithm - Enhanced Discrete Bees Algorithm to minimise Total Disassembly Time using EOL Products Components	Disassembly Efficiency	Optimal Disassembly Sequence	N/A
	Cheng et al. (2017) [69]	HRC, Fuses historical and real-time data using manifold algorithms for optimal scheduling and resource allocation in manufacturing workshop for disassembly using historical, real-time data: energy consumption, disassembly time and cost, safety	N/A	Assessment of manufacturing capabilities, recommendations for task allocation and scheduling	Electronic waste
	Xia et al. (2016) [70]	Teaching and learning based model for multi-objective disassembly sequence planning for LCD screen	Pareto-optimal frontier	Pareto-optimal disassembly sequences	Electronic waste
	Djogdom et al. (2024) [71]	Dynamic scheduling using behaviour trees to Improve collaboration between robots and humans in manufacturing operations through dynamic scheduling using manufacturing operation requirements and human-robot collab parameters	Robustness in task scheduling	Dynamic schedules for robot tasks	N/A
	Ghzouli et al. (2023) [72]	Comparative analysis of Behaviour trees and State Machines and exploring their applications to Analyze the application of Behaviour trees and state machines	N/A	Analysis and comparison results	N/A
	Pezzato et al. (2023) [73]	Active inference and Behaviour trees for pick and place to Enable reactive action planning and execution using task requirement	Reactivity, adaptability, execution success rate	Reactive actions and plans	N/A
Graph-based	Rastegarpanah et al. (2021) [74]	Behaviour-trees to Semi-automate disassembly of EV battery packs in an industrial environment using initial conditions of robotic system, start position, pre-defined tasks	Task time, success rate	Simulation, Dismantled EV components sorted, accomplishing task	EV battery
	Ding et al. (2019) [75]	Knowledge Graph to Enhance HRC in Disassembly	Disassembly Time, Idle Time, Human Workload	Knowledge-driven Disassembly Assistance	N/A
	Ebinger et al. (2018) [76]	Hierarchical disassembly graph to Flexible framework for RDLB adaptable to different disassembly tasks using a set of parts with defined geometries and positions within an assembly	Demonstration of 3D problems	Disassembly sequence, with collision-free removal trajectories for each part or sub-assembly	Automotive components
	Costa et al. (2018) [77]	Recursive Branch-and-Bound with Best-First Search to Disassembly of single cylinder engine using SolidWorks CAD data (including geometric mates and component information)	Cost and Efficiency	Disassembly Plan	Automotive components
	Paxton et al. (2017) [78]	Behaviour trees and vision for cobots CoSTAR to Instruct cobots using Behaviour trees and vision for more intuitive and flexible robot using collaborative task description, visual environment data	Task execution success rate, programming flexibility	Executable tasks by cobots	N/A
	Gammell et al. (2014) [79]	Informed RRT* is used, which is an enhancement of the Rapidly-exploring Random Trees (RRT*) algorithm, the objective is to improve the efficiency and effectiveness of the standard RRT* by introducing a method that focuses the search within an ellipsoidal area, defined by an admissible heuristic.	Solution cost and CPU time against the traditional RRT* method (Informed RRT* achieves quicker convergence to optimal solutions)	Reduced computational effort and faster convergence times & improving the quality of the path solutions	N/A

(continued on next page)

Table 3 (continued).

Approach	Author & Year	Objective	Performance evaluation	Output	Application
Tele-robotics	Hathaway et al. (2023) [80]	Tele-robotics (master-servant framework) for Disassembly of battery to recover high-value materials, operator commands through local inference (using haptic devices and cobot control)	Success rate, average task completion time	Successful completion of specific disassembly tasks, (unscrewing, sorting, and cutting)	EV battery
	Petereit et al. (2019) [81]	Variable-autonomy using tele-operation and VR for Robotic systems for decontamination tasks, focusing on autonomy in hazardous environments, improving safety and efficiency using Map of nuclear facilities, contaminated sites	Tested on decontamination tasks	multi-sensory 3D mapping, and VR tele-operation	Nuclear decommissioning
	Marturi et al. (2016) [82]	Tele-operation and autonomy with visual servoing and autonomous grasping algorithms to Explore novice human operators control of robots for basic manipulation tasks & improve task performance through autonomous robotics techniques using human commands, computer vision inputs	Comparison of direct teleoperation vs. human-supervised semi-autonomous control in terms of throughput rates, errors, safety enhancements	Semi-autonomous control of robot arms for manipulation tasks	Nuclear decommissioning

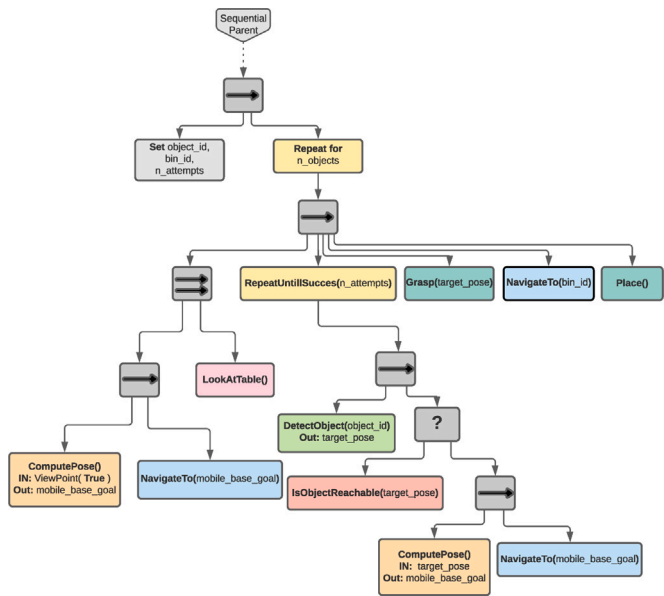


Fig. 3. Disassembly Sequence Behaviour Tree for Removing Specified Objects from a Workstation and Depositing Them into a Container Using a Mobile Manipulator [74].

the past decade, showcasing the diverse applications in which these technologies have been applied.

The body of work in robotic disassembly can be broadly categorised under two main problems: **Robotic Disassembly Sequence Planning** (RDSP) and **Robotic Disassembly Line Balancing** (RDLB). RDSP determines the optimal sequence of actions for disassembling a product. At the same time, RDLB focuses on distributing disassembly tasks across a production line to achieve optimal workload balance and efficiency.

RDSP can be categorised based on various factors. The chosen sequence may need to account for potential damage to parts during disassembly, affecting their value and necessitating a consideration of destructiveness. Objective functions vary, ranging from maximising profit or minimising carbon emissions to multi-objective models seeking Pareto solutions. Fig. 3 depicts the disassembly sequence architecture based on a **Behaviour tree** (BT) framework that enables the modular combination and monitoring of different robot capabilities to semi-automate the process of sorting the various components of an **Electric vehicle battery** (EVB) using a mobile manipulator [74].

On the other hand, RDLB, initially introduced by Gungor et al. [83], has become increasingly complex and is now a critical area of focus in the manufacturing sector, drawing significant interest from academia

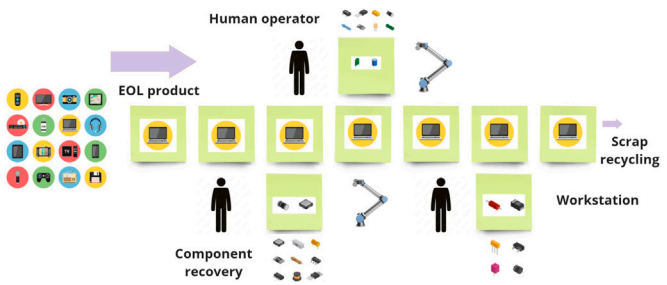


Fig. 4. An instance of disassembly line balancing.

and industry. The primary concerns of RDLB involve optimising the integration of human workers and multi-robot disassembly equipment to minimise time discrepancies and maximise efficiency, considering the technical skills, positions, rest periods, physical conditions, and personal efficiency of the staff [12]. Additionally, it addresses issues related to workstation cycles and efficiency, the time imbalances caused by varying disassembly times for different parts or products and the allocation of disassembly tasks among multiple products. **Reinforcement Learning** (RL) has been effectively applied to these challenges, with numerous successful case studies. For instance, one study by Mei et al. employed RL to tackle the multi-robot disassembly line balancing problem [84], evaluating the outcomes with three different **Deep Reinforcement Learning** (DRL) approaches. Another study by Tuncel et al. [36] used RL alongside Monte Carlo simulations to assess uncertainties on disassembly lines, showing the reliability and applicability of RL in resolving DLBP issues. Additionally, this [36] method can operate effectively in both deterministic and stochastic environments, thereby addressing the inherent uncertainties in DLBP. An instance of RDLB is shown in Fig. 4.

The general hierarchy of disassembly presents a structured approach to understanding and addressing the complexities of disassembling products at the end of their life cycle (can be seen in Fig. 5. This hierarchy facilitates detailed analysis and planning at various levels, from components' physical properties to managing multiple products as a batch [50,85]. Robotic disassembly hierarchy levels organise the process from specific interactions to broader operational concerns. The Physical Level assesses component properties like forces and materials. The Surface Level looks at component surfaces for separation ease. The Component Level considers component movement and disassembly sequence. The Modular Level focuses on disassembling functional subsystems. The Product Level plans the entire product's disassembly sequence. Lastly, the Batch Level manages the disassembly of multiple products for efficiency.

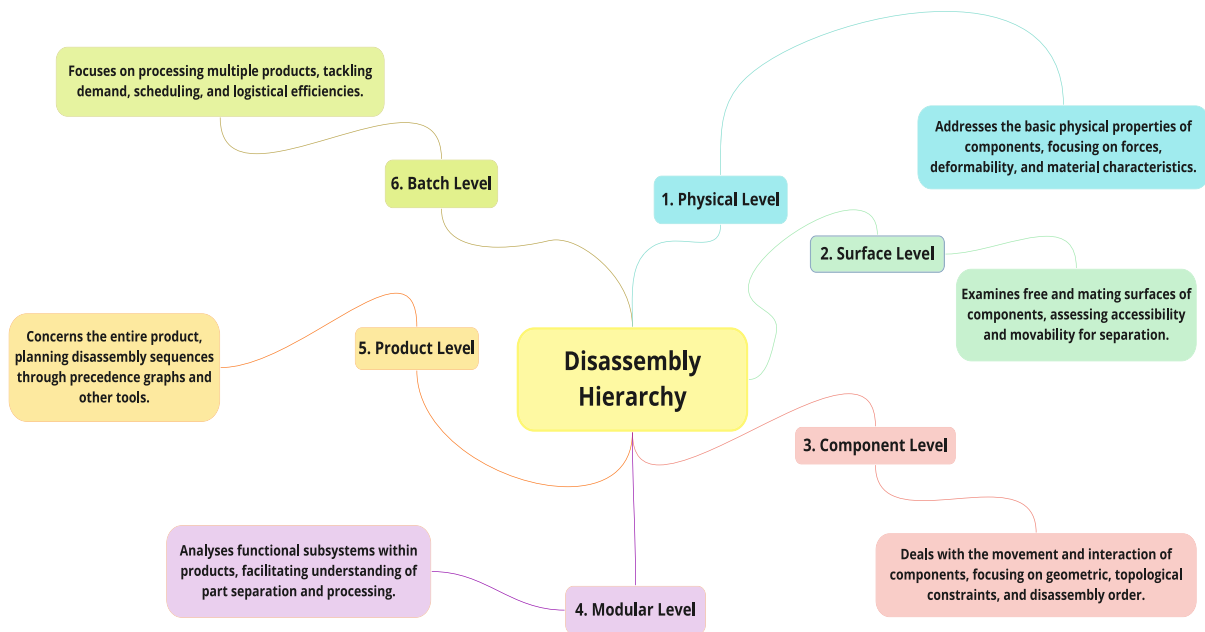


Fig. 5. Traditional disassembly hierarchy for end-of-life products.

2.1. Destructive vs non-destructive disassembly

The disassembly process can be either destructive or non-destructive, with the former focusing on material recovery where component integrity is not a priority and the latter preserving components for reuse or refurbishment [86]. Robotic systems excel in non-destructive disassembly, offering precise control that minimises damage to valuable components. Yet, the choice between destructive and non-destructive methods must be carefully considered based on the ultimate goal of the disassembly process, material recycling or component recovery [12]. Williams et al. [87] discussed the opportunities for computer-integrated de-manufacturing for materials recovery, emphasising the challenges posed by the proliferation of product designs and the need for detailed product information to support automated selective disassembly. This highlights the critical role of information availability in optimising the disassembly process, especially in non-destructive methods aimed at component preservation.

2.2. Level of autonomy

Robotic disassembly can be categorised by the extent of disassembly—full, selective, single target, or multi-target—with each category facing unique operational challenges. Full disassembly involves robots dismantling a product entirely, which can be lengthy and resource-heavy. Selective disassembly, in contrast, focuses on removing particular components, necessitating sophisticated decision-making from robots to determine the most effective disassembly sequence. Single-target and multi-target disassembly aim at retrieving one or several specific components or materials, demanding accurate identification and handling skills from the robots [88].

2.3. Robotic disassembly challenges

When comparing robot assembly and disassembly, one learns fascinating differences that exist during the processes, which are high due to the challenges brought about by disassembly. Contrary to what might be assumed, disassembly poses unique challenges that can surpass those encountered in assembly. While robot assembly may be automated because the target tasks are mostly precise and repeatable actions, disassembly involves tasks that have to adapt often and, in some

cases, have problem-solving competence whilst operating in harsh and unstructured environments. Disassembly must also be able to deal with the unknown conditions of used products, including wear, damage, and alterations, which complicate the identification and handling of parts [89]. This unpredictability demands flexible, adaptive robotic systems capable of learning and revising their actions to accommodate the range of product states they encounter [88]. This has motivated an HRC to become important in disassembly, leveraging human flexibility and robotic precision, hence balancing the variability for safety and efficiency [90].

2.3.1. Computational complexity

One of the primary challenges in RDSP is its computational complexity. As the number of disassembled components increases, the problem space expands exponentially, significantly increasing the time required to identify the optimal disassembly sequence. Hence, RDSP is classified as NP-Complete (Non-Deterministic Polynomial Complete), a type of problem known for its computational challenge. NP-complete problems are those for which no efficient algorithm is known and for which the time required to find a solution increases exponentially [91].

Mathematical definition

RDSP recognition as an NP-complete problem underlines its standing among the most significant challenges in computational theory. To elucidate RDSP complexity and its classification as NP-complete, we define it mathematically:

- Input: A directed graph $G = (V, E)$, where each vertex v_i represents a product component, and each edge (v_i, v_j) denotes the prerequisite that v_i must be disassembled before v_j .
- Objective: Identify a sequence S of all vertices such that for every edge (v_i, v_j) , v_i precedes v_j in S , minimising either total disassembly time or cost.

Demonstrating NP-completeness

To affirm RDSP NP-completeness, we examine:

1. NP Membership: Verifying a solution sequence S for adherence to dependencies E can be executed in $O(n+m)$ time, establishing the problem's membership in NP.

2. NP-Hardness: By reducing a known NP-complete problem (e.g., the Hamiltonian Path Problem) to RDSP in polynomial time, we illustrate RDSP NP-hardness. This indicates that solving RDSP is at least as hard as any NP-complete problem.

Mathematical infeasibility

The factorial complexity $O(n!)$ of exploring potential disassembly sequences with RDSP highlights the impracticality of brute-force methods for substantial instances. This complexity has been well-documented, showing the limitations of traditional mathematical approaches in efficiently solving RDSP [92], which necessitates alternative strategies, such as heuristic or approximation algorithms, for achieving near-optimal solutions within viable time-frame [92].

2.3.2. Complex geometries and interlocked parts

Robotic disassembly faces substantial challenges when dealing with complex geometries and interlocked parts commonly found in electronic and mechanical products. These challenges necessitate advanced manipulation techniques to ensure components are navigated and disassembled without damage. Initially designed to enhance product stability and durability, the interlocking mechanisms become significant obstacles during disassembly, requiring precise control and sophisticated disassembly strategies [93]. Recent research has focused on novel representation and planning models to address these issues. For instance, Wang et al. [94] introduced an approach using an assembly matrix combined with simple logic gates to manage interlocked structures effectively. This method facilitates the scheduling of disassembly sequences that can handle such complexities. Similarly, Lan et al. [93] developed automatic disassembly planning methods that employ a 'divide-and-conquer' strategy, identifying separable sub-assemblies to simplify the process. The dual approach involves recognising sub-assemblies that can be separated easily, simplifying disassembly. Overcoming interlocking challenges has the added benefit of reducing the computational load.

2.3.3. Simulation-to-reality (Sim2Real) problem

In robotic disassembly, a significant challenge is known as the **Simulation-to-reality** (Sim2Real) problem, or the reality gap. This difficulty arises from the gap between the skills robots learn in controlled, simulated environments and their ability to apply those skills in the dynamic real world. This gap stems from two main factors.

First, there are sensorimotor fidelity issues. Real-world sensors and robotic actions are inherently limited and prone to unexpected variations [95]. Unlike perfectly simulated counterparts, real sensors can provide noisy or inaccurate data, and robots may experience slight deviations in movement during operation. This distributional shift, where the real-world data deviates from the simulated data, creates a significant hurdle for developing robust and reliable robotic systems. Robots trained in simulations often struggle when faced with environmental perturbations, unexpected sensory feedback, or minor deviations in movement during real-world deployment.

Second, simulation environments prioritise specific tasks and objects, limiting a robot's ability to generalise skills. This limited scope can make it challenging for robots to adapt to new scenarios or variations in object properties and task requirements. Furthermore, the computational constraints of physics engines in simulations often necessitate simplifications or approximations. These limitations can lead robots trained under these conditions to exhibit unrealistic and ineffective behaviours when confronted with the complex dynamics of real-world physics [96]. A detailed overview of the contributions, limitations, and future directions of recent Sim2Real studies can be found in Table 4. The issue of Sim2Real is primarily driven by the **Simulation optimisation bias** (SOB), where policies optimised in simulation fails to transfer effectively to real-world settings due to differences in dynamics, physics, and environmental complexity. Recent approaches to addressing this challenge include the development of algorithms

like **Simulation-based Policy Optimisation with Transfer-ability Assessment** (SPOTA), which employs domain randomisation and a novel stopping criterion based on the estimation of SOB to enhance the generalisability of learned behaviours [97]. Additionally, iterative methodologies such as the Sim2Real2Sim strategy proposed by Chang et al. [95] refine simulations with real-world feedback to improve model fidelity. Moreover, the integration of Virtual Reality (VR) and digital twins, as explored by Alves et al. [98], offers immersive and adaptable platforms for remote robotics training and testing, mitigating data collection challenges imposed by scenarios like the COVID-19 pandemic. These solutions highlight a diverse approach to bridging the Sim2Real gap, emphasising the need for adaptive domain randomisation, continuous simulation refinement, and the leveraging of advanced technologies to enhance the accuracy and applicability of simulation-based learning in achieving real-world robotics competency.

2.3.4. Variability and uncertainty

Products arriving for EoL treatment exhibit significant variability and uncertainty, including differences in wear, tear, damage, and modifications from their original designs. This variability demands robotic systems that are flexible and adaptable, capable of adjusting their disassembly strategies in real time to manage the unpredictable nature of EoL products effectively. Achieving this level of adaptability necessitates sophisticated decision-making algorithms and possibly learning capabilities for robots [100,101].

A study by Foo et al. [40] addresses this challenge by presenting an artificial learning framework for robotic disassembly that enables a robotic system to learn new product structures and reason about existing knowledge of product structures. It works by relying on a database of past disassembly experience, which allows it to uncover the rule relations between disassembly components and concepts based on a knowledge base to generate further an ontology for future tasks of the disassembly process. The results show an impressive improvement in successful part identification, from 11% to 87%, after training on past disassembly experiences [40].

A fuzzy neural network control mechanism is developed with the adaptive to impedance learning to provide a new control strategy for constrained robotic systems by He et al. [100]. The author's method focuses on the interaction between the contact dynamics between the robot and the environment, emphasising where a robot needs to cope with physical constraints. Its capability is enhanced by fusing fuzzy logic with neural networks. In this case, impedance learning refines the control by adaptively adjusting the robot's behaviour so that the executed task develops a desired force and motion profile and allows the user to achieve compliant, smooth operation under changing conditions. This capability extends, in a broad way, the robotic manipulation capabilities to be applied for any industrial and exploratory task when delicate handling and flexibility are critical.

3. Disassembly approaches

Robotic disassembly approaches offer innovative automated recycling and maintenance solutions, addressing the increasing need for efficient resource management and waste reduction. These methodologies, detailed in Table 5, encompass various modelling strategies, each with unique advantages and challenges, facilitating diverse industrial applications.

3.1. Finite state machines

Finite State Machines (FSMs) mathematical models of computation have been Known for their simplicity, played a role in the development and application of robotic behaviours, offering several advantages that have made them a staple in traditional robotic applications [105]. Their simplicity and easy implementation stand out, as FSMs provided a

Table 4
Contributions, Limitations, and Future Directions from Sim2Reality papers.

Paper	Contributions	Limitations	Future Directions
Assessing Transfer-ability from Simulation to Reality for Reinforcement Learning, [97]	Introduces SPOTA, employing domain randomisation and SOB estimation for training control policies.	Relies on simulation accuracy and parameterisation. Needs validation across tasks and platforms.	Explore adaptive domain randomisation and integrate dynamic environmental factors.
Sim2real2sim: Bridging the Gap Between Simulation and Real-World in Flexible Object Manipulation, [95]	Proposes Sim2Real2Sim for iterative refinement of simulations based on real-world feedback. Demonstrated in the DARPA Robotics Challenge's Plug Task.	Task-specific solutions and heavy reliance on real-world data collection.	Apply to a broader range of tasks and develop automated model updating mechanisms.
The study introduces the Habitat-PyRobot Bridge and a new metric, Sim-vs-Real Correlation Coefficient, to evaluate if simulation performance predicts real-world robot performance, finding significant sim2real gaps due to agents exploiting simulator flaws. [99]	Offers a novel approach for simulation to reality robot transfer and quantifies simulation predictivity, revealing the necessity of tuning simulation parameters for accurate real-world performance prediction.	The agents' ability to exploit simulator imperfections leads to a discrepancy between simulated and real-world performance.	Future research could focus on refining simulation parameters for improved real-world predictivity and expanding the framework to a wider range of tasks and environments.
Developing a VR Simulator for Robotics Navigation and Human-Robot Interactions Employing Digital Twins, [98]	Uses VR and digital twins for remote robotics training and testing, offering a solution to pandemic-related restrictions.	Simulation fidelity and hardware constraints of VR technology.	Enhance realism of simulations and explore applications beyond navigation and fall detection.

Table 5
Detailed Summary of Modelling Approaches with Advantages, Disadvantages, and Key References.

Modelling approaches	Studies	Advantages	Disadvantages
Finite State Machine & Petri Nets	[102–106]	▷ Simple and clear for systems with limited states. ▷ Good for modelling concurrent processes.	▷ Becomes complex for large systems. ▷ Difficult to scale Petri Nets for large applications.
Matrix-Based Methods	[68,107]	▷ Structured for quantitative analysis and optimisation. ▷ Versatile for diverse problems.	▷ Computationally intensive for large systems. ▷ Abstract, less intuitive for dynamic behaviours.
Graph-Based Methods	[75,108]	▷ Visually intuitive and flexible. ▷ Suitable for various systems.	▷ Management becomes challenging for large networks. ▷ Requires sophisticated algorithms for analysis.
Graph-Based Methods (Focus on Tree-Based)	[72–74,78]	▷ Ideal for hierarchical structures. ▷ Efficient for specific operations like search and deletion.	▷ Cannot represent cyclic relationships. ▷ Trees can become unbalanced without careful management.
Other (PLC, Complexity)	[55]	▷ PLCs are robust and reliable for real-time industrial use. ▷ Scalable and handles complexity well.	▷ PLC programming can be inflexible. ▷ Complexity increases initial development time and cost.

straightforward method for modelling behaviour, enabling designers to quickly conceptualise and implement robotic actions, as noted by Endsley et al. [104]. The modularity complements this simplicity FSMs offer, allowing re-configurable manufacturing cell controllers to facilitate easier management and adaptation to various tasks or environments. Additionally, the formal modelling of affordances in human-machine cooperative manufacturing systems by Kim et al. [106], and the structured approach to discrete event systems modelling by Chen et al. [103] showcase the versatility and applicability of FSMs in enhancing robotic interactions and operations.

However, the limitations of FSMs become apparent as the complexity of robotic systems increases. Scalability issues emerge, with the state space of FSMs growing exponentially with the addition of components or behaviours, making them less suitable for managing complex systems efficiently [109]. FSMs sequential behaviour modelling presents challenges in handling concurrent processes, as discussed by Cortes et al. [110], complicating parallel disassembly actions' synchronisation. The limited dynamic behaviour adaptation of FSMs restricts their ability to spontaneously adapt to unforeseen environmental changes or task requirements without significant redesign [105]. Moreover, FSMs's inability to directly model uncertainties or probabilistic events and their

rigidity in behaviour modelling, despite attempts to introduce flexibility through parameterisation [111], limit their effectiveness in responding to real-time changes or incorporating continuous feedback. FSM limitations highlight the need for more advanced modelling techniques that offer modularity, maintainability, and reactivity.

3.2. Matrix-based

The matrix-based method effectively handles the disassembly of EoL products using a structured matrix to map the relationships among different product components. Known as a disassembly matrix, it organises the sequence of disassembly operations in a format quickly processed by computers and robotic systems [112]. In this matrix, rows and columns correspond to the product's components, and the entries detail the precedence relationships necessary for sequential disassembly-such as ensuring component A is removed before component B.

This method incorporates a disassembly precedence matrix to clarify the sequential relationships among operations [113]. The complexity of this matrix grows with the variable disassembly directions influenced by the components' spatial relationships in the product's **Computer Aided Design** (CAD) model, represented along the three Cartesian

coordinates [107]. Addressing this complexity are the conflict matrix, which identifies operational conflicts [114], and the dynamic disassembly priority matrix, adapted for changing conditions [85].

Moreover, matrix-based modelling manages uncertainties in part during disassembly. For instance, Ye et al. [115] investigated uncertain interference relationships between parts, establishing feasibility and confidence matrices to guide disassembly. Matrix-based methods are particularly beneficial in HRC contexts, as shown by Xu et al. [61] in their study on retired computer cases, where matrices defined stability, contact, and interference relationships, aiding in the optimisation of task sequences. Similar applications include [23], which introduced a method for generating disassembly solution spaces for Liquid-crystal display (LCD) screens, and [116], which used matrices to delineate the AND/OR relationships in disassembling an EoL bearing coupler. Wang et al. [94] presented a novel representation of representing physical contact and constraints using an assembly matrix and simple logic gates to generate a contact matrix, a space interference matrix, and a relation matrix, allowing the robot to schedule disassembly sequences that can handle interlocked mechanical structures.

Additional studies have demonstrated the application of operation priority matrices in battery disassembly processes. For instance, Wu et al. [57] used such a matrix for disassembling Tesla Model S battery modules. In contrast, Wegener et al. [22] applied a similar approach to the Audi Q5 hybrid battery system. Furthermore, [117] developed tailored disassembly strategies for specific **Lithium-Ion Battery (LIB)** components.

While this matrix-based approach requires substantial computational resources and precise data management for handling complex EoL products, it is very good at scalable representation and integration with AI for automated decision-making. Furthermore, the matrix-based approach can also handle uncertainties and smart reasoning in disassembly processes; however, the complexity of their setup and maintenance, besides dependence on data quality, are among the most significant challenges.

3.3. Graph-based

Graph-based approaches are popular for robotic disassembly, providing structured and efficient methodologies for decomposing EoL products into their constituent parts [118]. A graph-based model in robotic disassembly is a computational and visual framework used to represent the relationships and sequence of operations required for the disassembly of products. This model employs graphs to depict the components of a product as nodes and the connections or dependencies between these components as edges [118]. Standard techniques include Petri Nets, AND/OR Graphs, and tree-based approaches.

3.3.1. Petri Nets

The introduction of Petri Nets models addresses some of FSMs limitations by providing a more expressive framework for modelling complex, concurrent, and dynamic systems. Petri Nets, for instance, offer a graphical and mathematical modelling tool capable of representing parallel processes and enabling more straightforward incorporation of uncertainties through stochastic or fuzzy extensions. The Petri net model designed for robotic disassembly maps out the disassembly process through several vital components [102]. Places in the model represent states like component conditions, robot availability, and tool readiness for disassembly. Transitions indicate robots' disassembly actions, such as unscrewing, cutting, or part detachment. Arcs connect places and transitions, mapping component flow and setting conditions for robot actions. Tokens within places mark the status of components, robots, and tools. Their movement through the network, triggered by transitions, simulates the disassembly progress, showing how each action moves the process towards complete dismantling [102].

As the complexity of the product and the number of different ways it can be disassembled increase, the Petri net models become

more complex and more challenging to manage [119]. This can make them less practical for real-time planning [120], where quick and efficient computational processes are needed. Additionally, Petri nets may not be the best choice for dealing with the uncertainties of disassembling used products, such as unknown conditions or configurations, because they typically require more flexible modelling approaches [101]. As computational techniques have advanced, newer and more dynamic algorithms have been developed, offering more scalable and computationally efficient solutions.

3.3.2. AND/OR graph

The AND/OR graph is a tool used in robotic disassembly to model all possible disassembly sequences, optimising the process by identifying the most efficient sequence based on various criteria. AND/OR graphs are essential for modelling disassembly operations, with multiple studies enhancing their application. In the context of robotic disassembly, the nodes represent the components or sub-assemblies of a product, and the hyper-arcs represent the disassembly operations that can be performed [64]. Min et al. [114] introduced a weighted version to represent disassembly costs. In contrast, Tian et al. [121] and Chen et al. [122] used transformed AND/OR graphs to clarify priority relationships in EoL products. Bentaha et al. [123] and Ren et al. [124] applied AND/OR graphs for planning and analysing disassembly in different products, with Ren et al. developing an improved genetic algorithm for optimisation.

Similarly, Xu et al. [125] and Liu et al. [126] constructed AND/OR graphs for HRC disassembly, incorporating Q-learning and dynamic transformations to manage uncertainties. However, representing complex disassembly with AND/OR graphs can be challenging, especially in expressing priorities simultaneously and avoiding combination explosions, as seen in the Chevrolet Bolt battery with its numerous parts [127]. However, AND/OR graphs struggle to include broader objectives for comprehensive decision-making, as there is inherent complexity in large graphs, which limits their adaptability to changes and increases preprocessing time. Additionally, AND/OR graphs require detailed prior knowledge of the product structure and all possible disassembly sequences. In environments where robots encounter products with unknown or variable configurations (different models of EVB), creating and maintaining these graphs can be challenging and resource-intensive.

3.4. Tree-based approaches

Interestingly, a subcategory of graph-based approaches, which includes techniques such as **Rapidly exploring random Trees (RRT)**, **Decision trees (DT)** and **Behavior trees (BTs)** offer a more dynamic and flexible approach to modelling and executing disassembly tasks. These methods streamline the integration of decision-making and adaptability to fluctuating conditions, making them ideal for settings with expected uncertainties and variations in EoL product conditions. Their hierarchical structure, marked by transparent parent-child relationships, makes it easier to decompose disassembly tasks into smaller, manageable sub-tasks, improving the decision-making and oversight of disassembly processes.

Building on this, Torres et al. [128] developed an automatic disassembly system using DTs for robot task allocation. This foundational study demonstrated how tree structures can enhance decision-making and efficiency in robotic task distribution. The DT task planner organised robot activities based on their operational areas and specific features, showing the benefits of cooperative robotic systems in reducing disassembly time and handling complex tasks that single robots cannot perform alone. Further refining tree-based methods in disassembly planning was done by Ebinger et al.

Notable tree architectures include RRTs and the parallel RRT-based path planning introduced by Aguinaga et al. [108], which were two essential contributions to path planning. Lambert et al.'s research focused

on minimising sequence-dependent costs, offering a strategic viewpoint on cost reduction in disassembly processes. Although computationally intensive, this approach showed a reduction in operations.

Aguinaga et al.'s application of parallel RRT showcases an innovative method for navigating complex disassembly spaces, enhancing the adaptability and efficiency of robotic systems in dynamic environments [108]. Another study by Gammell et al. [79] introduced the informed RRT*; this variant improves the efficiency of path planning by focusing the search on an ellipsoidal subset of the search space, which is determined by the cost of the current best path and the start and goal positions. The algorithm's focus on a specific subset of the search space, based on the current best path and the particular start and goal positions relevant to the disassembly process, could significantly enhance the efficiency and speed of robotic arms or machines in identifying optimal disassembly sequences [79].

Moreover, integrating human expertise with robotic precision, introduced by Ding et al. [75] through the development of a robotic task-oriented knowledge graph, illustrates a forward-thinking approach to HRC in disassembly tasks. HRC capabilities open new avenues for tackling complex disassembly operations. However, the success of [75] heavily depends on the structure and maintenance of the underlying knowledge graph.

Further refining tree-based methods in disassembly planning was done by Ebinger et al. [76], who presented a robust disassembly planning framework that demonstrates improved performance by quickly finding solutions and efficiently managing parallel task processing through hierarchical planning while incorporating adjustments to the standard RRT method to mitigate issues in constrained settings, such as optimising the sampling strategy to prevent thrashing in tight spaces. However, the framework struggles with highly constrained configurations and relies on precise sub-assembly identification, impacting its overall efficiency. Additionally, the flexibility to switch between various search strategies and separation techniques introduces computational overhead, which can be challenging in scenarios requiring fast decision-making.

BTs have been recognised for their advantages over traditional sequential architectures (FSMs), especially in modularity and re-usability. Colledanchise et al. [109] study serves as a guide, offering insights into designing BTs for complex robotic behaviours/ Their study detailed several case studies on implementation for robotic disassembly tasks, showing improvements in system flexibility and responsiveness to environmental changes. Recently, Ghzouli et al. [72] compared BTs and state machines, which offered insights into why BTs are more suited for dynamic and complex disassembly tasks. Their research presented a detailed theoretical analysis supplemented by practical experiments comparing the two methods in various robotic applications, highlighting BTs superior flexibility and scalability. Similarly, Pezzato et al. [73] took a novel approach by combining active inference with BTs, detailing an advanced framework for robotic decision-making that anticipates and reacts to changes in real-time. Their study offered a deep dive into the theoretical underpinnings of active inference, its integration with BTs, and real-world applications in robotic disassembly, illustrating marked improvements in adaptability and responsiveness to unforeseen challenges.

A recent advancement in **collaborative robotics** (cobots) merges BTs with computer vision proposed by Paxton et al. [78]. Their system, **Collaborative System for Task Automation and Recognition** (CoSTAR), allows non-experts to program and interact with cobots intuitively. CoSTAR task plans combine high-level perceptual data with spatial reasoning and robot actions for adaptability and reliability. Additionally, it simplifies complex tasks into manageable structures, improving maintainability. However, while adaptable, CoSTAR setup in new environments can be challenging, requiring inputs like **Support Vector Machines** (SVMs) and 3D models. Performance depends on accurate object detection and pose estimation, with errors potentially

hindering execution. Despite limitations, CoSTAR makes robotic programming more accessible. Future work should focus on improving vision robustness under diverse conditions and developing more precise sensing and perception algorithms.

Similarly, Rastegarpanah et al. [74] built a novel semi-autonomous framework for sorting EVB components using a mobile manipulator, using BT for cognitive task execution and monitoring. Integration of the different capabilities of a robot, say, navigation, object tracking, and motion planning, to name a few, can easily be done in a modular way through simulation testing in both static and dynamic environmental scenarios. The study [74] shows a high rate of success when applying BTs to the sorting EVB. Considering the diversity in battery types and their varied conditions, the framework needs to be expanded to additional experimentation. The BT framework used is designed with decision nodes specifying their actions towards identifying and sorting different battery components, along with performance metrics for the system in demonstrating the effectiveness and efficiency gains over manual sorting methods. While the framework demonstrates promising results in simulation, transferring these capabilities to real-world scenarios may present additional challenges, including dealing with more unpredictable environmental conditions and ensuring the reliability of object detection and manipulation as highlighted in Section 2.3.3. Additionally, the semi-autonomous framework requires user input to initialise the object tracking system, which could limit the system's autonomy and efficiency. Future improvements might include more advanced, autonomous object detection and pose estimation methods to enhance the system's independence.

The introduction of a **Large Language Model** (LLM)-based BTs using **Natural Language Processing** (NLP)-enhanced Dialogue was proposed as **LLM-Multi-Agent Robot Systems** (LLM-MARS) by Lykov et al. [129], facilitating dynamic human–robot interactions and allowing robots to comprehend and execute complex commands with high accuracy. With an average task execution accuracy of 79.28% for compound commands and exceeding 90% for commands containing up to two tasks, LLM-MARS demonstrates the potential for significantly improving the efficiency and productivity of multi-agent robotic systems. This increase in task execution accuracy reduces the likelihood of errors and rework, streamlining operations and potentially lowering operational costs. While the practical trials within the Eurobot 2023 game rules provide a focused application scenario, the underlying technologies of LLM-MARS have broader applicability. Beyond logistics, such systems could enable autonomous exploration missions in hazardous or inaccessible environments and support complex manufacturing disassembly processes.

Graph-based studies demonstrate much potential for furthering robotic disassembly, particularly in sequence optimisation and strategic planning. However, challenges such as computational demands, model complexity, and adaptability remain critical areas for improvement. Their flexibility allows for easy adaptation to new tasks, quick fixes, and dynamic adjustments to unexpected situations.

3.5. Machine learning and AI approaches

Machine learning in robotics falls into supervised, unsupervised, and **Reinforcement learning** (RL) categories. Supervised learning leverages labelled data, fuelling advancements in robotic perception, sensing, and navigation (primarily through deep learning and computer vision). Unsupervised learning finds patterns in unlabelled data, aiding robotics in complex scenarios. RL trains goal-oriented behaviours with reward systems, driving progress in robotic adaptability; a simplified instance of this can be seen Fig. 6.

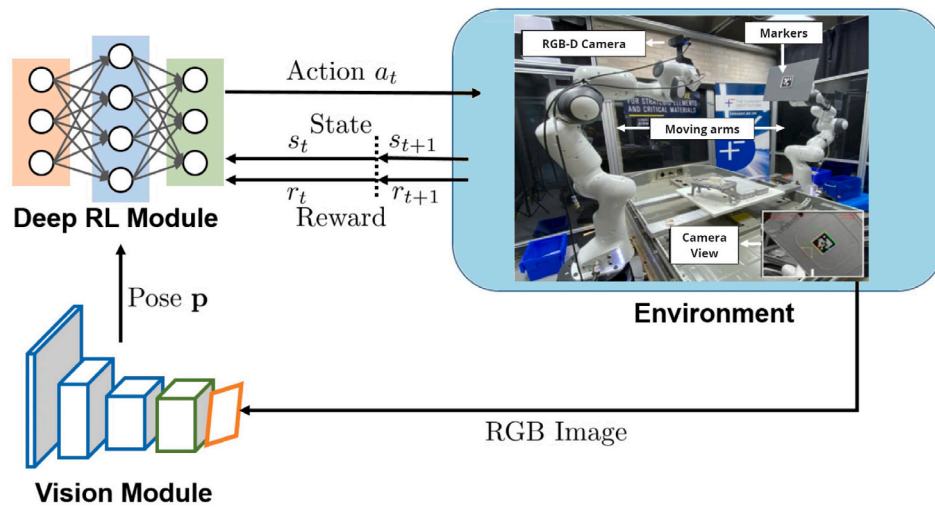


Fig. 6. A schematic diagram of an RL framework, where an agent senses its environmental states s and takes action a according to a policy π that is optimised for future reward r . This specific scenario combines computer vision and robotic control using a deep policy network.

3.5.1. Deep learning and Computer vision

Deep learning and computer vision have seen remarkable advancements in recent years, significantly impacting various applications, including automated disassembly [130]. Machine vision integration, which encompasses object detection, localisation, and segmentation, has become pivotal in industrial applications. These technologies achieved remarkable accuracy and efficiency. Research employing deep learning models such as **Convolutional Neural Networks (CNNs)** and their derivatives, including Faster R-CNN, Mask R-CNN, and **You Only Look Once (YOLO)** [131], along with innovative methods for 3D perception [41], have showcased the profound impact of AI integration in tasks that were once considered difficult for automation. The deep learning architectures have empowered machines to analyse and interpret visual data with unparalleled precision.

The advancement in deep learning, precisely Computer vision, primarily uses two types of input data: 2D RGB images and 3D point clouds, each with distinct methodologies and applications tailored to the challenges of disassembling electronic waste, optimising railway fastener detection and enhancing structural health monitoring, among others. Table 6 presents a concise comparison of 2D RGB image and 3D point cloud processing methods. For 2D RGB images, region proposal-based two-stage methods have been developed to improve the accuracy of object detection tasks, such as screw detection in electronic waste disassembly. Studies like [42–45,48] exemplify the use of deep learning models, like Faster R-CNN, combined with techniques such as gamma correction and rotation edge similarity, to achieve high precision in identifying and classifying objects for disassembly. Despite their high accuracy, these methods are noted for their slower inference speeds and higher computational costs, which may limit their suitability for real-time applications.

Another electronic waste disassembly proposed by Few et al. [45] used screw detection and gamma correction to standardise image brightness and leverage a deep learning model to detect screws visually. Their method not only improved detection accuracy but also employed logical reasoning to identify likely undetected screws based on common location patterns, further refining the model with re-training on missed detections. Su et al. [42] developed a Mask R-CNN-based laptop parts detector, focusing on the disassembly scenario. By generating mask images of parts and predicting their rotation angles, the system applied specific rules based on quantity and space constraints to enhance accuracy, demonstrating the potential of deep learning in automating the disassembly process. Zhang et al. [48] validated machine vision and deep learning for structural health monitoring, specifically for autonomous bolt loosening detection. By training a Faster R-CNN model

on a dataset of images depicting various bolt states, they achieved an average precision of 0.9503 in bolt damage detection, showcasing its capability to detect bolt loosening with high accuracy under diverse conditions.

Similarly, Bai et al. [43] proposed an optimised railway fastener detection method, combining a modified Faster R-CNN with **Support Vector Data Description (SVDD)** algorithms. This two-stage classification model significantly improved the efficiency and precision of fastener detection, highlighting the effectiveness of deep learning models in enhancing the accuracy of object detection tasks. On the other hand, Li et al. [44] introduced an accurate screw detection method that combines Faster R-CNN with a **Rotation edge similarity (RES)** algorithm for automatic screw disassembly. This method achieved remarkable location deviation and classification accuracy, significantly improving the success rate and speed of automated screw disassembly for mobile phone mainboards. Despite these methods' high accuracy, they are characterised by slower inference speeds and higher computational costs, posing challenges for real-time applications. The computational intensity and time required for processing may limit their practicality in scenarios where rapid decision-making and action are essential. However, the detailed exploration and application of these advanced techniques underscore the potential of deep learning models to revolutionise the field of robotic disassembly, offering pathways to more efficient, accurate, and automated processes. Building upon advancements in object detection, Lee et al. [132] explore an HRC task planner that incorporates **Human Behaviour Prediction (HBP)** using Faster R-CNN in disassembly tasks with UR-5e robots. While the previous study focused on automating screw disassembly through accurate detection, the aforementioned study investigates how robots can collaborate with humans in disassembly tasks by predicting human actions.

In contrast, region proposal-based one-stage methods have emerged as a strategy for enhancing the speed and efficiency of real-time tasks. This approach is used in the works of [38,46,133], which leverage advanced models such as Tiny-YOLOv2 and an improved version of YOLOv5. These models are distinguished by their ability to perform object detection tasks directly from full images in a single evaluation, thereby streamlining the detection pipeline and significantly reducing the computational burden. Wong et al. [134] employed Mask R-CNN and Conditional Random Fields (CRFs) for object and keypoint detection, achieving high accuracy (sub-centimeter error). However, their method requires training on both backgrounds and specific objects. Another approach developed by Yildirim et al. [135] is a custom neural network-based object detection algorithm optimised for refuelling

Table 6
Comparison of 2D RGB Image and 3D Point Cloud Processing Methods.

2D RGB Image Format		
Method	Advantages	Disadvantages
<i>Region Proposal-Based Two-Stage Methods</i> [130]	▷ High accuracy in object detection tasks ▷ Precise detection results ▷ Handles a wide variety of object scales	▷ Slower inference compared to one-stage ▷ Computationally expensive ▷ Not suitable for real-time applications
<i>Region Proposal-Based One-Stage Methods</i> [130]	▷ Designed for speed and suitable for real-time ▷ Simplifies pipeline ▷ Efficient in memory and computation	▷ Lower accuracy for small objects ▷ Struggles with class imbalances ▷ More false positives
3D Point Cloud		
Method	Advantages	Disadvantages
<i>Point-Based Methods</i> [136]	▷ Directly processes raw data ▷ Robust to noise and partial occlusions ▷ Handles unordered data	▷ Computationally intensive, Requires much memory ▷ Sensitive to point cloud density
<i>Multi-View Based Methods</i> [137]	▷ Leverages 2D CNN architectures ▷ Efficient in computation and memory ▷ Improves performance with 2D techniques	▷ Loss of spatial information ▷ May not capture intricate details ▷ Requires alignment of views

tasks. Applied to a 6 **Degree of freedom** (DoF) robot arm, it achieved a 96.6% accuracy rate. While adaptable to other object localisation scenarios, its reliance on extensive, object-specific data labelling limits its effectiveness in tasks with high variabilities, such as battery pack disassembly.

Similarly, screw detection in electronic devices was explored by Brogan et al. [46] using Tiny-YOLOv2. The authors achieved an optimal compromise between detection performance and resource utilisation by tuning the input resolution and mini-batch size parameters; this resulted in high average precision scores across different testing datasets, highlighting the effectiveness of one-stage methods in faster detection tasks, which are common in robotic disassembly and servicing applications. Building on updated versions of the YOLO approach, Chen et al. [38] enhanced the YOLOv5 model for classifying and positioning circuit board components, addressing the challenge of automatic dismantling and recycling of electronic components. The combined approach significantly improved recognition accuracy and precision by integrating a hierarchical classification algorithm, demonstrating a 38% increase in algorithm effectiveness compared to the original YOLOv5 model. This is particularly relevant for disassembling complex electronic waste, where identifying and localising small components are essential. While one-stage methods offer substantial benefits in speed and computational efficiency, they do come with a trade-off in accuracy, particularly for smaller objects. This is because the model's design aims to simultaneously evaluate all possible locations in an image, potentially leading to a higher **false positive rate** (FPR). Despite this, the advancements in one-stage detection methods represent a significant step forward in developing real-time robotic disassembly systems and efficient robotic vision capabilities.

The domain of 3D point clouds introduces point-based and multi-view-based methods for handling the unique challenges of three-dimensional data. The processing of 3D point clouds is critical for accurately identifying and manipulating objects. Point-based methods, which directly process raw point cloud data, are shown in the following works [39,41]. These methods are particularly advantageous for their robustness to noise and capability to handle unordered data, which is inherent in point clouds. For instance, Zheng et al. [39] used PointNet, a deep neural network, to identify components of turbochargers from 3D point clouds. Their method could consistently train classifiers to recognise objects accurately, even in sensor imprecision. Wu et al. [59] investigated Sim2Real transfer learning for point cloud data in screw

localisation for disassembly. They proposed a novel patch-based attention network to address imbalanced learning, achieving effective screw localisation in a simulated environment. However, the method's accuracy dropped by 12.95% when applied to real-world scenarios, highlighting the limitations of sim-to-real transfer and the need for larger datasets for training. Similarly, Braadland et al. [41] described a design based on 3D perception methods for capturing point clouds and segmenting instances of cabling on electric vehicle battery packs, demonstrating graph-based deep-learning algorithms' scalability and robust performance.

Multi-view-based methods project 3D point clouds into 2D images to leverage established 2D CNN architectures for efficiency. This approach, as seen in the works of [138–140], can lead to faster processing times where Pang et al. in [138] proposed a method that uses a CNN for 2D detection tasks, handling all viewpoints and rotations for the same class of object together, which resulted in a significant speed-up compared to traditional 3D point cloud detection methods. Similarly, Jaritz et al. [139] introduced **Multi-view PointNet** (MVPNet), aggregating 2D multi-view image features into 3D point clouds and then using a point-based network to predict 3D semantic labels, showing superior robustness to varying point cloud density. Another recent study by Bilal et al. [140] focused on the automation of EV LIB pack disassembly, merging images from multiple viewpoints using Multiple-View Super Resolution to achieve higher accuracy in screw detection tasks. While point-based methods offer high accuracy and robustness, they are often computationally intensive and require significant memory, limiting real-time applications. Multi-view-based methods, although more efficient, may compromise the richness of spatial information due to the projection process, potentially affecting the detection accuracy. These trade-offs highlight the importance of selecting the appropriate method based on the specific requirements of the robotic disassembly task, whether it be the need for high precision, computational efficiency, or real-time processing capabilities.

3.5.2. Reinforcement learning

Parallel to the strides made in deep learning and computer vision, reinforcement learning (RL) has emerged as a pivotal framework for optimising decision-making processes in dynamic environments. This branch of machine learning, characterised by its ability to learn optimal strategies through trial and error, has been instrumental in advancing robotic disassembly, HRC, and other automated tasks. In RL, an agent senses the state of the environment and learns to take appropriate

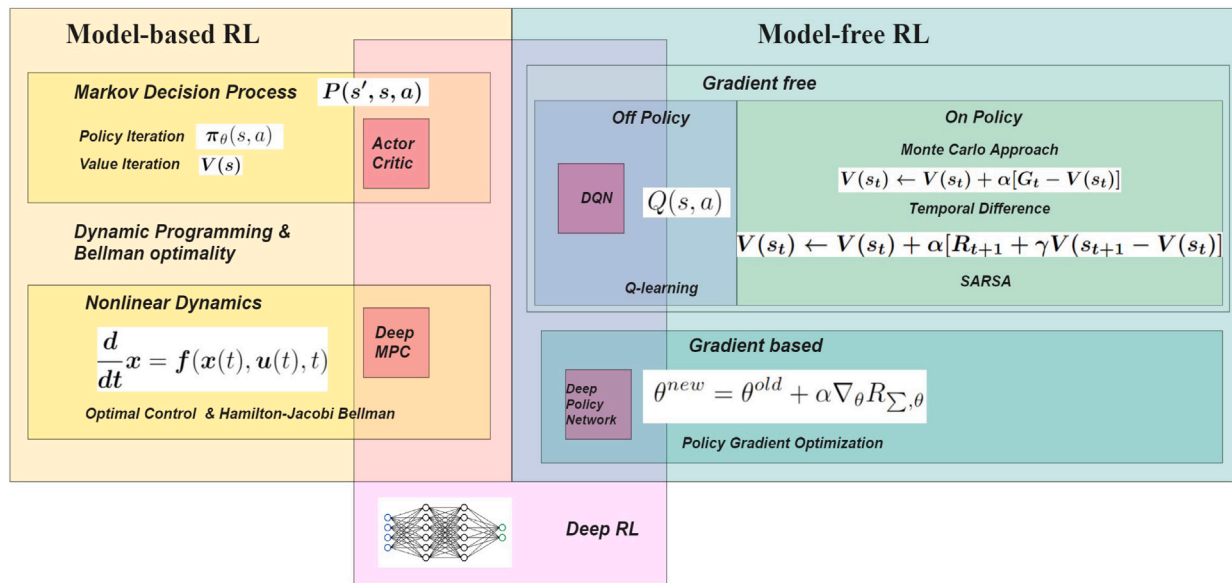


Fig. 7. This taxonomy provides a broad overview of RL techniques. A fundamental distinction in this classification is between model-based and model-free RL techniques [141].

actions to optimise future rewards. The ultimate goal is to learn a set of control strategies through positive or negative reinforcement. An RL agent interacts with the environment by taking actions based on a policy that is optimised through learning to maximise future rewards. Given a policy, we define a value function that quantifies the desirability of being in a particular state. Developing an optimal policy can range from closed-form optimisation to high-dimensional, unstructured optimisation.

A significant classification in RL is model-based versus model-free RL. In model-based RL, where a known model of the environment exists, strategies such as policy iteration and value iteration—forms of dynamic programming using the Bellman equation—are used to learn the optimal policy or value function. When there is no model for the environment, alternative strategies like Q-learning are employed. An overview of this categorisation is illustrated in Fig. 7.

Recent studies employing Q-learning [142], **Twin Deep deterministic policy gradient** (TD3) [143], **Deep Q-Networks** (DQN), and other RL algorithms highlight the method's versatility in adapting to various operational challenges [141], from precision disassembly to efficient task scheduling. RLs integration into automated systems has enhanced process efficiency and safety, contributing to adaptive learning machines. RLs application spans discrete and continuous action domains, from AlphaGo's achievements to autonomous driving advancements, showcasing its versatility. Robotic disassembly tasks benefit from RLs fine-grained control in continuous action spaces, essential for precise disassembly manipulations [144,145]. RLs has been crucial in driving innovation within autonomous systems. For a detailed breakdown of RLs methods, refer to Table 7.

Robotic disassembly tasks often require sophisticated control mechanisms to handle the complexity and variability of the operations involved. These tasks can be effectively addressed by designing specialised controllers, such as velocity, position, torque, and force controllers, which operate based on inputs from a continuous space [146]. While significant progress has been made in applying RL within discrete action spaces, the nuanced requirements of robotic disassembly tasks make continuous action spaces a more fitting choice. Disassembly processes require precise manipulations that can only be achieved through continuous action-based RL. Research demonstrates that RL can facilitate efficient and adaptable robotic disassembly tasks, as studies like [144,145] illustrate.

A data-driven, model-free inverse rRL algorithm developed by Lian et al. reconstructs the unknown cost function of discrete-time (DT)

dynamical systems with antagonistic disturbances using only demonstrated trajectories without requiring knowledge of system dynamics or control policy gains [147]. This approach employs an inverse RL policy iteration scheme to derive an off-policy inverse Q-learning algorithm that includes Q-function evaluation, state-penalty weight improvement, and action policy updates, ensuring unbiased estimates even with exploration noises. The method is validated through examples, demonstrating robustness and accuracy in handling disturbances and uncertainties, making it highly applicable in robotic disassembly scenarios requiring optimal control under unknown conditions. Similarly, the study by Nguyen et al. investigated On-policy and Off-policy Q-learning algorithms for time-varying linear discrete-time systems (DTSs) under complete dynamic uncertainties, transforming the original DTS into a time-invariant DTS using the lifting method to ensure convergence without the conventional controllability condition [148]. Both algorithms are model-free, effectively using collected data, with the Off-policy technique enhancing data efficiency by reusing data across iterations. These methods can be used to optimise robotic disassembly by handling dynamic uncertainties, learning model-free and efficient disassembly sequences, and ensuring safety and stability through continuous data-driven adaptation.

The integration of RL in robotic disassembly has seen significant advancements, with studies focusing on optimising DLBP, **Selective disassembly sequence planning** (SDSP), and addressing tool deterioration. Furthermore, Tuncel et al. [36] successfully applied a Monte-Carlo-based RL technique to solve the DLBP, demonstrating the method's effectiveness in deterministic and stochastic environments. This study highlighted the robustness of RL in complex, large-scale problems but also noted the necessity for extensive sampling to avoid sub-optimal solutions. Similarly, [30] demonstrated the efficiency of an improved Q-learning algorithm in solving two-sided DLBP, explicitly developing a DQN-SDSP, outperforming the **State-Action-Reward-State-Action** (SARSA) algorithm, especially in large-scale cases. The study introduced a novel Q-table designed for DLBP involving robots and human workers, proving its efficiency across various case scales without impacting the algorithm's running time. The authors combine RL with a **Multi-level selective disassembly hybrid graph model** (MSDHGM), and the formulation of SDSP as a finite Markov Decision Process enables the adaptation to structure uncertainty. Despite its success, the study acknowledges the need for more sophisticated deep reinforcement learning approaches to tackle increased complexity in DLBP for complex models or large-scale problems. Additionally, Gu

Table 7
Reinforcement Learning (RL) Categorisation.

Category	Definition	Algorithms	Studies	Advantages	Disadvantages
Model-free RL	Learns without a model of the environment.	Q-learning, State-Action-Reward-State-Action (SARSA), Deep Q-Networks (DQN)	[30,31]	▷ Does not require a model of the environment, making it simpler and more robust. ▷ Can adapt to the environment directly from experiences.	▷ Higher sample complexity, requiring more interactions to learn effectively. ▷ Can be data-hungry and computationally intensive due to the need for storing and processing every experience.
Model-based RL	Involves learning or having access to a model of the environment.	Dyna-Q, Monte Carlo Tree Search, Model-based policy optimisation (MBPO)	[35]	▷ Can reduce sample complexity by using a model to simulate outcomes and plan. ▷ Enables more efficient learning from fewer interactions. ▷ Facilitates transfer learning to new tasks or environments.	▷ model bias and error risk can affect performance. ▷ Requires building and updating a complex model of the environment. ▷ Limited by the quality and availability of data for model accuracy.
Policy-based RL	Focuses on directly learning the policy.	REINFORCE, Proximal Policy Optimization (PPO), Actor-Critic methods, Deep deterministic policy gradient (DDPG), Twin-DDPG (TD3)	[27,28]	▷ Directly learns the policy, potentially leading to more efficient policies in high-dimensional or continuous action spaces. ▷ Easier to integrate with stochastic policies.	▷ Can suffer from high variance, requiring techniques to reduce it. ▷ May converge to local rather than global optima.
Value-based RL	Focuses on learning the value function.	Q-learning, Deep Q-Networks, Value Iteration	[141]	▷ Efficient in learning optimal policies in discrete action spaces. ▷ Simpler to implement and understand compared to policy-based methods.	▷ Difficulty handling high-dimensional or continuous action spaces. ▷ May struggle with estimating accurate value functions in complex environments.

et al. in [145] expanded the application of RL into complex 3D manipulation tasks, employing an asynchronous off-policy training of deep Q-functions. This approach significantly reduced training times and demonstrated the potential of deep RL algorithms in real physical robots despite the high sample complexity and initial limitations to more straightforward tasks. Both studies highlight the potential of RL in overcoming the challenges of large-scale and complex disassembly tasks. However, they also acknowledge limitations such as the need for extensive exploration to avoid sub-optimal solutions and the high sample complexity of direct DRL algorithms.

Implementing RL frameworks, particularly those based on **Actor-critic** (A2C) architectures can significantly enhance robotic disassembly processes by optimising control actions and ensuring collision avoidance and prescribed performance, even in the presence of modelling uncertainties [149]. Using RL to adaptively control robots enhances stability and robustness, guaranteeing semi-global uniform ultimate boundedness (SGUUB) of the system throughout the learning process [150]. Additionally, integrating RL with neural networks allows for continuous optimisation of control actions, improving the system's ability to handle dynamic and uncertain environments [151]. Moreover, coordinated control of dual-arm robots using (A2C) designs further enhances the efficiency and accuracy of complex disassembly tasks [152]. Finally, learning from expert demonstrations accelerates training and improves sample efficiency, making RL a practical approach for teaching robots intricate disassembly operations in dynamic environments [153].

Furthermore, the study by Cai et al. [28] applied the advantage **Actor-critic** (A2C) algorithm to solve DLBP considering tool deterioration, showing superior results compared to the actor-critic algorithm. A2C algorithm integrates an actor-network (Policy Network) and a critic network (Value Network) to simultaneously find feasible disassembly sequences, allocate tasks to workstations in an optimal manner, and minimise the cycle time of workstations. This marked the first application of A2C to DLBP, discussing states, actions, and environments in the context of tool deterioration. Despite its success, the study suggests further work to optimise the spatial complexity and adaptability of the algorithm, indicating room for improvement in algorithm efficiency and adaptability.

Another e-waste disassembly task that used RL was introduced by Kristensen et al. [34], who developed a framework for training and testing algorithms for e-waste disassembly, focusing on unscrewing tasks. Using a Q-learning algorithm within a simulation architecture that includes ROS, Gazebo, and MoveIt, this study provided a pathway from simulation to real-world application, albeit with limited framework complexity. **Reinforcement Learning Assigning Policy** (RLAP) introduced by Chen et al. [35] for multi-project scheduling in cloud manufacturing, achieving significant improvements in schedule quality. RLAP improved schedule quality by 32.1% and 5.7% compared to NSGA-II and Q-learning, respectively. It also effectively adjusted the service load distribution and scheduled stochastically arriving projects. This shows the dynamic state representation algorithm as a key innovation that enhances service load distribution. [35] demonstrates the RL approach's ability to improve schedule quality significantly compared to traditional methods. However, it also points to the potential for further improvement by incorporating heuristic knowledge.

DQN used for adaptive disassembly sequence planning in **virtual reality** (VR) maintenance training by Mao et al. [31], showcasing the method's applicability in on-site maintenance where faults are uncertain. The study integrates a fitness function from the genetic algorithm to replace the long-term return in DQN, marking a significant innovation and improving the efficiency of the training process. Applying RL into complex 3D manipulation tasks, with substantial contributions, was done by Gu et al. [145]. Their methodology involves an asynchronous off-policy training of deep Q-functions, which reduces training times and showcases the effectiveness of DRL algorithms in operating real physical robots. Despite the complexities involved and initial restrictions to simpler tasks, their work illustrates DRLs capability in managing large-scale and complex disassembly tasks. Nonetheless, the approach requires extensive exploration to sidestep sub-optimal solutions and grapples with the high sample complexity typical of direct deep RL algorithms.

In another advancement, Aflakian et al. [27] developed a method to optimise DRL by applying an offline workspace bounding technique inspired by human demonstrations. This method effectively confines the agent's search area to the most advantageous regions, accelerating the learning curve. Their findings highlight the utility of incorporating

a convex hull to define workspace boundaries, which significantly improves metrics such as slicing accuracy, path deviation, and the management of contact and force interactions. Despite these advancements, the restricted exploration space may hinder the agent's discovery of innovative solutions, a critical aspect of ongoing RL research. Similarly, Rastegarpanah et al. [146] fused RL, **Learning from Demonstration** (LfD), and Ensemble Learning to accelerate the RL refining process through an online hyper-volume action bounding approach. This approach leverages insights from multiple controllers to restrict the agent's action space, facilitating quicker policy improvements. Their work applied to a **Visual Servoing** (VS) task showcased advancements in training efficiency and superior outcomes in error convergence and manipulability metrics. A limitation of this method is its reliance on having multiple expert algorithms at hand, which might not always be feasible. The authors propose that future endeavours could integrate this technique with **Multi-Agent Reinforcement Learning** (MARL) to diminish the number of required training episodes [146]. Xiao et al. [24] introduced a MARL method for optimising disassembly sequences in HRC disassembly tasks, particularly for EVB recycling tasks. Their approach navigates the complexities and uncertainties inherent in EVB disassembly, using MARL to dynamically optimise disassembly sequences as environmental conditions change. They test their MARL on a case study on battery modules.

Nonetheless, the study acknowledges challenges in obtaining optimal sequences due to fluctuations in the real-time status and conditions of the components. The authors suggest that future research could enhance disassembly optimisation for EVB packs and incorporate digital twin technology to minimise human fatigue and improve safety [24]. The conventional method of handling EoL products has predominantly been a manual process, where human labour is used to dismantle items for recycling, remanufacturing, or disposal. This approach is marked by its labour-intensive nature, time-consuming operations, and economic inefficiencies, particularly in complex products such as electronics or vehicles. Additionally, the manual disassembly process poses significant health and safety risks to workers, from exposure to hazardous materials and the handling of sharp objects.

3.6. Human Robot Collaboration

HRC in disassembly processes leverages the strengths of both humans and robots, enhancing efficiency and safety. Recent research has shown significant progress in this area, with advancements in computer vision techniques and optimisation strategies playing a crucial role [12,24,57,154,155]. For instance, Huang et al. [58] optimised the disassembly sequence and reduced cycle times in a collaborative cell for disassembling complex automotive components like turbochargers. Studies such as [57] applied multi-objective optimisation for disassembling waste power battery modules from the Tesla Model S, aiming to reduce workstation idle time.

Researchers combine human dexterity with robotic precision to address high labour costs and safety risks in manual disassembly. For example, using **Particle Swarm Optimisation** (PSO) to obtain Pareto optimal solutions, Zhou et al. demonstrated enhanced screw removal processes [56]. Similarly, Kay et al. employed semi-automated robotic disassembly to tackle LIB modules from EVs [55]. This approach strategically assigns high-risk, precise tasks to robots while leveraging human expertise for complex decision-making.

Moreover, integrating human expertise with robotic precision, introduced by Ding et al. [75] through developing a robotic task-oriented knowledge graph, illustrates a forward-thinking approach to HRC in disassembly tasks. HRC capabilities open new avenues for tackling complex disassembly operations. However, the success of [75] heavily depends on the structure and maintenance of the underlying knowledge graph.

To optimise task allocation and ensure safety in HRC disassembly, a sequence planner can be employed that dynamically assigns tasks

based on real-time human motion prediction [132]. This approach by Lee et al. incorporates a cost function that factors in both movement distance and time, along with safety constraints, to protect the human operator [132]. Experiments with disassembling a wooden toybox and a used hard disk drive (HDD) demonstrated the planner's ability to generate optimal sequences, adapt to changing human actions, and uphold disassembly and safety rules. Future research could improve the model's predictive capabilities and test its suitability for a wider variety of disassembly situations.

Further advancements in HRC for the disassembly of EVB have been made by Yuan et al. [54] and Chu et al. [53], who introduced advanced algorithms to optimise disassembly strategies. Yuan et al. proposed a new heuristic for multi-criteria resilience assessment in spent lithium-ion battery recycling, incorporating human-robot collaboration parameters. Chu and Chen et al. applied a hybrid PSO with Q-learning to minimise disassembly completion times, demonstrating superior efficiency and robustness compared to other meta-heuristic approaches [53,54].

3.7. Telerobotics

Recent advancements in telerobotics and human-robot collaboration (HRC) significantly enhance disassembly processes in hazardous and complex environments. For example, Marturi et al. demonstrated the potential of visual servoing and autonomous grasping techniques in nuclear decommissioning, enabling novice users to guide robotic systems remotely [82]. Similarly, Petereit et al. combined teleoperation with virtual reality for enhanced 3D mapping and safer robotic decontamination in nuclear facilities [81]. Additionally, Bustamante et al. [156] used a Master-Slave teleoperated system with adaptable levels of autonomy for successful object manipulation with a 95% success rate; however, the approach lacked haptic feedback and did not account for joint space limits. Finally, [80] employed a master-servant framework using haptic devices and collaborative robot control for precise disassembly tasks, such as bolt removal and material cutting in EV battery recycling. Examples of such tasks are illustrated in Fig. 13.

Researchers are also investigating various teleoperation controllers for effective telerobotics. Kebria et al. [157] compared Cartesian and joint space control methods, suggesting joint space control is better suited for master-slave setups with mismatched degrees of freedom. To address force feedback, Singh et al. employed a joint torque coupling controller with twin robot arms, but their experiments were limited to simple object manipulation [158].

These studies collectively show the benefits of HRC and telerobotics, emphasising the necessity for ongoing research as we transition towards greater autonomy in disassembly; maintaining human involvement in the loop remains a critical first step.

3.8. Optimisation

In robotic disassembly line operations, there are five principal characteristics to consider: (1) the objective functions, which may include maximising profit, enhancing material recovery, minimising carbon emissions, or reducing energy consumption [159]; (2) the variable recovery rates of materials from disassembled EoL products, which introduce uncertainty [159,160]; (3) the need for distinct disassembly sequences and technologies tailored to products with different geometrical characteristics [161]; (4) the impact of disassembly line layouts on work efficiency, where different configurations can lead to varying productivity levels [160]; and (5) the challenge of balancing the disassembly line and structuring workstation cells, necessitating diverse models to effectively address these issues [159,160].

3.8.1. Single objective and Multi objective

Robotic disassembly processes can be enhanced through either **Single-Objective** (SO) or **Multi-Objective** (MO) optimisation strategies. SO optimisation zeroes in on a single criterion, such as minimising time or cost, which is ideal for straightforward disassembly tasks but may not capture the complexity inherent in such processes. In contrast, MO optimisation shows its flexibility through methods like combining all objectives (aggregate method) and finding the best balance between competing goals (Pareto approach) [92]. The choice between SO and MO optimisation hinges on the disassembly task's specific requirements, with MO offering a nuanced method that better suits the multifaceted nature of these tasks.

To optimise end-of-life product disassembly, a robotic-driven disassembly sequence generator was developed by Elsayed et al. [162]. This method promotes efficient disassembly processes, facilitating recycling and e-waste management. By automating disassembly, it reduces human involvement in potentially hazardous tasks. However, its primary focus on electronic products currently limits its broader application across various EoL products. A similar approach with enhanced precision was proposed by Friedrich et al. [51], where the authors created the **Robotic Visual Disassembly and Assembly System** (RoViDiAsS) control architecture for robotic visual disassembly/assembly systems [51]. RoViDiAsS control aims to minimise errors and streamline complex disassembly tasks where accurate identification and manipulation are critical. However, the study's focus on control architecture may limit its application to systems already equipped with advanced visual and robotic capabilities, potentially excluding lower-tech environments or more straightforward disassembly tasks [51]. Another SO study by Laili et al. [64] focused on improving robotic disassembly re-planning through a two-pointer detection strategy combined with a super-fast bees algorithm. Their approach addresses the dynamic nature of disassembly tasks, allowing for quick adjustments and optimisations in the disassembly process. This ability to rapidly re-plan and execute disassembly sequences minimises downtime and increases overall efficiency. However, the specific focus on re-planning through this particular algorithmic approach may not be universally applicable across all disassembly scenarios, especially those that do not exhibit the same level of dynamism or complexity. Similarly, Zhang et al. [63] explored the application of active compliance in peg-hole disassembly, presenting a method that enhances disassembly tasks' precision and success rate. This contribution is significant for tasks requiring high levels of accuracy and delicate handling. The study's limitation is its narrow focus on peg-hole tasks, representing just a subset of disassembly activities.

The studies focusing on (MO) optimisation in robotic disassembly highlight the significant advancements over (SO) optimisation by integrating comprehensive approaches that address the multifaceted challenges of robotic disassembly. For instance, Alshibli et al. [19] introduced a robust disassembly sequence using orthogonal arrays and task allocation, demonstrating MO optimisation's capability to balance efficiency with task complexity, offering a more holistic solution to disassembly challenges [19]. Similarly, Malekkhouyan et al. [163] showcased the integration of vehicle routing with job-shop-type scheduling in e-waste management, illustrating MO optimisation's application in navigating complex logistical and operational tasks, thereby enhancing both efficiency and effectiveness. Furthermore, Hartono et al. [92] employed the Bees Algorithm to optimise disassembly sequences with an emphasis on sustainability, effectively balancing profits, energy savings, and greenhouse gas emission reductions. This approach underlines MO optimisation's potential in aligning disassembly processes with broader sustainability objectives, offering a path towards more sustainable and efficient disassembly operations. Collectively, these studies overcome the limitations of SO studies by providing a nuanced methodology that accounts for the complexity of disassembly tasks, enabling a comprehensive optimisation of the process.

Furthermore, RDSP optimisation includes three main approaches: heuristic methods, which use algorithmic rules to find quick solutions but may fail with unexpected product classes; meta-heuristic methods, such as **Genetic algorithms** (GA) and **Neural networks** (NNs), which handle complex problems and large models by converging towards a solution without guaranteeing optimality, often requiring specialised software and more computational time; and exact methods, which use mathematical programming and search algorithms, like simplex and branch and bound, to systematically find the optimum solution, offering transparency and flexibility but struggling with nonlinear problems and high computational time, making them suitable for optimising moderate-sized models or evaluating heuristic/meta-heuristic approaches [92].

3.8.2. Meta-heuristic algorithm

Meta-heuristic methods have become increasingly popular for their flexibility and effectiveness in tackling complex, large-scale optimisation problems typical in disassembly and other industrial applications. These high-level, problem-independent frameworks guide heuristic algorithms to find satisfactory solutions to complex issues, often drawing inspiration from Nature. While heuristic algorithms apply problem-specific rules or intuition to quickly identify acceptable solutions without ensuring optimality, meta-heuristics offer broader applicability. They are extensively used in disassembly sequence planning to determine the optimal order of operations, in disassembly line balancing to enhance task distribution efficiency, and in task and motion planning for coordinating robotic actions and movements. Algorithms like **Genetic Algorithms** (GAs) [164], **Bee colony algorithm** (BA) [64, 68], **Simulated Annealing** (SA) [165], **Particle Swarm Optimisation** (PSO) [85], **Ant Colony Optimisation** (ACO) [15], and **Tabu Search** (TS) in disassembly operations offers practical [166], good-enough solutions within a reasonable time-frame. For instance, GAs can evolve optimal disassembly sequences for complex products, while SA navigates the vast search space to minimise time and maximise material recovery. Similarly, PSO and ACO can optimise collaborative efforts in disassembly operations, enhancing resource allocation and task scheduling among teams or robotic systems. For a detailed comparison of these algorithms' advantages and disadvantages in the context of disassembly operations, refer to Table 8 below, which highlights key strengths and challenges associated with each meta-heuristic approach.

3.8.3. Fuzzy Logic

Fuzzy logic is a computational paradigm that emulates human decision-making processes through approximate reasoning instead of binary reasoning. It uses IF-ELSE rules to guide robots through unstructured environments, essential in robotic disassembly tasks [121]. Moreover, fuzzy logic facilitates collaborative optimisation between humans and robots by accounting for variables such as cost, safety, and complexity, thus enhancing the efficiency and safety of disassembly operations [171]. Additionally, fuzzy logic supports optimising robotic disassembly sequence planning and line balancing, employing advanced methods like the improved discrete Bees algorithm to manage multiple objectives and constraints [61].

The study by Kim et al. [170] employs fuzzy methods to address the challenges of selective disassembly sequencing in environments where multiple components may be removed simultaneously. Additional notable research includes studies on intelligent fuzzy rule-based sequence planning [172] and fuzzy modelling of disassembly lines [161]. These methods leverage tools such as fuzzy logic, genetic algorithms, and Petri nets to manage the uncertainties and dynamic challenges inherent in robotic disassembly tasks.

Table 8
Advantages and Disadvantages of Meta-Heuristic Algorithms in Disassembly Operations.

Algorithm	Studies	Advantage	Disadvantage
Genetic Algorithms (GAs)	[164]	▷ Evolve optimal disassembly sequences for complex products through natural selection and genetic operators. ▷ Adaptable to multi-objective optimisation problems.	▷ Require significant computational resources and time for convergence, especially for vast search spaces.
Bee Colony Algorithm (BA)	[64,167]	▷ Mimics the foraging behaviour of bees to find efficient solutions. ▷ Good for optimising multi-objective and environmental impact tasks.	▷ Can get trapped in local optima; performance heavily dependent on parameter tuning.
Simulated Annealing (SA)	[165]	▷ Effective in minimising time and maximising material recovery by simulating the process of annealing. ▷ Robust for complex disassembly sequences.	▷ Cooling schedule and temperature parameter tuning are critical ▷ Convergence may be slow
Particle Swarm Optimisation (PSO)	[85,168]	▷ Enhances task scheduling through simulation of social behaviour. ▷ Known for its simplicity and fast convergence.	▷ Struggles with discrete optimisation and may converge prematurely to local optima.
Ant Colony Optimisation (ACO)	[15]	▷ Optimises complex disassembly tasks based on the behaviour of ants. ▷ Efficiently solves problems mimicking pheromone trail-following.	▷ Computationally intensive with potential for slower convergence in larger-scale problems.
Tabu Search (TS)	[166,169]	▷ Uses memory structures to escape local optima. ▷ Effective in navigating complex solution spaces.	▷ Management of memory structures and tabu lists is critical to prevent cycling and premature convergence.
Integer Linear Programming	[170]	▷ Supported by well-established techniques. ▷ Suitable for linear relationships and precision-critical problems.	▷ Limited by its assumption of linearity ▷ May not capture the complexity of disassembly problems.
Fuzzy Methods	[121,160]	▷ Offers flexibility and an intuitive approach to handling uncertainty in disassembly planning.	▷ Requires expert knowledge for designing fuzzy rules; solution quality varies based on fuzzy element definitions.
Reinforcement learning (RL)	[26–28,34]	▷ Enables algorithms to learn optimal strategies through trial and error, interacting with the environment, ideal for adaptive and complex decision-making tasks.	▷ Requires a large amount of data for training and can be slow to converge, making it challenging to apply in environments where simulations are costly or impractical.

3.8.4. Reinforcement learning

Reinforcement learning methods have emerged as alternatives to pure mathematical models [108] and heuristic algorithms [68], as highlighted in Section 3.8.2 and Section 3.8.3. Many of these traditional methods only consider static environments. Hence, the subfield of RL, **Policy Search** (PS) attempts to solve the problems through direct optimisation in the parameter space [144]. Studies like those by Qu et al. [25] have demonstrated the feasibility of training robots to perform disassembly tasks and transfer skills across different precision levels, emphasising RL’s adaptability to diverse scenarios. Similarly, Gullapalli et al. [173] showcased RL’s capability in acquiring skills for intricate operations like peg-in-hole insertion, overcoming challenges posed by sensor noise. Apolinarska et al. [174] extended the application of RL into the architectural construction domain, specifically in the automated assembly of timber joints, highlighting its potential in complex, contact-rich tasks. A cognitive robotics approach, incorporating learning and revision processes, was explored for efficient and adaptable automated disassembly [88]. Another method to optimise DSP, both in partial and full disassembly scenarios, an RL solution, was presented by Allagui et al. [26]. The study’s innovation is introducing a fitness function within the RL algorithm to optimise five disassembly parameters, leading to time-cost efficiency. The approach was applied to a demonstrative example and compared with other methods from the literature, showing its efficiency. The success of this approach is evident in the time savings achieved compared to GA and ACO [26]. However, the study recognises that implementing RL algorithms requires a high level of computer expertise, which could be a barrier to widespread adoption. Collectively, these studies underline the promise of RL in robotic disassembly, from enhancing adaptability and efficiency to reducing manual programming needs and handling uncertainty. However, they also reflect on the challenges, including high computational demands, significant data requirements, the risk of suboptimal policies, and issues in transferring skills from simulated to real-world environments.

3.8.5. Optimal nonlinear control

Optimal nonlinear control is significant in various applications where systems exhibit complex, nonlinear behaviours that linear control methods cannot adequately manage. Unlike linear control, which assumes a direct proportionality between variables, optimal nonlinear control can handle a broader range of dynamics and uncertainties. This is crucial for accurately modelling and controlling real-world systems such as robotics, aerospace, and automated vehicles, where linear approximations fall short.

The **Hamilton–Jacobi–Bellman** (HJB) equation (shown in Eq. (1)) is essential in optimal nonlinear control because it characterises the value function $V(x)$, which represents the minimum cost from any state x , derived from Bellman’s Principle of optimality. In this context, x represents the state variable, and u represents the control variable, with $f(x, u)$ describing the system dynamics and $L(x, u)$ being the running cost. The HJB equation effectively handles nonlinear dynamic systems, deriving optimal feedback control laws for continuous time and state spaces. It offers a general framework for various control problems, including those with constraints and stochastic elements, making it crucial for determining the best control policies in complex real-world applications.

$$\min_u \left\{ \frac{\partial V(x)}{\partial x} f(x, u) + L(x, u) \right\} = 0 \tag{1}$$

In reinforcement learning (RL), the principles of the HJB equation are utilised to find optimal policies, particularly in systems modelled as Markov Decision Processes (MDPs). RL algorithms, such as Q-learning and policy gradient methods, implicitly solve the HJB equation by iteratively improving the policy and value function based on observed data. This connection allows RL to handle various control problems involving nonlinear dynamics and uncertainties.

The significance of the HJB equation in RL is highlighted by its ability to provide a theoretical foundation for understanding and improving

Table 9
Key Innovations and Applications of Hamilton–Jacobi–Bellman Equation in RL Studies.

Study	Key innovation	Comparison to traditional methods	Application in robotic disassembly
[176]	Entropy regularisation MDPs used to improve exploration and robustness	Traditional methods often do not include entropy regularisation, which can improve exploration and policy robustness	Develop robust policies for handling variability in object states and configurations
[177]	Accelerates convergence with Successive Over-Relaxation using Q-learning	Traditional Q-learning does not use SOR, leading to slower convergence	Speed up learning for optimal disassembly sequences
[178]	Introduced an asymptotically optimal feedback planning algorithm integrating Hamilton–Jacobi–Bellman equation approximations, the Fast Marching Method, and adaptive mesh refinement	Outperforms PRM* and RRT* with faster, more accurate convergence and reduced computational cost through targeted mesh refinement	Optimises robotic disassembly by ensuring efficient, precise feedback planning for complex, nonholonomic constraints
[179]	Reinforcement learning solution for HJB equation arising in constrained optimal control problem	Traditional methods for HJB equations are typically on-policy and less flexible	Implement control strategies that respect physical and safety constraints
[180]	Bridging Physics-Informed Neural Networks with Reinforcement Learning: Hamilton–Jacobi–Bellman Proximal Policy Optimisation (HJBPPPO)	Traditional PPO methods do not integrate HJB equations or physics-informed networks	Enhance disassembly precision and efficiency by integrating physical knowledge

RL algorithms. By leveraging the HJB framework, RL can achieve better performance and stability in complex environments, driving advancements in fields ranging from autonomous systems to advanced robotics. Studies such as those referenced in Table 9 demonstrate the practical applications and effectiveness of using HJB equations in RL, highlighting their importance in developing robust and efficient control strategies.

The study by Nguyen et al. [175] presents a novel RL-based formation tracking control structure for multiple uncertain surface vehicle (SV) systems, highlighting the use of the Bellman equation and Lyapunov approach. By integrating an online actor-critic architecture, each RL agent learns an optimal policy to track desired trajectories, achieving lower cumulative costs and negligible tracking errors compared to non-RL methods. The Bellman equation minimises a cumulative cost function, while the Lyapunov function candidate ensures the stability of the closed formation control system, achieving uniformly ultimately bounded stability (UUB). Simulation results demonstrate the scalability of the proposed control structure, showcasing its computational efficiency in various formation scenarios. The study shows the potential of RL-based control approaches in real-world multi-agent systems. The actor-critic RL architecture facilitates real-time learning and optimisation, enabling robots to adapt to new tasks and improve performance without extensive reprogramming. Enhanced coordination among multiple robots ensures synchronised actions. This could greatly facilitate complex disassembly tasks, ultimately making the process more robust.

4. Task and Motion Planning

Task and Motion Planning (TAMP) is a crucial component of robotic disassembly and other complex manipulation tasks that require a series of sub-tasks to achieve a final goal. TAMP integrates high-level task planning—deciding on a sequence of actions to achieve a goal—with low-level motion planning, which calculates feasible motions for executing these actions in the physical world. For instance, in a disassembly task, the robot’s goal could be to dismantle a device into its constituent parts. High-level tasks might include identifying the parts to be removed, determining the removal sequence to minimise effort or damage, and planning the final placement of each part. Low-level motion planning involves the robot physically executing these

tasks, including approaching the target object, sensing its precise location, grasping it, and searching for trajectory paths to achieve the desired outcome. However, there is significant scope for improvement in planning solutions, especially concerning collision avoidance, environmental uncertainties, and optimisation strategies. Hence, the subject of TAMP has received a significant amount of interest from the robotic and AI community.

In the field of Task and Motion Planning (TAMP), research can be divided into several categories based on the main types of task planners used: Hierarchical-based (Hb-TAMP) focuses on breaking tasks into smaller subtasks, Belief-based (Bb-TAMP) emphasises the robot’s understanding of the environment, Optimisation-based (Ob-TAMP) seeks the most efficient solutions, Constraint-based (Csb-TAMP) defines task requirements and limitations, and Sampling-based (Sb-TAMP) utilises random sampling for solution exploration.

The popular **Programming Domain Definition Language (PDDL)** is extensively employed in various types TAMP for robotic manipulation. It represents an advanced form of traditional planning; since its publication, there have been several updated versions, including PDDL2.1, PDDL3.0, and PDDL+ [181]. Fig. 8 contains the PDDL description of the relevant code to unscrewing a bolt. It outlines the types of objects present in the domain (bolts and wrenches), predicates that describe the state of these objects (e.g., whether a bolt is tightened or loosened, whether a wrench is being held), and actions that can be taken (e.g., unscrewing a bolt, picking up a wrench). The actions are defined with parameters, preconditions that must be true for the action to be taken, and the effect of the action on the state of the domain. Similarly, Fig. 9 presents a specific problem instance within this instance (a particular bolt and wrench), the initial state of the environment (the bolt is tightened, and the wrench is available to be picked up), and the goal state to be achieved (the bolt is loosened). This problem instance is what the planning system aims to solve, using the actions and predicates defined in the domain to generate a sequence of actions (a plan) that transitions from the initial state to the goal state.

4.1. Hierarchical-based Task and Motion Planning

Disassembly environments have significant uncertainties (Section 2.3.4), affecting perception systems and limiting visibility. These issues

```

(define (domain unscrewing-bolt)
  (:requirements :strips :typing)

  (:types
    bolt wrench - object)

  (:predicates
    (isTightened ?b - bolt)
    (isLoosened ?b - bolt)
    (hasWrench ?w - wrench)
    (isHoldingWrench ?w - wrench))

  (:action unscrew
    :parameters (?b - bolt ?w - wrench)
    :precondition (and (isTightened ?b)
      (isHoldingWrench ?w))
    :effect (and (isLoosened ?b) (not (
      isTightened ?b))))

  (:action pickWrench
    :parameters (?w - wrench)
    :precondition (hasWrench ?w)
    :effect (isHoldingWrench ?w))
)

```

Fig. 8. PDDL Domain Definition for Unscrewing a Bolt.

```

(define (problem unscrew-bolt-problem)
  (:domain unscrewing-bolt)

  (:objects
    bolt1 - bolt
    wrench1 - wrench)

  (:init
    (isTightened bolt1)
    (hasWrench wrench1))

  (:goal
    (and (isLoosened bolt1))))

```

Fig. 9. PDDL Problem Definition for Unscrewing a Specific Bolt.

are more pronounced in dynamic industrial settings with frequent, unpredictable changes. Overcoming these challenges for long-term planning in such environments is difficult. Disassembly tasks can often be structured as hierarchical trees, with nodes representing disassembly stages and branches representing possible actions. Table 10 shows several **Hierarchical-based TAMP** (Hb-TAMP) studies.

Studies have looked into Hb-TAMP approaches to handle these uncertainties [109,182]. Hb-TAMP combines high-level planning with low-level motion control, creating a more adaptable and reliable framework for disassembly tasks.

The foundational work by [183] introduced a sound and complete hierarchical planning method, contributing significantly to the development of **Hierarchical Task Network** (HTN) planning. It functions as a task planner, using domain knowledge to search for solutions and manage the execution of each operation. This method was crucial for structuring complex tasks efficiently. An HTN is structured around two types of tasks: compound and primitive tasks. Starting with a task description, HTN systematically decomposes each task into its primitive components until no compound tasks remain. Each primitive task is transformed into a standard operator, complete with preconditions and effects defined by a set of literals. If a precondition proves unsatisfiable

during execution, the HTN allows for replanning at that specific point rather than requiring a return to the root task with adjustments to the domain. Kaebbling et al. [184] further improved upon this by integrating Task and Motion Planning (TAMP), introducing a top-down decision-making process within a detailed, continuous framework. This innovation reduced the search space required for planning, offering a scalable and efficient method tailored to robotic disassembly's complex and variable tasks.

Recent advancements in Hb-TAMP of robots in industrial settings include Alford et al. [185], which linked task and goal decomposition with task sharing, enhancing collaborative robotic disassembly lines. Similarly, Suarez et al. [186] emphasised the interleaving of hierarchical task planning with motion constraint testing for dual-arm manipulation, which is beneficial in disassembly tasks requiring precise and coordinated movements.

To streamline the setup and operation of robotic disassembly systems, Kast et al. proposed a set-theoretic model-based hierarchical planner to automate the automation process itself by Kast et al. [202]. This approach suggests a meta-level optimisation with significant efficiency potential. Castaman et al. [203] introduced a **Receding horizon-task and motion planning** (RH-TAMP) algorithm that can adjust to varying product conditions on the fly. RH-TAMP for a given task uses a geometric reasoning module that evaluates the feasibility of a sequence of a actions selected from a total of n actions by the task planner. The motion planner then directs the robot to the first viable configuration, q_i . The process advances by shifting the action window forward by one step. If the plan is not feasible, an action fails, or the system's state changes unexpectedly, the task planner is. This ability to deal with dynamic environments is very significant to robotic disassembly. Similarly, Newaz et al. [154] studied DRL with hierarchical planning, which effectively addresses uncertainties in TAMP. This method develops an optimal high-level tasking policy for multiple robots in a discrete state space. At the same time, a motion planner generates a precise low-level control policy for a single robot, accommodating physical constraints and uncertainties in a continuous state space.

To enhance the efficiency and robustness of robotic path planning, a hierarchical path planning algorithm can be employed [182]. This approach, proposed by Mesesan et al. [182], merges a global planner with multiple local planners working in parallel through a workspace decomposition strategy. The global planner generates collision-free paths for the robot's end-effector based on workspace information. The local planners then verify these paths within the configuration space by expanding a task space RRT. This strategy mitigates the shortcomings of purely sampling-based or optimisation-based planners, balancing exploration and exploitation while overcoming non-convergence or local minima.

The hierarchical path planner surpasses the performance of single-tree and parallelised bidirectional Configuration-space planners, particularly with high degrees of freedom in robots [204]. The algorithm decomposes the workspace into disjoint convex shapes, enabling parallel exploration and offering a scalable solution. It facilitates efficient task space exploration across various configuration subspaces while consistently succeeding in complex path-planning scenarios.

Stock et al. [201] leveraged **Conflict-driven Hierarchical Meta-CSP Planner** (CHIMP) with HTN planning, offering nuanced strategies for robotic manipulation and automating disassembly planning. Subsequently, Wang et al. [199] introduced space manipulator network planning using HTN with an improved A^* algorithm, catering to multi-arm robots for on-orbit servicing, demonstrating the adaptability of Hb-TAMP in spatially constrained and complex task environments.

Hierarchical Task Planning in robotics has evolved through various architectures, which include Behaviour Trees (BT) and **Extended Behaviour Trees** (EBTs), further advancing the framework for decomposing complex tasks into manageable components. BTs, initially used in the gaming industry and later adopted in robotics, use a two-way control transfer to enhance modularity and system reactivity without

Table 10
Studies on Hierarchical-based task and motion planning.

Author & Year [Study]	Task definition	Application
Luo et al. (2024) [187]	Hierarchical imitation learning	Multi-stage robotic manipulation tasks specific focus on cable routing
Gregoric et al. (2024) [188]	Hierarchical-graphs with E^*	Path planning in large environment
Miao et al. (2024) [189]	HTN with LLMs using SkillKG (knowledge graph dataset)	UR5 manipulation in a real environment
Shah et al. (2024) [190]	HSMP with Zero-Shot Options	
Banfi et al. (2022) [191]	HTN for multi-robot task allocation with GNNs based estimator	Multi-robot (UAV and UGV) planner efficiently computes high-quality routing and task allocation plans for up to 45 robots and 20 areas.
Rico et al. (2021) [192]	PDDL2.1 formulation and BTs executor, PlanSys2	Dynamic robotic manipulation
Lee et al. (2021) [193]	Modified Vector Field Histogram formulation and A^* search algorithm	Robotic manipulation under uncertainties (MoveIt!)
Kast et al. (2020) [194]	Hierarchical planning using HTN	Online re-planning using whilst dual-arm manipulation
Görner et al. (2019) [195]	Using MoveIt! along with a hierarchical tree structure	Robotic manipulation under uncertainties
Munoz et al. (2018) [196]	VRP solver with a symbolic HTN planner	Aerial robotic manipulation
Mesesan et al. (2018) [182]	HTP with RRT	Robotic manipulation
Suarez et al. (2018) [186]	HTN formulation	Barrett WAM robots, dual-arm robotic manipulation
Rovida et al. (2017) [197]	PDDL, HTN and EBT	Dynamic robotic manipulation
Johannsmeier et al. (2016) [198]	HTN with HRC, AND/OR graphs as the representation of plans (for parallel execution), A^* algorithm and Fibonacci heap	Human–robot collaboration
Wang et al. (2016) [199]	Space manipulator network using HTN with Improved A^* algorithm	Multi-arm robot for on-orbit servicing
Hayes et al. (2016) [200]	CC-HTN, with an algorithm to autonomously build from graphical task representation's topological properties	Human–robot teaming Inferring collaborator goals through state estimation and task learning
Stock et al. (2015) [201]	CHIMP with HTN	Automate disassembly planning

compromising program clarity or maintainability [109]. Several studies highlight how task-planning methods can enhance robotic disassembly. BTs prove valuable when combined with hierarchical planning for robotic manipulation under uncertainty, as demonstrated by Rovida et al. (2020) [197]. For rapid and adaptable task definition, EBTs were proposed as a flexible framework that allows dynamic reconfiguration to adapt to disassembly needs [197]. Rovida et al. [197] also combined PDDL with HTNs. HTNs used a hierarchical structure of partially or totally ordered actions constrained by state variables. This approach has been shown to achieve superior expressive performance compared to traditional planning methods. Furthermore, Rico et al. (2021) demonstrated the optimisation of PDDL plan execution using BTs, effectively bridging high-level planning and low-level robotic actions [192].

Building on tree-search methods for obstacle rearrangement, Lee et al. [193] incorporated both prehensile (grasping) and non-prehensile (pushing/sliding) manipulation strategies. This was achieved by introducing a tree-search-based disassembly planner that determines the minimum number of obstacles needing relocation. Their algorithm uses a Modified Vector Field Histogram to evaluate the obstacle blockage of a target. This capability for diverse manipulation actions is essential for cluttered disassembly environments, allowing robots to effectively clear paths for component access.

Additional contributions to the research on hierarchical planning include a study by Hadfield et al. [205] explores the optimisation of execution costs in **Hierarchical planning now** (HPN) with **Stanford Research Institute Problem Solver** (STRIPS) formulations, emphasising execution under uncertainty. It focuses on improving the efficiency of decision-making processes in dynamic environments where immediate and long-term planning is required. The problem is formulated using PDDL, and the study involves the manipulation of PR2 under uncertainties. The approach can reduce the energy consumption and time required for robotic disassembly tasks by focusing on cost efficiency. However, there are instances where the approach generates suboptimal plans regarding task completion time or quality. Researchers leverage MoveIt! with hierarchical tree structures for dynamic robotic manipulation and online re-planning. Görner et al. (2019) employed this framework in dual-arm manipulation [195], while Kast et al. (2020) applied hierarchical planning with HTNs [194]. Furthermore, in complex multi-robot task allocation scenarios, Banfi et al. (2022) demonstrate the effectiveness of combining hierarchical planning with **Graph Neural Networks** (GNNs) for optimisation purposes [191]. Their approach

significantly accelerates the computation of high-quality routing and assignment plans, addressing a critical challenge in multi-robot collaboration. While their study does not directly focus on disassembly, the same principles could be applied to robotic disassembly processes. The hierarchical planning structure is ideally suited for representing complex disassembly sequences. For instance, the GNN-based estimator could learn to predict the suitability of different robot teams for specific disassembly actions like unscrewing, cutting, or sorting a scenario used by [74]. This would optimise the allocation of robots within the disassembly line, maximising efficiency and minimising damage or downtime.

In the context of HRC disassembly, Hayes et al. [200] proposed **Clique/Chain Hierarchical Task Networks** (CC-HTNs), which build from topological properties for human–robot teaming and goal inference. This method, notable for its broad applicability, does not require specific insights into motor primitives or environmental representations, making it versatile across various human–robot interaction scenarios. By applying CC-HTNs in goal inference and task planning, the research addresses significant challenges in interpreting and adapting to human actions in real time, enhancing collaborative efficiency. The approach also leverages prior experiences, incorporating internal nodes representing multiple motor primitives into an agent's action dictionary to accelerate task-planning solutions. CC-HTNs could be used to improve dynamic task allocation as disassembly lines change. Goal inference could enable robots to react to disassembly actions done by humans, enhancing collaboration. Prioritising the reuse of previously learned plans could make the process faster. However, Building and applying CC-HTNs autonomously might be very complex for real-time disassembly. Similarly, Johansmeier et al. [198] explored HTN with HRC and AND/OR graphs for parallel task execution in collaborative disassembly. The framework uses two abstraction levels: a high-level abstract world model generates coordinated skill sequences for human and robot agents using distinct cost functions. In contrast, a lower level employs hierarchical state machines for real-time robot behaviour adaptation. This approach suits dynamic, partially known industrial environments with human presence. The system's focus on decision-level unpredictability makes it valuable for physical human–robot interaction. Further improvement could come from dynamic cost functions adapting to human and robot capabilities. Planning robot and human disassembly steps together using [198] approach would optimise efficiency. This would require designing appropriate cost functions for disassembly tasks, which specifically require fine-tuning.

Combining hierarchical task planning with constraint solvers, Munoz et al. (2018) demonstrated a method for dynamic task scheduling in multi-robot assembly environments [196]. While simulations indicated potential efficiency improvements, full system integration and scalability pose ongoing challenges. This approach has potential applications in disassembly, optimising robot scheduling and enhancing collision avoidance strategies.

With the advance of applying large language models to robotic planning, studies like [189] investigate HTN planning with LLMs using SkillKG in UR5 manipulation in a real environment, showcasing the integration of advanced language processing models with hierarchical planning for robotic manipulation tasks. Gregoric et al. [188] combined LLMs with hierarchical planning for addressing the complex challenges inherent in robotic disassembly and other robotic tasks, pointing towards more efficient, adaptable, and intelligent robotic systems. Luo et al. [187] studied hierarchical imitation learning for multi-stage robotic manipulation, focusing on cable routing. This research highlights the potential of learning-based hierarchical planning in improving the efficiency and precision of complex robotic manipulation tasks. Whereas Shah et al. [190] discuss **Hierarchical Stochastic Motion Planning** (HSMP) with Zero-Shot Options, delving into the future possibilities of applying hierarchical planning in uncertain and dynamic environments to enhance robotic adaptability and decision-making capabilities, this would be particularly useful for hazardous conditions that may arise in disassembly.

4.2. Belief-based Task and Motion Planning

Belief-based Task and Motion Planning (Bb-TAMP) in robotic disassembly addresses the challenges of inherent uncertainties and dynamic conditions in disassembly tasks. This is a probabilistic framework where the system's state is not completely known. Instead, a “belief” about the state is maintained, essentially a probability distribution over all possible states. This distribution updates as new observations are made, typically through actions the robot performs. This method employs probabilistic models of the environment, enabling robots to adaptively, efficiently, and safely disassemble components even with incomplete information. Bb-TAMP this method focuses on **Partially Observable Markov Decision Processes** (POMDPs) focusing on high-entropy (high uncertainty) environments. These are extensions of **Markov Decision Processes** (MDPs) that include a layer of uncertainty about the state of the world. POMDPs help make statistically optimal decisions, factoring in both the immediate rewards and the future state uncertainties. Bb-TAMP enhances robot capabilities by continuously updating beliefs and decisions as new data becomes available, which is crucial in robotic disassembly's variable and uncertain domain. Table 11 shows several Belief-based TAMP studies.

In Bb-TAMP, the integration of task-level (high-level goals and plans) and motion-level (the physical movement actions required to achieve these goals) planning is crucial. The approach often uses frameworks like **Simultaneous Task and Motion Planning** (STAMP) [206] and methodologies like **Task Motion Multigraph** (TMM) [207], which help in translating the uncertainties and probabilistic outcomes of motion planning into the task planning domain.

Probabilistic and MDP-based planning methods are pivotal for navigation and manipulation in such contexts. Significant contributions include studies by Guo et al. [208] and Yang et al. [209], which developed MDP-based planners for iRobot and Gazebo robotic navigation, respectively. These studies utilise probabilistic planning to navigate through dynamic and uncertain environments effectively. Further, Shah et al. [207] expanded this approach to robotic manipulation under uncertainties, applying STAMP and MDP planning to adapt strategies in real time. This adaptability is essential for robotic disassembly tasks, where conditions and parameters may change unpredictably. Similarly, Zhao et al. [210] demonstrate the use of **Monte Carlo Tree Search** (MCTS) algorithms in automating disassembly planning, offering a

forward-thinking approach to handle the complexities associated with such tasks.

Behaviour Trees are versatile in handling belief spaces through their control flow nodes and probabilistic actions. Safronov et al. proposed **Belief Behaviour Trees** (BBTs) enhance BTs by incorporating an “unknown” status into the condition node, allowing them to address applications that are only partially observable [211]. In this framework, task planners operate directly within the belief space to construct BBTs. Here, the belief state is defined as the probabilistic distribution of various physical states. Action nodes within BBTs are categorised into actuation and perception actions, each with probabilistic outcomes. The framework by a previous study by Safronov et al. [212] introduces a Skipper node as a novel control node in BTs to manage uncertainties in the current state and specifies actions with preconditions and post-conditions to address future state uncertainties.

Integrating PDDL with Belief-based Task and Motion Planning strategically aligns high-level task planning with the challenges of operating under uncertain conditions. The works of Dantam et al. [213] with the Task-Motion Kit and the work by Garrett et al. [214] used PDDL to enhance robotic manipulation, highlighting the advantages of declarative planning languages in managing the unpredictability in robotic tasks. The proposed approaches can benefit disassembly by providing a structured framework that can adapt to changing conditions, improving the accuracy of robotic operations.

A novel approach called **Search with Task Relevant Uncertainty Guidance** (STRUG) was proposed by Curtis et al. [215]. The research addressed online POMDPs within continuous domains. This method focuses on reducing uncertainty relevant to the task, showing promise for finding long-horizon solutions. While STRUG has been initially demonstrated on simulated toy POMDPs problems, it is essential to acknowledge that further testing is crucial for evaluating its performance before transitioning to real-world scenarios with partial observability [215].

For navigation in belief space, Thomas et al. [216] introduced **Motion-Planning-aware Task Planning** (MPTP). This approach considers environmental uncertainty to improve navigation efficiency by integrating motion planning into task planning. Similarly, Rafendran et al. [155] proposed a unified ROS-based framework for robot task and motion planning, enabling seamless integration between task-level decisions and motion-level executions for improved robot operation. Both studies demonstrate the crucial link between task and motion planning for effective robotic navigation in complex, uncertain environments.

4.3. Optimisation-based Task and Motion Planning

Optimisation-based TAMP (Ob-TAMP) is crucial for enabling robotic disassembly systems to perform intricate tasks necessitating both strategic decision-making and precise motion execution. Ob-TAMP methods aim to find feasible solutions that minimise planning costs or time. These approaches strive to identify optimal solutions while maintaining computational efficiency in TAMP. Additionally, the specifications of operators in these methods are linked to temporal logic and cost characteristics. Table 12 shows several Optimisation-based TAMP studies.

A promising approach for robotic manipulation involves using a multi-level optimisation framework. This approach breaks down complex problems into more manageable sub-tasks, namely task planning, action planning, and motion planning. By iteratively optimising task-level goals and incorporating feedback from lower-level optimisations, superior solutions can be achieved while ensuring feasibility and time efficiency, as shown by Zhang et al. [218]. This framework demonstrated similar solution costs to established algorithms like Lin-Kernighan 2-opt. This highlights the potential to surpass traditional top-down methods. However, applying this model to the dynamic scenarios encountered in disassembly tasks would be a valuable direction for further research.

Table 11
Studies on Belief-based task and motion planning.

Author & Year [Study]	Task definition	Application
Zhao et al. (2024) [210]	LLMs and MCTS search algorithm for task planner	Automate disassembly planning
Curtis et al. (2023) [215]	POMDP with STRUG	Tackling the tiger problem
Castaman et al. (2021) [203]	RH-TAMP with PDDL formulation	UR5 and Panda manipulation under uncertainties
Safronov et al. (2020) [211]	BBTs executor	Robotic manipulation under uncertainty
Shah et al. (2020) [207]	STAMP and MDP planner	Robotic manipulation under uncertainties
Yang et al. (2020) [209]	MDP planner and ASP	Robotic navigation whilst detecting uncertainties in Gazebo
Phiquepal et al. (2019) [217]	CTAMP under partial observability with LGP	Plan policies that combine exploratory and exploitative actions in simulation
Guo et al. (2018) [208]	LTL and MDP planner	iRobot navigation

Table 12
Studies on Optimisation-based task and motion planning.

Author & Year [Study]	Task definition	Application
Mukherjee et al. (2022) [223]	MPLP with graph-based formulations and parallel algorithm	PR2 manipulation in Gazebo
Cheng et al. (2021) [224]	HTP from human demonstrations, with optimised-based planner	Human-Aware Robot Task Planning Based
Faroni et al. (2020) [225]	PLATINUm task planner with MoveIt!	Human-robot collaboration in robotic manipulation
Saha et al. (2017) [219]	MTL formulation and MILP solver	Baxter robot manipulation
Zhang et al. (2016) [218]	Heuristic solver as weighted graph	Pick and place experiments

Logic-geometric programming (LGP) and **Answer Set Programming (ASP)** both address complex planning challenges within **TAMP**. **Combined Task and Motion Planning (CTAMP)** using (LGP) was introduced by Phiquepa et al. [217], whereas applied Ding et al. [75] used ASP for autonomous driving. LGP excels in environments where information is incomplete, such as disassembly tasks, by integrating logical and geometric layers for adaptive planning. This adaptability is crucial for robots operating in dynamic settings. Conversely, ASP leverages logic programming for deliberative task planning, effectively managing uncertainties and complexities through formal logic to represent and infer knowledge. This capability is particularly beneficial in autonomous driving, where navigating urban uncertainties is important [75].

To integrate high-level task specifications with low-level optimisation, task planning can combine **Metric Temporal Logic (MTL)** with gradient descent optimisation. This was achieved by Saha et al. [219]. For efficient resolution of complex action sequences under constraints, **Mixed-Integer Linear Programming (MILP)** can be used to identify optimal low-level tasks that meet the MTL specifications [219]. In the context of disassembly, MTL could precisely define the steps involved, while MILP could optimise plans based on time or energy efficiency while ensuring safety. A potential challenge is the computational complexity of MILP solutions, which might hinder real-time applications. Integrating geometric information into task planning by leveraging Aliakbar et al. [220], who used a FastForward heuristic state-space task planner, illustrating how geometric constraints can guide the search towards more feasible states. Similarly, Zimmermann et al. [221] and Leu et al. [222] further contributed to this discourse with methodologies focused on multi-level constrained optimisation and hybrid planning algorithms, respectively, emphasising the potential of optimisation techniques in enhancing planning efficiency and reliability, particularly in the context of robotic disassembly.

Researchers have proposed a layered approach to human-aware task and motion planning (Ob-TAMP) that leverages a Human-Aware task planner. This planner operates at different abstraction levels. At the task planning level, the focus is on high-level operations without intricate motion considerations. Action planning then refines these operations based on real-time human state and geometric data. This hierarchical framework allows for bottom-layer feasibility feedback to inform top-layer process re-optimisation. The effectiveness of this method was demonstrated in an HRC scenario for mosaic assembly, as shown by Faroni et al. [225]. This study used the *PLATINUm* [226] framework for the task planning layer to model the collaborative task, generate flexible plans that balance workload between the human and robot, and adapt the plans dynamically based on feedback.

Further research in HRC has explored integrating human actions into robotic task planning using hierarchical models learned from

demonstrations as explored by Cheng et al. [224]. Here, optimisation-based planners aim to minimise collaborative costs, including factors like completion time, human fatigue, and spatial conflicts, as explored by Gombolay et al. [227]. A potential improvement to this approach could involve data augmentation to increase the size and diversity of the demonstration dataset. This data enrichment can enhance model robustness and generalisability, helping overcome challenges posed by the limited availability of high-quality demonstrations.

Traditional lazy search algorithms in robotics balance searching a graph with evaluating connections (edges), typically for collision checking, which is computationally expensive. However, research by Mukherjee et al. [223], and Mandalika et al. [228] introduced modified lazy search algorithms for task and motion planning. Mukherjee et al. [223] introduced the **Massively Parallelised Lazy Search Algorithm (MPLP)**, which addresses this by performing both searching and edge evaluation tasks completely asynchronously in parallel. This means they can happen simultaneously, significantly improving planning speed. Whereas, Mandalika et al. [228] propose the **Generalized Lazy Search (GLS)** framework, which balances search effort and the evaluation of action feasibility. These approaches are highly relevant for robotic disassembly as they often involve complex search spaces and expensive feasibility checks for robot actions in cluttered environments. By speeding up the planning process, these methods could enable real-time adaptation of robot actions in response to unexpected situations or obstacles encountered during disassembly.

Another notable approach included using gradient descent of cost functions by treating the path as a function and optimising the trajectory at each iteration. The **Covariant Hamiltonian Optimisation for Motion Planning (CHOMP)** algorithm proposed by Ratliff et al. [229] processes an input trajectory and generates a new trajectory that minimises the objective functionals. Unlike methods that optimise one point at a time, CHOMP applies a gradient step to every point in the trajectory during each iteration. It projects the free gradient onto the Riemannian manifold of smooth trajectories to ensure smooth transitions between paths. The primary focus of the functional is on path length, but the authors also introduced techniques to account for obstacles and robotic constraints. While CHOMP has demonstrated effectiveness across various robots, its time-intensive nature has led to more efficient methods. Kalakrishnan et al. [230] introduced **Stochastic Trajectory Optimisation for Motion Planning (STOMP)**, which uses stochastic gradient descent and handles non-differentiable objectives and reducing the optimisation process's time requirement [230].

Additionally, STOMP incorporates torque at every joint throughout the motion as an element of the objective function. Similarly, **Incremental Trajectory Optimisation for Motion Planning (ITOMP)** introduced by Park et al. [231], is designed for dynamic environments,

Table 13
Studies on Constraint-based task and motion planning.

Author & Year [Study]	Task definition	Application
Bustamante et al. (2022) [156]	PDDL with CATs	Variable autonomy in daily activities
Dantam et al. (2018) [213]	IDTMP with STRIPS formulation and SMT solver	Baxter robot manipulation
Lagriffoul et al. (2016) [232]	ASP task planner	Robotic manipulation and culprit problem detection
Mirabel et al. (2016) [233]	Constrained based graph representation with CSP solver	Baxter robot manipulation
Lagriffoul et al. (2014) [234]	CTAMP with CSP solver	Pick and place experiments
Srivastava et al. (2014) [235]	CTAMP with PDDL formulation	PR2 restaurant manipulation and navigation
Mansouri et al. (2014) [236]	CSP with Constraint-Based Calculi	PR2 Robotic manipulation

addressing the possibility of collisions occurring after a trajectory has been planned. By iteratively conducting the planning and execution phases, ITOMP enables robots to navigate around moving obstacles without prior knowledge of their presence.

4.3.1. Constraint-based TAMP

Constraint-based Task and Motion Planning (Csb-TAMP) planner integrates the cognitive aspects of task planning with the precision of motion planning, ensuring that robot actions are feasible, efficient, and tailored to operational limits. It proves particularly beneficial in robotic disassembly, navigating complex manoeuvres and strict safety requirements to bridge the gap between high-level disassembly goals and the feasibility of physical actions [232]. Studies of Constraint-based TAMP can be seen in Table 13.

Integrating temporal reasoning with sampling-based motion planning, Edelkamp et al. (2018) used PDDL to address multi-goal robotic tasks involving dynamics and time constraints [237]. This approach directly incorporates sequencing and timing considerations into the disassembly planning phase, which is highly valuable in disassembly scenarios. However, computationally efficient optimisation remains challenging due to the complexity of merging temporal and motion-related planning.

Research into **Cooperative Motion Planning** (CMP) has yielded promising tools that allow users to specify motions at a cooperative task-space level. This simplifies the programming process by automatically calculating individual arm movements, scaling trajectories, and managing synchronisation. These advancements have significant potential for simplifying complex disassembly tasks involving multi-arm robotic work cells, as demonstrated by Basile et al. [238]. However, reliance on traditional robotic languages might introduce limitations in adaptability to rapidly evolving or undefined tasks.

Addressing the challenges of Csb-TAMP **Combined Task and Motion Planning** (CTAMP) was introduced by Srivastava et al. [235], a novel system has been introduced that employs the PDDL. This approach removes the need for task-specific heuristics and seamlessly integrates with off-the-shelf planners. The result bridges high-level planning and detailed motion control, a significant step for navigation and manipulation tasks. It is worth noting, however, that PDDLs abstraction level could sometimes mask the fine-grained details necessary for precise manipulation, suggesting a potential area for further refinement in translating high-level plans into detailed motion strategies.

To reduce planning complexity, researchers have employed CTAMP with **Constraint Satisfaction Problem** (CSP) solvers [234]. Geometric constraints are leveraged to reduce the search space, focusing attention on feasible disassembly paths. This approach has proven effective at reducing planning time for complex disassembly tasks. However, in dynamic environments where the constraints are not fully known a priori or can change during task execution, CSP-based methods could face limitations. Furthering this research, [232] integrate **Answer Set Programming** (ASP) task planner mechanisms for culprit detection, identifying spatial inconsistencies and unavoidable collisions early in the planning process. Such mechanisms lead to more robust and efficient disassembly sequences, allowing for preemptive adjustments that circumvent unnecessary stops and re-planning.

To account for the physical and temporal limitations of robotic actions, a framework for planning to incorporate spatial, temporal, and resource models was introduced by Mansouri et al. [236] using Constraint-Based Calculi along CSP solver. This approach is crucial for optimising disassembly sequences with limited space and tight schedules. However, the system's reliance [236] on detailed and accurate modelling of environments and tasks could pose practical challenges in highly dynamic or unpredictable scenarios where rapid changes occur. Another helpful technique involves using constraint-based graph representation with a CSP solver to unify task and motion planning. This is especially well-suited for robotic disassembly scenarios where robots must skilfully navigate and manipulate objects in confined environments. By planning manipulation paths that consider grasps and object placement, tasks can be executed with increased efficiency, as demonstrated by Mirabel et al. [233]. A primary challenge with this CSP-based method is the computational complexity, which has implications for real-time performance as task complexity grows.

Research in **Incremental Dynamic Task and Motion Planning** (IDTMP) has introduced a framework using STRIPS formulation with **Satisfiability Modulo Theories** (SMT) solvers [213]. This constraint-based approach can potentially improve scalability and performance for disassembly planning. Geometric information is integrated directly into the task planning level, allowing plans to be adjusted dynamically based on real-world constraints. While this increases adaptability, the incremental approach may introduce additional computational demands, potentially affecting efficiency and scalability in highly dynamic environments [213]. Finally, Bustamante et al. [156] introduced **Constraint Action Templates** (CATs), which combine the advantages of Action Templates and Shared Control Templates. CATs enable robots to execute plans with shared control and supervised autonomy, adaptable to user inputs during task execution. They demonstrate its application in daily living tasks using the wheelchair robot EDAN.

4.4. Sampling-based TAMP

Sampling-based TAMP approaches concentrate on identifying viable solutions within a constrained search space, offering probabilistic completeness. These methods typically sample within the intersecting regions of various constraints, effectively narrowing the hybrid search space. Sampling-based Task and Motion Planning (Sb-TAMP) signifies a shift towards the adoption of probabilistic approaches for addressing intricate planning challenges in the field of robotics. Sb-TAMP planners are recognised for their rapid and proficient iterative techniques that construct a trajectory based on the robot's configuration space. In each cycle, they typically perform a duo of actions: selecting a random point from the configuration set and then connecting this new point to the previously assessed points. These planners maintain a record of every point in a tree-like structure, which facilitates the addition of new elements and the search within the tree with logarithmic efficiency, as detailed in [239] work on planning algorithms. Studies of Sampling-based TAMP can be seen in Table 14.

The most widely utilised Sb-TAMP method in planning is known as Rapidly-exploring Random Trees (RRT). This method employs a **K-dimensional Tree** (KD-Tree) structure to effectively tackle problems with high-dimensional spaces, swiftly finding collision-free routes that

Table 14
Studies on Sampling-based task and motion planning.

Author & Year [Study]	Task definition	Application
Fang et al. (2023) [240]	DiMSAM with PDDLStream	Robotic manipulation
Rajendran et al. (2022) [155]	PDDL2.1 formulation with RobMAP	Robotic navigation
Thomas et al. (2021) [216]	MPTP with PDDL2.1 formulation	Robotic navigation under uncertainty
Wang et al. (2020) [241]	Non-uniform sampling in likely path areas, Voronoi graphs	Efficiency and obstacle avoidance in robotic navigation
Migimatsu et al. (2020) [242]	Extended LGP and STRIPS	Robotic manipulations with tower of Hanoi problem
Thomason et al. (2019) [243]	PDDL formulations	PR2 manipulation in Gazebo
Garrett et al. (2018) [244]	EAS extended by SAS+	PR2 manipulation
Lee et al. (2017) [245]	RRT with GMM-based configuration space representation	Adaptive motion planning in dynamic environments
Salzman et al. (2016) [246]	Lower Bound Tree-RRT (LBT-RRT)	Finding shorter paths with near-optimal solutions
Zito et al. (2012) [247]	RRT adaptability for push manipulation	Local planning and action maps for object manipulation
Rios-Martinez et al. (2011) [248]	Risk-RRT for socially aware navigation	Service robots considering human interactions
Karaman et al. (2011b) [249]	Enhanced RRT*, introduction of 'rewiring' step	Optimisation in collision-free routing

connect two points and offering a systematic approach for planning. Nevertheless, the basic RRT algorithm may produce paths that are not optimal. To address this issue, Karaman et al. [249] introduced an enhanced version of RRT, designated as RRT*, which incorporates a novel step termed 'rewiring.' This process involves reassessing the connections within the tree in light of newly acquired information, ensuring that, with an infinite number of iterations, the algorithm is guaranteed to converge to the optimal path. Enhancements to RRT and RRT* are documented in [246,250]. RRT* was refined by Melchior et al. [251] for better post-path optimisation. RRT-Connect proposed by [250], is a single-query planner that uses dual parallel RRTs from start and goal, merging them to form a path; this probabilistic method by simulates points multiple times for safer paths amidst uncertainties. **Lower Bound Tree-RRT (LBT-RRT)**, introduced by Salzman et al. [246], generalises RRTs to find shorter paths using an approximation metric, ensuring convergence to near-optimal solutions. For object manipulation, Zito et al. [247] use RRTs adaptability for push manipulation, creating action maps for local planning. Wang et al. [241] improves efficiency by non-uniform sampling in likely path areas, using Voronoi graphs and heuristics for obstacle avoidance. Socially aware navigation for service robots is developed in [248] with Risk-RRT, considering human interactions. [241] enhances RRT* for better performance through domain-specific sampling and optimisation strategies. Other notable studies include the paper by Lee et al. [245], which introduces a novel approach for adaptive motion planning in robotic manipulation, particularly useful in dynamic environments where changes between the robot base and workspace can render pre-calculated motion trajectories obsolete. This adaptive approach uses high-dimensional **Gaussian Mixture Models (GMMs)** to learn and predict collision and collision-free regions in a robot's configuration space. The proposed method uses RRT in conjunction with the GMM-based representation of the configuration space. The collision and collision-free predictions from the GMMs are used to guide the expansion of the RRT, effectively biasing the tree growth towards viable paths that avoid obstacles. This integration allows for rapid adaptation to environmental changes, as the RRT can quickly adjust its pathfinding based on the updated GMMs without starting from scratch. The approach in [245] is compared with traditional sampling-based planning methods, demonstrating significant improvements in computational efficiency and adaptability to changing environments. This makes it a promising tool for enhancing the capabilities of robots in industrial settings, where quick adaptation to new tasks and environments is essential.

The paper by Fang et al. [240] introduces **Diffusion Models as Samplers (DiMSam)** for TAMP under Partial Observability. The paper enhances TAMP by using PDDLstream and diffusion models to learn and generate constraint-satisfying samples for robot manipulation tasks in partially observed environments. The authors demonstrate that their method, which operates on low-dimensional latent embeddings of object states, can effectively handle articulated object manipulation tasks

requiring multi-step reasoning, even with previously unseen objects. Experiments in simulation and real-world settings show that the approach can generate diverse samples that satisfy constraints, improving the robustness and flexibility of autonomous robotic systems in complex and dynamic environments. Similarly, Mendez et al. [252] introduced lifelong learning TAMP using various generative models. These models produce candidate continuous parameters for a planner, enhancing planning success in both 2D and BEHAVIOR domains. (To learn more about diffusion models, address the following review [253].)

An approach to extending the **Explicit-Space Action Selector (EAS)** with the SAS+ formalism [254] is explored for PR2 robot manipulation tasks was proposed by Garrett et al. [244]. This combines the robustness of sampling methods with the structure provided by explicit action spaces, aiming to improve efficiency in solving these tasks. However, the reliance on sampling could lead to performance variability and potentially increase computation time, especially in complex tasks or environments with many obstacles. Migimatsu et al. [242] further this domain by extending **Logic-geometric Programming (LGP)** and STRIPS to tackle the classical Tower of Hanoi problem with robotic manipulators. Their method demonstrates a sophisticated application of sampling-based planning to decompose and solve a well-defined problem space. However, the challenge of generalising this approach to less structured environments remains. Instead of using STRIPS, Thomason et al. [243] used PDDL formulations to address PR2 manipulation in the Gazebo simulation environment, highlighting the potential of integrating traditional planning languages with sampling-based techniques. While this demonstrates versatility, translating simulated success to real-world application necessitates additional considerations regarding the fidelity of simulation environments and physical robot dynamics.

4.5. Innovative application and frameworks

The study by Wang et al. [199] presents a hierarchical task planning method for multi-arm space robots, focusing on managing complex and various on-orbit servicing missions. The process involves two main hierarchies: mission profile analysis and task node planning. In the mission profile analysis, the approach defines several types of primitive tasks and operators, allowing complex tasks to be decomposed into sequences of primitive tasks using an HTN approach. The task node planning uses an improved A^* algorithm to adapt to the continuous motion of manipulators, further breaking down some primitive tasks into task nodes when direct execution is constrained. The paper validates the effectiveness and feasibility of the proposed method through simulation, demonstrating how it can optimise resource allocation and ensure task feasibility by incorporating constraints and efficient collision detection algorithms [199]. This work contributes to the field by offering a structured approach to planning and executing tasks for multi-arm robots in space, addressing the challenges posed by

space's complex and constraint-laden environment. However, it could benefit from improvements in adapting to dynamic environments, real-time decision-making, human–robot interaction, energy efficiency, and scalability.

To go beyond simple task execution and incorporate a human-like understanding of tasks into robotic manipulation, Miao et al. (2024) introduced a hierarchical knowledge-based framework [189]. The authors used a knowledge graph dataset (SkillKG) comprising 13,154 triples representing various entities and relations associated with manipulation tasks. This dataset is enriched with knowledge extracted from text datasets and an external knowledge base, DBpedia, with the aid of LLMs. This enables multi-source knowledge updates and task execution plan inference, promoting manipulation skill transfer. The approach includes a hierarchical structure that categorises manipulation factors into static and dynamic layers, including objects, agents, scenes, tasks, skills, and actions. These layers are designed to accommodate the complex nature of robotic manipulation, encompassing both continuous and discrete data and static and dynamic factors. The framework's reliance on semantic knowledge and a graph embedding method for reasoning tasks such as entity alignment and link prediction enable it to generate meaningful and actionable insights for robotic manipulation, leading to more adaptive and intelligent robotic behaviours [189]. This research leverages the power of knowledge graphs and LLMs; it opens new avenues for robotic manipulation, offering a more nuanced and flexible approach to task planning and execution. The successful application of this framework in real-world scenarios demonstrates its potential to improve robotic manipulation, which extends to disassembly and makes robots more versatile and efficient in performing complex tasks.

Focusing on reducing computational complexity in path planning, Gregoric et al. (2024) developed a method to generate a hierarchical graph autonomously from floor plans. This approach allows for handling large, intricate environments [188]. This method handles large and complex environments and uses the E* algorithm for finding partial paths at the most detailed level, ensuring near-optimal path planning, which is crucial for practically applying robots in real-world environments. Furthermore, the flexibility to adjust the hierarchy's depth based on environmental features allows for tailored path-planning strategies that can suit various applications, from indoor navigation to urban exploration. However, despite its advantages, the approach has limitations that warrant further exploration. First, relying on floor plans for initial graph generation may not always be feasible, especially in dynamic or unstructured environments where real-time data and adaptability are crucial. However, the computational overhead of generating and updating the hierarchical graph in response to environmental changes could impact the method's efficiency in highly dynamic settings [188].

5. Expanded applications

The following section will expand on the various robotic disassembly applications, including Electronic waste disassembly (which includes multiple consumer electronics). We discuss automotive components, moving to a recent application that has gained much interest both in the industry and academia, which is the robotic assembly of EV LIB; finally, we close the section by proposing future applications in robotic disassembly that could be helpful. Fig. 10 illustrates the distribution of different applications of robotic disassembly explored in our study. Fig. 11 presents the publication timeline of the papers included, highlighting the evolution and focus areas in the field.

5.1. Electronic waste

Electronic waste, commonly called e-waste, encompasses a broad range of end-of-life electronic products subject to disposal or recycling processes. As digital technology advances and increases, the volume of

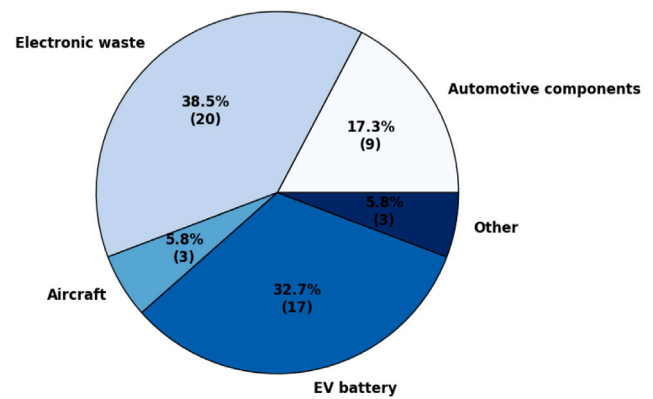


Fig. 10. The number of different applications of robotic disassembly explored.

e-waste has surged, presenting challenges and opportunities for waste management and environmental sustainability [2,5]. E-waste includes various items such as computers, smartphones, televisions, and other electronic devices that have reached the end of their useful life. The disposal of these products has become a significant environmental concern due to the toxic substances they contain, such as lead, mercury, and cadmium, which can leach into the soil and water, causing contamination and health risks [5]. The United States alone generates millions of tons of electronic equipment that will end up in landfills each year, contributing to a large percentage of toxic waste despite accounting for a small fraction of total trash produced by humans. Globally, the situation is even more dire, with estimates suggesting that e-waste generation could reach 74 million tons by 2030 if left unchecked. Robotic disassembly is a promising solution to the e-waste challenge [10]. Companies like Google and HP have embraced circularity by creating durable products built using recyclable materials and designed to minimise the downstream impact from recycling and reusing [255].

Europe's Circular Economy Package and the ambitious goals of the European Green Deal are pivotal in driving the transition towards a more sustainable and resource-efficient economy [3]. The legislation aims to improve waste management and recycling capacities and encourage producers through financial responsibilities and eco-design initiatives to minimise electronic waste generation. Despite achieving notable recycling targets, the paper by [256] points out the persisting challenges in elevating reuse rates and addressing consumer scepticism towards second-hand products. Plastics, a predominant component in electronic products due to their versatile properties, present substantial environmental challenges. With current global recycling rates lingering around 14% and the EU aiming for a 50% recycling target by 2030, the study emphasises the critical need for advancing recycling technologies and regulatory support to enhance the economic viability of recycled plastics. Moreover, Alvarez et al. [256] sheds light on the complexities surrounding recycling plastics mixed with hazardous substances like flame retardants, necessitating sophisticated separation techniques to ensure recyclable and environmental safety. Additionally, the recycling of plastics derived from LCD screens, which maintain robust mechanical properties such as Young's modulus and tensile strength, alongside the extraction of valuable metals such as copper, zinc, and nickel from PCB and PC components through chemical leaching, further illustrate the tangible benefits of disassembly [257].

Recent advancements in robotic disassembly of electronic waste recycling can be attributed to the discussion in Section 3.5.1 on machine learning and computer vision; these methods have improved the efficiency and accuracy of disassembling processes. Studies like those conducted by [45] Foo et al. 2021, and [44] Li et al. 2021, have shown that deep learning models and computer vision systems, such as Faster R-CNN combined with Rotation Edge Similarity, are pivotal in

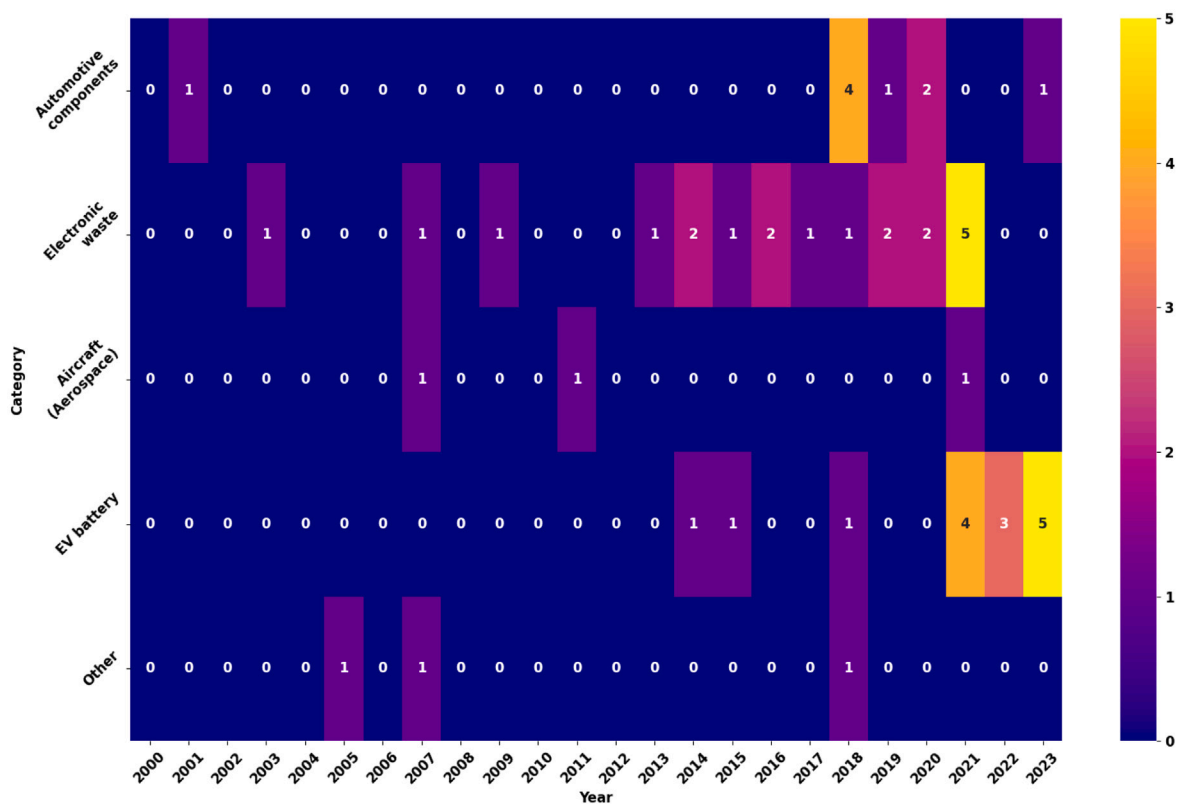


Fig. 11. The publication date of the different robotic disassembly application papers included.

identifying and classifying screws in LCD monitors and mobile phone main-boards. These technological approaches aim to automate disassembly by accurately locating critical components, thereby preventing damage to valuable parts and streamlining recycling operations. Additionally, optimisation algorithms, as applied by [61] Xu et al. 2020, using a Modified Discrete Bees Algorithm, aim to minimise both cost and time in the disassembly of personal computers, highlighting the importance of efficiency in recycling processes. On the fastener detection front, [46] Brogan et al. 2021, and [42] Su et al. 2021, employed Tiny-YOLO v2 and Mask R-CNN, respectively, to quickly and accurately identify disassembly points in various electronic devices, including the challenging task of detecting laptop parts and their orientations.

For electronic waste, the approach often centres on developing and applying robotic systems to automate the disassembly and sorting of electronic devices to recover valuable materials or components. The objective across multiple studies is typically to improve recycling processes' efficiency, safety, and environmental sustainability. Performance evaluation often involves metrics such as disassembly time, accuracy of component identification and extraction, and recovery rate or quality (can be seen in Table 3). These studies collectively illustrate the significant role of advanced algorithms in improving the disassembly process, suggesting a shift towards more automated, precise, and cost-efficient recycling methodologies. This technological evolution promises to address the complexities of e-waste components more effectively, leading to enhanced sustainability and environmental protection.

5.2. Automotive components

A similar approach has been taken in the robotic disassembly of automotive components. Studies like Wang et al. 2018, have used vision-guided control and matrix manipulation techniques, respectively, to identify and efficiently disassemble used car wheels and 4-stroke engines. These approaches highlight the critical role of visual data and

precise identification in automating the disassembly process for recycling automotive components.

Further innovations by Costa et al. 2018 [77] and Ebinger et al. 2018 [76] explored the use of detailed CAD models and hierarchical disassembly graphs to plan the disassembly of single-cylinder engines and adapt robotic disassembly line balancing to various tasks. These studies demonstrate the potential of detailed planning and flexible frameworks to automate complex disassembly tasks, significantly enhancing operational efficiency and cost-effectiveness.

Additionally, Ren et al. 2018's [66] application of genetic algorithms for optimising the disassembly of a mechanical transmission system emphasises the importance of strategic planning in minimising disassembly time. By optimising the disassembly sequence planning, these methodologies offer significant efficiency gains, showcasing the advancement in robotic disassembly processes tailored for the automotive recycling industry.

Overall, for automotive components, the research focuses on automating vehicle parts' disassembly or recycling processes, such as engines and other vital components. The objective is similar to that of electronic waste (Section 5.1), aiming to enhance efficiency, safety, and sustainability. However, there is an added emphasis on handling more complex and significant components specific to automotive applications. Performance evaluation for this type of disassembly often includes metrics like disassembly time, cost-effectiveness, accuracy of part identification, and the success rate of material recovery, which can be seen in Table 3.

5.3. Electric vehicle batteries

Many countries are transitioning from internal combustion engine vehicles to EVs to meet decarbonisation goals. By 2022, the global EV fleet will have surpassed 20 million, with sales expected to increase 18-fold by 2030 [258]. This surge in EVs highlights the critical need for recycling eOL LIBs, which pose significant environmental risks when

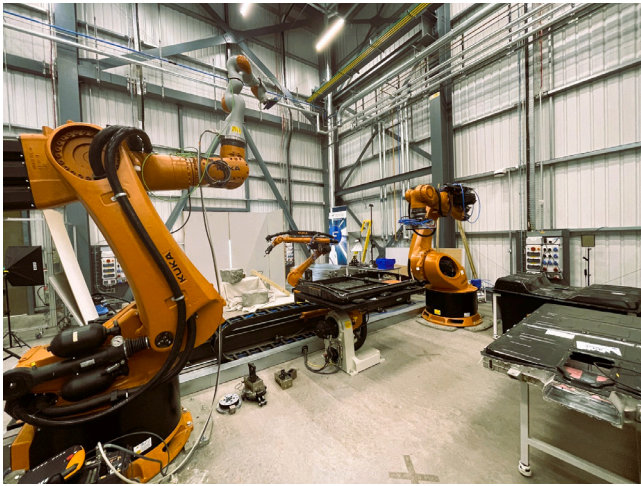


Fig. 12. A robotic test bed, located at the Birmingham Energy Innovation Centre, is designed for disassembling complex products, including EV batteries. The test bed features a KR500 robot equipped with a heavy-duty vacuum suction gripper, a long-reach manipulator achieved by attaching an iiwa LBR robot arm to the KR500's end-effector, a KR20 mounted on a 5 m rail track, and several advanced custom-designed tools for disassembling various types of EV batteries.

landfilled or improperly processed. Recycling these “urban mines” to recover valuable materials such as lithium, cobalt, manganese, and nickel is essential, spurred by initiatives like Europe’s Battery 2030+ and new recycling regulations.

This interest has catalysed notable government-funded projects, including REBELION [259], a comprehensive initiative funded by the European Union aimed at enhancing the circular economy for LIBs used in EVs. REBELION addresses the waste management challenges of these batteries, incorporating advanced robotic disassembly at their life cycle’s end to promote recycling as both an environmentally friendly and economically advantageous practice. This is particularly significant given the EU’s reliance on imported critical raw materials and the substantial market share of EVs within Europe. The project’s primary goals are to improve battery efficiency and safety, maximise second-life applications, and boost recycling processes.

The increase in EV usage and the consequent demand for efficient battery recycling highlight the necessity for innovative solutions in battery recycling and resource recovery. This emphasises the role of automation in overcoming the disassembly challenges inherent in mass production. As countries implement sustainable battery recycling practices to protect the environment and conserve resources, optimising the disassembly process to facilitate material recovery becomes increasingly crucial. This aligns ecological goals with technological advancements in recycling and remanufacturing batteries [260].

Recent research has focused on diagnostic and prognostic indicators used in battery management systems, including indicators like state of health, state of charge, and remaining useful life [261]. These metrics help maximise a battery’s operational life until EoL. However, these batteries become highly unstable once depleted and require careful recycling, demonstrating the need for robotic disassembly. Challenges in the disassembly process, crucial for LIB recycling, continue due to the complexity and diversity of LIB models. There have also been advances in automating several tasks which range from sorting, unbolting, pick and place, cutting, and nut removal [80]. These developments are crucial for enhancing the efficiency and safety of battery recycling processes (Fig. 13 illustrates common tasks involved in EV battery disassembly).

Robotic disassembly of EV LIBs has seen significant advancements through various innovative approaches aimed at improving efficiency, safety, and material recovery (this evolving and ever-increasing trend

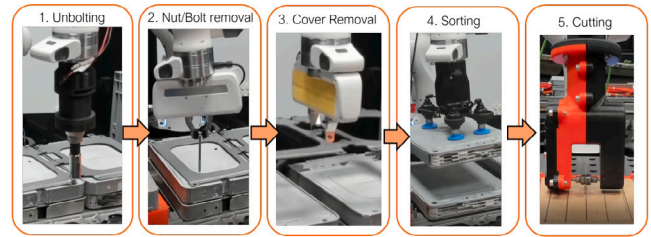


Fig. 13. Common tasks that are involved in EV battery disassembly [80].

can be seen in Fig. 1). This approach leverages the strengths of both human and robotic capabilities, advanced algorithms, and machine learning techniques to address the challenges associated with disassembling EV batteries. Fig. 12 shows an example of an industrial-scale battery disassembly test bed.

The foundational work by Wegner et al. [22] introduced a planning approach for the disassembly of EV batteries, explicitly focusing on the Audi Q5 Hybrid system. This approach uses matrix manipulation and dynamic programming to generate a disassembly graph, highlighting the importance of planning the disassembly depth from pack-to-module and module-to-cell levels. However, the evaluation metrics focus on the planning approach rather than the execution, which may not fully capture the practical challenges of robotic disassembly. Further advancements in planning and optimisation were made by [57], who presented a multi-objective optimisation model for cell-level disassembly of waste power battery modules from the Tesla Model S in a human-machine hybrid mode. This model achieved optimal allocation of disassembly tasks by balancing objectives such as minimising the number of workstations, idle time, workers, and disassembly cost. Additionally, [56] developed a human–robot collaboration disassembly technique using a Stackelberg model and a PSO-Pareto algorithm, focusing on removing screws from EV batteries to enhance cooperation, safety, and disassembly efficiency. Concurrently, [53] explored HRC-enabled disassembly, employing a hybrid of PSO and Q-learning for disassembling Tesla Model S batteries, covering both pack-to-module and module-to-cell disassembly depths.

Exploring the synergy between human dexterity and robotic precision, [52] investigated the integration of human–robot collaboration in the disassembly of EV battery disassembly, leveraging the strengths of both humans and robots to increase safety and efficiency potentially. However, the evaluation of this method is not detailed in the provided text, making it difficult to assess its effectiveness compared to fully automated systems. [55] focused on developing a robotic system for EV battery module disassembly, highlighting the benefits of human–robot collaboration by comparing the robotic system’s performance with manual disassembly and identifying areas for further optimisation. The evaluation compares the robotic system’s performance with manual disassembly, highlighting the benefits of human–robot collaboration. However, the slower pick-and-place operations indicate areas for further optimisation. [80] investigated the application of telerobotics in EVB disassembly, comparing a traditional haptic-“cobot” master–slave framework with identical master and slave cobots, suggesting that identical cobots reduce task completion time but with a slight decrease in first attempt success rate. The findings indicate that while force feedback’s realism is less crucial for some tasks, accurate force mapping enhances tactile cues for more precise operations.

Advancements in computer vision discussed in Section 3.5.1 have a pivotal role in automating the disassembly process of EV batteries. Zorn et al. [262] highlight how instance segmentation enables precise identification of components like busbars and battery modules, while 3D pose estimation facilitates accurate robotic manipulation. The adaptable vision algorithms accommodate the diverse designs of battery packs, ensuring non-destructive disassembly and potential reuse

of components. Robots can accurately identify and sort materials based on their composition by integrating computer vision with a labelling system and sensor-based material detection, improving sorting processes and ensuring high-quality recycling. This automated approach enhances disassembly efficiency and safety, reduces manual labour, and mitigates hazards like thermal runaway. The proposed vision pipeline, validated through robotic grasping of busbars, combines depth imaging, instance segmentation, and point cloud registration, offering a comprehensive solution for identifying, localising, and manipulating battery components [262].

A vision-based task planner is proposed to disassemble EV battery components, generating a dismantling plan and robot movements by Choux et al. [133]. The system measurement errors were lower than 5 mm. It could perform all steps in the order with an average time of 34 s. However, the evaluation focuses solely on measurement accuracy and time efficiency, neglecting the impact of varying battery designs on the process. This limitation restricts the generalisability of the approach to real-world scenarios with diverse battery configurations. Similarly, Rastegarpanah et al. [74] developed a hybrid task planner using behaviour trees to sort EV battery components with a multi-robot setup. This framework, assessed through simulation, may not perfectly mirror real-world conditions. The focus was on disassembling 1st generation Nissan Leaf batteries from the pack to the module level. Moreover, while promising, the framework's performance in dynamic environments indicates a need for enhancements to ensure its efficacy and robustness in real-world applications.

Advanced image recognition algorithms, such as modified FasterRCNN and SVDD, have been adapted for EV battery disassembly and have shown improvement in fastener detection and screw loosening tasks. This adaptation is promising for identifying critical disassembly points and enhancing task-specific approaches for unbolting and unscrewing as explored by [43,48]. Building on this, Hellmuth et al. [127] evaluated the automation potential for disassembling a 2017 Chevrolet Bolt battery, suggesting that while unscrewing could be automated, lifting operations should remain manual, providing a structured method to increase automation. Additional use of advanced image recognition algorithms is promising for identifying disassembly points as demonstrated by [41,49]. Still, it must be noted that these study's evaluation metrics are more relevant to fastener detection, not the broader disassembly process.

A novel approach that employed fuzzy logic to estimate disassembly times for plug-in hybrid electric vehicle (PHEV) battery systems was introduced by [60], aiming to provide more accurate predictions under uncertain conditions. However, this focus on time estimation might not fully address other crucial aspects, such as safety and precision. Additionally, a hybrid disassembly framework was introduced by [263], which uses modified robotic arms and specialised tools at a hybrid workstation. This framework aims to optimise the disassembly process of electric vehicle (EV) batteries by focusing on design, safety, and cost parameters, potentially offering a significant improvement in disassembly efficiency. A **multi-agent reinforcement learning** (MARL) method was also proposed by [24] to optimise the disassembly sequence of retired EV batteries in a human–robot collaboration (HRC) setting, aiming to enhance recycling efficiency and safety. This approach, tested on a Samsung MS372P5s battery module, seeks the optimal disassembly paths for specific battery modules, showcasing the potential of MARL in improving the recycling process. Furthermore, a novel heuristic algorithm for multi-criteria resilience assessment in HRC disassembly was introduced in [54] to support the recycling of spent lithium-ion batteries. This method combines fuzzy Bayesian fusion with analytical network processing to evaluate HRC disassembly resilience, considering stability, redundancy, efficiency, and adaptability factors.

5.4. Future application

The aerospace industry stands on the brink of a transformative era where robotic disassembly and assembly can significantly enhance the efficiency, safety, and sustainability of aircraft maintenance, recycling, and manufacturing processes. Studies like those by [264] Aguinaga et al. [168] Zhong et al. and [31] Mao et al. represent a significant evolution in the aerospace sector's approach to robotic disassembly, offering innovative methodologies and technological advancements. Aguinaga et al. introduced an integration of virtual reality with Transition-based rapidly exploring random Trees to automate disassembly sequences, specifically for aircraft engine components, showcasing a method that enhances efficiency and precision. Zhong et al. used Genetic Algorithms and PSO to refine disassembly sequences for civil aircraft systems, aiming to streamline maintenance procedures and uphold high safety standards [168]. Mao et al. furthered this progression by employing Deep Q-Networks for adaptive disassembly planning in virtual reality training environments, enhancing learning outcomes and operational safety for maintenance personnel [31]. These studies highlight the aerospace industry's movement towards adopting more sophisticated, safe, and sustainable maintenance and recycling practices through advanced robotics and simulation technologies.

This integration can increase sustainability and support circular economic values in the aerospace sector by enhancing the efficiency and precision of disassembling processes, thereby maximising the reuse and recycling of valuable materials while minimising waste. As the global commercial aircraft disassembly, dismantling, and recycling market continues to grow, projected to reach \$14.72 billion by 2033 [265], these technological advancements will play a pivotal role in shaping the future of aerospace industry practices, making them more sustainable and economically viable. In March 2023, the aviation sector disposed of over 40,000 tons of EoL composite materials into landfills [265].

Innovative robotic systems like Boeing's "Flex-track" and Airbus's "Futura", which are currently used for drilling, fastening, and inspection tasks, highlight the feasibility and advantages of integrating robotic technologies in aerospace manufacturing and maintenance. These examples also show how robotic disassembly can be used for complex tasks, paving the way for a more sustainable and efficient approach to aircraft life-cycle management [266].

Beyond the current applications, the future of robotic disassembly in aerospace presents vast opportunities for expansion, particularly in the salvage of high-value aircraft components. Advanced composites and metals used in aircraft wings, cabin equipment, and other integral parts offer significant potential for reuse and recycling [267]. The durability and valuable materials can be recycled for second-life use cases, improving the sustainability benefits of employing robotic disassembly on a broader scale within the industry.

6. Future directions

6.1. Digital twin technology

Digital twin research has surged in popularity, showing promise as a transformative tool in robotics (Fig. 14). This technology creates a virtual replica of a physical system and comprises various aspects such as sensing and control models, statistical models, big data analytics, and artificial intelligence, enabling real-time monitoring, simulation, and control. The adoption of digital twins is becoming more widespread in the manufacturing sector, driven by applications such as product development, preventive and prescriptive maintenance, performance optimisation, and fault detection. Integrating digital twins with robotic disassembly processes can enhance efficiency, adaptability, and intelligence. By simulating disassembly tasks virtually, digital twins allow for optimising sequences, motion planning, and identifying potential issues before they occur physically. This reduces errors, improves efficiency, and contributes to sustainable recycling solutions for end-of-life

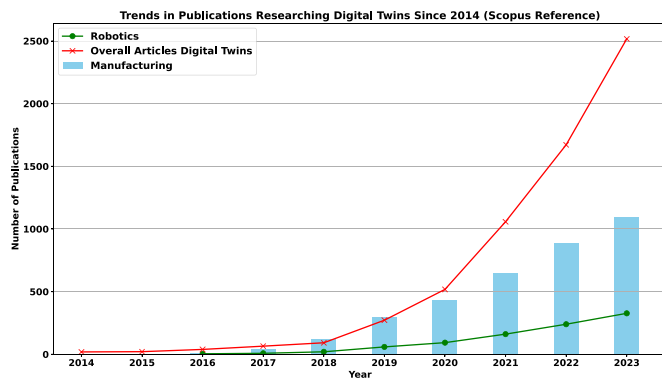


Fig. 14. Number of publications researching digital twins since 2014, based on data from Scopus. A clear trend of increasing publication numbers each year reflects growing interest and research activity in this field.

products. Several studies outline the use of digital twins in robotics can be seen Table 15 (Digital twin Research Area) provides a summary of these studies and outlines potential future developments in the context of robotic disassembly.

Current issues

Current limitations of digital twins include the complexity of creating and integrating them into existing systems, particularly for complex robotics [268]. Additionally, the lack of standardised definitions for digital twins complicates their design and application across different sectors. Furthermore, digital twins demand high levels of data accuracy and real-time processing capabilities, which can be technologically challenging and resource-intensive to achieve [269].

6.2. Improving industrial robot communication

The study presented by Mineo et al. [279] introduces the **Interfacing Toolbox for Robotic Arms (ITRA)**, a C++ language-based toolbox designed to facilitate the integration of industrial robotic arms with server computers, sensors, and actuators. ITRA is highlighted for its flexibility and extensibility, capable of controlling multiple robots simultaneously and supporting sophisticated manufacturing operations. It achieves fast adaptive robotic systems with latency as low as 30 ms and is cross-platform compatible, offering great flexibility between different computer architectures. Initially focused on KUKA hardware, ITRA's architecture allows for future support for robots produced by other manufacturers by customising the communication protocol and templates for communication packets. This hardware-agnostic approach means faster industry adoption and a smoother transition of academic research into industrial solutions.

ITRA's integration into a digital twin would benefit robotic disassembly at an industrial scale. This integration would optimise task planning and motion paths and simulate various disassembly scenarios for improved efficiency, safety, and cost-effectiveness. ITRA's real-time adaptive capabilities would enable the digital twin to respond dynamically to variations in battery types or conditions. This would lead to more precise robotic disassembly operations and support sustainable, scalable recycling solutions for electric vehicle batteries and other complex products.

6.3. Building on foundation models for robotic control

Following the introduction of the seminal Transformer architecture by Vaswani et al. [280], and subsequent research in NLP and Computer Vision have yielded powerful models like **Bidirectional Encoder Representations from Transformers (BERT)**, **Generative Pre-trained**

Transformer-4 (GPT-4) [281], **Vision Transformer (ViT)**, and **Contrastive Language-Image Pre-training (CLIP)** [282]. These models have significantly advanced the state of the art by enabling machines to process and understand language and visual information in ways that closely mimic human cognitive abilities. The convergence of these technologies has given rise to a new class of systems that integrate **Vision-language models (VLMs)** and **Large language models (LLMs)** for comprehensive scene understanding and the generation of actionable instructions for robotic applications [283,284]. This synergy between foundation models and robotics has paved the way for innovative architectures to improve robotics and enhance robots' capabilities to perform various tasks with greater autonomy and adaptability.

Recent studies have showcased various approaches to integrating foundation models with robotics, each contributing unique advancements to the field. Khandelwal et al. [282] introduced **Context Optimisation (CoOp)** to adapt CLIP-like models for image recognition, significantly outperforming hand-crafted prompts and demonstrating robust domain generalisation. Chen et al. [283] developed **Pathways Language and Image model (PaLI)**, a multilingual language-image model that excels in vision, language, and multimodal tasks by leveraging the joint scaling of pre-trained language models and ViTs. Driess et al. in [284] presented **Pathway Language model- Embodied (PaLM-E)**, an embodied multimodal language model that incorporates sensor data into language models, enabling robots to perform tasks that require complex reasoning. Similarly, Brohan et al. introduced the **Robotics Transformer RT-1** and its successor **RT-2** [285], which learn from diverse data sources for scalable, real-world control and the transfer of web knowledge to robotic control, respectively. Building on this, Leal et al. [286] proposed **Self-Adaptive Robust Attention-RT (SARA-RT)**, which enhances the efficiency of RT models through self-adaptive robust attention and "up-training" to reduce computational demands. [287] described **AutoRT**, a system that orchestrates multiple robots to gather training data, leveraging VLMs and LLMs to understand environments and suggest tasks. Lastly, Xie et al. [281] explored the application of ChatGPT for human-robot interaction and task planning, introducing an architecture that enables ChatGPT to plan tasks based on the robot's state. These studies collectively advance robotic capabilities, allowing them to understand and interact with their surroundings more effectively and perform complex tasks with increased generalisation and efficiency.

Disassembling structures in hazardous environments – think toxic waste sites or collapsed buildings – is dangerous for humans. New research integrating advanced foundation models with robotics offers a glimpse into a safer future. Building robots that can adapt to the unique visual cues of a hazardous environment using CoOp [282] to identify and disassemble materials accurately. PaLI [283] allows robots to understand multilingual instructions, facilitating global collaboration in disaster response. PaLM-E [284] equips robots with real-time decision-making abilities crucial for handling unexpected changes. RT models [285] excel in unpredictable environments due to their ability to learn from diverse data. SARA-RT [286] ensures efficient decision-making despite limited resources. AutoRT [287] allows robots to continuously learn from new tasks, moving closer to lifelong learning and enhancing their adaptability. Finally, ChatGPT for robots [281] improves communication and coordination between robots and human operators, which is crucial for safe and efficient operations. These advanced foundation models could empower robots with the intelligence and adaptability needed to navigate the complexities and dangers of hazardous disassembly, paving the way for a future where robots take the lead in these high-risk tasks.

6.4. Improvement in multi-agent task planning for disassembly

Cooperative **Multi-agent Planning (MAP)** is a burgeoning research field that integrates the AI Planning community with the Multi-agent Systems community, focusing on optimally distributed problem-solving

Table 15
Categorised Future Research Directions in Robotic Disassembly focusing on Digital Twins.

Digital twin research area	Study	Description and future work
Safety and Assurance		
Safety-aware Human-Robot Collaboration	[270]	Integrated mixed reality using deep learning instance-based segmentation (3D point cloud data to track skeletal information) and digital twin technologies to establish safe distances in human–robot collaboration. Future work could focus on enhancing the real-time capabilities of these systems and expanding the deep learning models to better predict and mitigate potential safety issues in more complex environments.
Safety Assurance in Collaborative Robotics	[271]	The study presented a modular digital twinning framework for safety assessment in collaborative robotics. Their framework virtually validated safety claims for the robot controller and was successfully tested with a robotic arm in a dynamic environment. Future enhancements could include AI, mixed reality, IoT sensors, and secure 5G/6G wireless technologies. This framework has the potential for broader application in high-risk disassembly environments and increased adaptability across diverse robotic manufacturing processes.
Simulation and Planning		
High-Fidelity Digital Twins for Industrial Robots	[269]	Discussed the development of high-fidelity digital twins for industrial robots, focusing on their usability in manufacturing. Future research could expand on this by exploring the integration of these high-fidelity digital twins with cloud computing platforms to enhance data analysis and real-time decision-making capabilities or combining with novel approaches for disassembly scheduling as discussed in [35] or other previously published cloud robotic studies like [272].
Enhancing Gesture-Control and Digital Twin Simulation	[273]	Explored the integration of gesture-controlled robotics and digital twin simulations for industrial disassembly tasks. To enhance this work, more advanced gesture recognition algorithms should be developed. Currently, the system relies on an RGB sensor for spatial HRC and a random forest-based depth estimator. Future directions include applying deep learning techniques or instance-based segmentation for hand gesture recognition, allowing for a wider range of commands. Refining the digital twin models would further improve the accuracy and efficiency of disassembly sequence planning.
Adaptive Planning for Human-Robot Collaboration	[274]	Highlight the potential of adaptive planning in HRC disassembly, especially for complex products like lithium-ion batteries, based on genetic algorithm. Future work could focus on extending these adaptive planning techniques to other types of LIB batteries or different EoL products, incorporating more dynamic and uncertain conditions into the digital twin environment.
Addressing Uncertainties in Robotic Disassembly	[275]	Proposed a framework for dynamic planning under uncertain conditions, using deep learning to adapt disassembly sequences on the fly. Further research could aim to integrate this approach with more complex disassembly operations, considering additional uncertainties such as varying material properties or unexpected component failures.
Collaborative Systems		
AR-assisted Digital Twin for Multi-Robot Collaboration	[276]	This research introduces an AR-assisted digital twin-enabled system for collaborative manufacturing with multiple robots and human-in-the-loop control. The system maps digital twins of industrial robots onto their physical counterparts, visible through Augmented reality (AR) glasses. A multi-robot communication mechanism is used to synchronise the state of robots in the digital twin environment alongside integrating an RL algorithm for enhanced motion planning. This approach replaces conventional kinematics-based robot movement, enabling three interactive modes for real-time motion control, planned motion control, and robot monitoring, which operators can switch. Future research could explore further scalability of this system in various industrial settings, enhance the intuitive control features, and integrate additional machine learning techniques to optimise multi-robot coordination and task efficiency in more complex manufacturing scenarios.
Exploiting VR and Digital Twins for Task and Motion Planning	[98]	Demonstrate the use of VR and digital twins for robotics navigation and human–robot interactions. Future research could leverage these technologies to create detailed virtual simulations of disassembly processes, optimising task and motion planning in a risk-free environment.
Learning and Optimisation		
Implementing Deep Reinforcement Learning Frameworks in Industrial Robots	[277]	The study introduces a novel approach where a digital twin is integrated with a physical robotic system for assembly tasks. This integration allows for dual training environments where virtual and real robotic systems can operate in parallel. The digital twin system uses virtual inputs to predict and correct the real robot's grasping points, thus ensuring accuracy and reducing the real-world training time and associated risks. Future works can explore disassembly tasks and look into life-long learning algorithms
Optimising Robotic Disassembly Lines with Deep Reinforcement Learning	[278]	Provide insights into the dynamic balancing of U-shaped robotic disassembly lines using DRL. The next steps in research could involve applying these deep learning approaches to various disassembly line configurations and exploring their effectiveness in different industrial settings.

as opposed to the classical single-agent planning paradigm [288]. This field has evolved significantly with the advancement of TAMP, which has been adapted into MAP in three ways. First, even though many studies focus on sequential, single-arm manipulations in single-robot dual-arm manipulation, dual-arm robots like the PR2 require symbolic planning for both arms during bi-manual tasks [288]. Second, in multi-robot cooperation, a comprehensive TAMP approach must address high-level task decomposition and allocation, which challenges the scalability of traditional TAMP methods [289]. Third, in human–robot cooperation, TAMP is increasingly employed in unstructured human

environments where effective communication is essential, and robots must be capable of anticipating human intentions [56].

A promising approach for collaborative task execution in multi-robot systems is the **decentralised task and path planning (DTPP)** framework [290]. This framework uses MOMDPs to optimise task assignment to individual robots, while localised forward dynamic programming ensures collision-free operation. The decentralised nature allows robots to adapt their paths independently in response to changing environments. Future work could focus on applying the DTPP framework to the challenges of robotic disassembly tasks, which would

require tailoring task definitions and the decision-making process to handle the unique requirements of dismantling objects.

6.5. Domain adaptation

Generalisability refers to the ability to use previous knowledge from a source environment to achieve a desired performance in a target environment; it is widely seen as a necessary step to produce artificial intelligence that behaves similarly to how human [291]. Generalisation may improve the characteristics of learning target tasks by increasing the starting reward on the target domain, the rate of learning for the target task, and the maximum reward achievable [292]. Studies in robotics grasp [293] have shown innovation in attempting generalisability hand-eye coordination; however, as highlighted in Section 2.3.3, the methods struggle not to obtain accuracy for real application. Other approaches for generalisability are known as meta-learning, also known as learning to learn [293].

Another interesting future direction is the field of domain adaptation strategies used in computer and robotic vision. Domain adaptation, traditionally a mainstay in computer vision, aims to address the challenge of applying models trained on one domain (source) to another distinct domain (target), where direct transferability of learned features and models is often limited due to differences in data distribution. These methods may overcome domain shift issues, and knowledge transfer between various data distributions can be facilitated [294]. Three domain adaptation techniques exist classical, deep learning-based, and hybrid. Traditional approaches, like **Transfer Component Analysis (TCA)** and **Maximum Mean Discrepancy (MMD)**, concentrate on statistical feature space alignment, whereas deep learning-based approaches, such as **Simulation Generative Adversarial Network (SimGAN)** [295], **Domain Adversarial Neural Networks (DANN)** and **CycleGAN**, take advantage of neural networks to develop domain invariant representations [296]. Hybrid systems like DAN use traditional and deep learning algorithms to exploit their complementary capabilities.[296] introduced **Joint domain alignment and discriminate feature learning for unsupervised deep domain adaptation (JDDA)**, which enhances transfer learning by integrating domain alignment with discriminative feature learning, specifically targeting improvements in intra-class compactness and inter-class separability within a shared feature space. At the same time, in parallel, deep learning techniques enhance this alignment by delving into the intricate, non-linear relationships present within the data. This synergy improves robustness, particularly when facing limited labelled data or noisy datasets. These domain adaptation strategies are effective in various robotic and computer vision applications. For instance, when transferring from a synthetic domain to a real-world environment, domain adaptation strategies have increased accuracy in object identification tasks from 60% to 80% [296,297].

Additionally, domain adaptation approaches have demonstrated a 15% reduction in classification error in robotic vision scenarios when adapting to unfamiliar settings [297,298]. Domain adaptation techniques are becoming increasingly valuable for real-world situations, making them essential tools for enhancing the generalisation capacities of computer and robotic vision systems [298]. These strategies enable the smooth deployment of vision models in various real-world applications, including autonomous cars, healthcare systems, and industrial automation, by efficiently resolving the domain shift problem [299].

Domain adaptations applications in computer vision are well-documented and extensive, ranging from image classification to object detection and segmentation; the principles and techniques of domain adaptation hold potential for broader applicability across various fields, including robotic control. In robotic control, systems rely on an array of sensor data—including force, torque, and velocity measurements—to interact with their environment precisely and intelligently. These sensors provide real-time data that inform control

algorithms, enabling robots to adjust their actions based on the observed physical properties of the objects they interact with and the tasks they perform. However, the efficacy of these control algorithms can significantly diminish when a robot is transitioned to a new environment (similar to the problem in Section 2.3.3) or given a new task, primarily due to changes in the data distribution of sensor readings—akin to the domain shift problem in computer vision.

Expanding domain adaptation techniques to robotic control involves several innovative adaptations and considerations are needed to push the field of robotic generalisability:

1. **Sensor Data Adaptation:** Domain adaptation techniques can be tailored for sensor data time-series nature to facilitate seamless policy transfer between different conditions with minimal performance loss.
2. **Cross-Modal Domain Adaptation:** Robots can better generalise behaviours across different environments and tasks by developing strategies to align feature spaces across various sensor modalities.
3. **Simulation to Real World Transfer:** Sim2Real domain adaptation employs techniques like domain randomisation to effectively transfer models from simulation to real-world applications.
4. **Adversarial Training and Feature Disentanglement:** This method isolates task-relevant features from those specific to the environment, enabling consistent robotic control across different domains.
5. **Interactive Learning and Adaptation:** Robots dynamically adapt their control strategies in real-time through feedback, which is crucial for operating in unpredictable environments.

By expanding domain adaptation techniques to robotic control and other fields beyond computer vision, researchers and practitioners can leverage the underlying principles of transfer learning to tackle the challenges of data variability and distribution shift across diverse application areas. This cross-disciplinary approach enriches the toolbox available for tackling domain adaptation problems and opens new avenues for innovation and application in robotics and beyond.

7. Conclusion

This review paper thoroughly examines the integration of advanced AI planning methodologies, especially TAMP, into industrial robotic disassembly settings. Our analysis addresses critical research questions, providing insights into current applications and potential future directions.

While TAMP is increasingly used in robotic disassembly, there is a gap between theoretical advancements and practical implementation due to the complexities of real-world scenarios. We must enhance model robustness and focus on domain adaptation to bridge this gap. Existing disassembly systems are often static and rigid; integrating TAMP could transform them by boosting adaptability for long-horizon tasks and improving operational efficiency. This shift aligns with the growing interest in digital twins for waste management optimisation.

Recent AI advancements, like RL and computer vision, complement TAMP to enhance the flexibility and effectiveness of robotic disassembly through optimised real-time decision-making. Expanding robotic disassembly into areas like electronic waste, lithium-ion batteries, and aerospace components requires sophisticated AI planning methods for safe, efficient, and sustainable operations.

Integrating sophisticated AI planning methodologies with robotic disassembly operations promises significant benefits in safety, efficiency, and sustainability. This is especially true in the handling of complex and hazardous materials like electronic waste and LIBs. Additionally, we envision considerable potential for applying this integrated approach within the aerospace industry.

Finally, a roadmap for future research is outlined, focusing on domain adaptation, digital twins, integration of symbolic planners

with LLMs, and multi-agent systems. These strategies will help close the gap between theory and practice, improve decision-making, and optimise collaboration in robotic disassembly systems. The convergence of TAMP with cutting-edge AI offers exciting opportunities to revolutionise robotic disassembly and drive sustainable industrial practices.

Overall, the convergence of TAMP with cutting-edge AI technologies offers exciting opportunities for advancing robotic disassembly and enhancing its impact on sustainable industrial practices.

Abbreviations

The following abbreviations are used in this manuscript:

ASP	Answer Set Programming
BA	Bee colony Algorithm
BBT	Belief Behaviour Trees
Bb-TAMP	Belief-based Task and Motion Planning Bb-TAMP
BT	Behaviour Tree
CC-HTNs	Clique/Chain Hierarchical Task Networks
CMP	Cooperative Motion Planner
CNN	Convolution Neural Network
CSP	Constraint Satisfaction Problems
CHIMP	Conflict-driven Hierarchical Meta-CSP Planner
CHOMP	Covariant Hamiltonian Optimisation for Motion Planning
CTAMP	Combined Task and Motion Planning
Cob-TAMP	Control-based Task and Motion Planning
Csb-TAMP	Constraints-based Task and Motion Planning
DDL	Deep Learning
T	Decision Trees
DTPP	Decentralised Task and Path Planning
EAS	Extended Action Specification
EoL	End Of Life
FSM	Finite State Machine
GA	Genetic Algorithm
GNN	Graph Neural Network
HARP	Hierarchical Abstraction-guided Robot Planner
HSMP	Hierarchical Stochastic Motion Planning
Hb-TAMP	Hierarchical-based Task and Motion Planning
HTN	Hierarchical Task Network
HTP	Hierarchical Task Planning
HBP	Human Behavior Prediction
IDTMP	Incremental Task and Motion Planning algorithm
ITOMP	incremental Trajectory Optimization Motion Planning
LGP	Logic-geometric Programming
LIB	Lithium-ion Batteries
LLM	Large Language Model
LTL	Linear Temporal Logic
MARL	Multi-Agent Reinforcement Learning
MCTS	Monte Carlo Tree Search
MDP	Markov Decision Process
MPLP	Massively Parallelised Lazy Search Algorithm
M-VFH	Modified Vector Field Histogram
MILP	Mixed Integer-Linear Programming
Nb-TAMP	Navigation-based Task and Motion Planning Nb-TAMP
OMPL	Open Motion Planning Library
Ob-TAMP	Optimisation-based Task and Motion Planning
PDDL	Planning Domain Definition Language
PETLON	Planning Efficiently for Task-Level-optimal Navigation
POMDP	Partially Observable Markov Decision Process

PR2	Personal Robot 2
PSO	Particle Swarm Optimisation
RDSP	Robotics Disassembly Sequence Planning
RL	Reinforcement Learning
RRT	Rapidly-exploring Random Tree
RobMAP	Robotic Motion and Action Planning
SA	Simulated Annealing
SAS	Simple Action Specification
Sb - sTAMP	Sampling-based Task and Motion Planning
SMT	Satisfiability Modulo Theories
STAMPS	Simultaneous Task and Motion Planning Scheduling
STOMP	Stochastic Trajectory Optimisation for Motion Planning
STRIPS	Stanford Research Institute Problem Solver
TAMP	Task and Motion planning
TS	Tabu Search
TMKit	Task Motion Kit
TMPUD	Task and Motion Planning for Urban Driving
VRP	Vehicle Routing Problem

CRedit authorship contribution statement

Mohammed Eesa Asif: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Alireza Rastegarpanah:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Rustam Stolkin:** Writing – review & editing, Supervision, Resources, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the “Research and Development of a Highly Automated and Safe Streamlined Process for Increased Lithium-ion Battery Repurposing and Recycling (REBELION)” project, supported by Grant 101104241. We extend our appreciation to the Manufacturing Technology Centre (MTC), particularly to Karol Janik and Thomas Legleu.

References

- [1] Liu Z, Li C, Fang X, Guo Y. Energy consumption in additive manufacturing of metal parts. *Procedia Manuf* 2018;26:834–45.
- [2] Kumar A, Holuszko M, Espinosa DCR. E-waste: An overview on generation, collection, legislation and recycling practices. *Resour Conserv Recy* 2017;122:32–42.
- [3] Commission E. Circular economy action plan. 2020, [Online]. Available: https://environment.ec.europa.eu/strategy/circular-economy-action-plan_en; environment.ec.europa.eu.
- [4] Kerin M, Pham DT. Smart remanufacturing: a review and research framework. *J Manuf Technol Manag* 2020;31(6):1205–35.
- [5] Cucchiella F, D’Adamo I, Koh SL, Rosa P. Recycling of WEEEs: An economic assessment of present and future e-waste streams. *Renew Sustain Energy Rev* 2015;51:263–72.
- [6] Council RI. Reman council. 2023, [Online]. Available: <https://remancouncil.org/>; Remanufacturing Industries Council.
- [7] Ismail H, Hanafiah MM. A review of sustainable e-waste generation and management: Present and future perspectives. *J Environ Manag* 2020;264:110495.
- [8] Karvonen I, Jansson K, Behm K, Vatanen S, Parker D. Identifying recommendations to promote remanufacturing in Europe. *J Remanuf* 2017;7:159–79.
- [9] Yasin S, Ghafoor A, Lodhi AS, Ahmed M, Kausar R. Green marketing: A study of consumers’ attitude towards environment friendly products. *Lasbela Univ J Sci Technol* 2015;4:109–16.

- [10] Bogue R. Robots in recycling and disassembly. *Industr Robot: The Int J Robot Res Appl* 2019;46(4):461–6.
- [11] Gao K-Z, He Z, Huang Y, Duan P-Y, Suganthan PN. A survey on meta-heuristics for solving disassembly line balancing, planning and scheduling problems in remanufacturing. *Swarm Evol Comput* 2020;57:100719.
- [12] Hjorth S, Chrysostomou D. Human-robot collaboration in industrial environments: A literature review on non-destructive disassembly. *Robot Comput-Integr Manuf* 2022;73:102208.
- [13] Garrett CR, Chitnis R, Holladay R, Kim B, Silver T, Kaelbling LP, Lozano-Pérez T. Integrated task and motion planning. *Ann Rev Contr, Robot, Autonom Syst* 2021;4:265–93.
- [14] Guo H, Wu F, Qin Y, Li R, Li K, Li K. Recent trends in task and motion planning for robotics: A survey. *ACM Comput Surv* 2023.
- [15] Çil ZA, Mete S, Serin F. Robotic disassembly line balancing problem: A mathematical model and ant colony optimization approach. *Appl Math Model* 2020;86:335–48.
- [16] Paprocka I, Skołod B. A predictive approach for disassembly line balancing problems. *Sensors* 2022;22(10):3920.
- [17] Guo X, Zhou M, Abusorrah A, Alsokhry F, Sedraoui K. Disassembly sequence planning: a survey. *IEEE/CAA J Autom Sin* 2020;8(7):1308–24.
- [18] Alartsev S, Stellmacher S, Ortmeier F. Robotic task sequencing problem: A survey. *J Intell Robot Syst* 2015;80:279–98.
- [19] Alshibli M, ElSayed A, Kongar E, Sobh T, Gupta SM. A robust robotic disassembly sequence design using orthogonal arrays and task allocation. *Robotics* 2019;8(1):20.
- [20] Wang Y, Lan F, Pham DT, Liu J, Huang J, Ji C, Su S, Xu W, Liu Q, Zhou Z. Automatic detection of subassemblies for disassembly sequence planning.. In: *ICINCO (1)*. 2018, p. 104–10.
- [21] Chen S, Yi J, Zhu X, Jiang H, Ju W. RFID-based integrated method for electromechanical products disassembly decision-making. *Int J Comput Integr Manuf* 2017;30(2–3):229–54.
- [22] Wegener K, Andrew S, Raatz A, Dröder K, Herrmann C. Disassembly of electric vehicle batteries using the example of the audi Q5 hybrid system. *Proc CIRP* 2014;23:155–60. <http://dx.doi.org/10.1016/j.procir.2014.10.098>, URL <https://www.sciencedirect.com/science/article/pii/S221282711401155X>, 5th CATS 2014 - CIRP Conference on Assembly Technologies and Systems.
- [23] Jin G, Li W, Wang S, Lu X. Solution space generation for disassembly research on liquid crystal displays televisions. In: *Proceedings of the 2014 IEEE 18th international conference on computer supported cooperative work in design. CSCWD, 2014*, p. 35–40. <http://dx.doi.org/10.1109/CSCWD.2014.6846813>.
- [24] Xiao J, Gao J, Anwer N, Eynard B. Multi-agent reinforcement learning method for disassembly sequential task optimization based on human-robot collaborative disassembly in electric vehicle battery recycling. *J Manuf Sci Eng* 2023;145(12):121001.
- [25] Qu M, Wang Y, Pham DT. Robotic disassembly task training and skill transfer using reinforcement learning. *IEEE Trans Ind Inf* 2023.
- [26] Allagui A, Belhadi I, Plateaux R, Hammadi M, Penas O, Aifaoui N. Reinforcement learning for disassembly sequence planning optimization. *Comput Ind* 2023;151:103992.
- [27] Afkarian A, Stolkin R, Rastegarpanah A. Integrating multi-demonstration knowledge and bounded workspaces for efficient deep reinforcement learning. In: *2023 IEEE-RAS 22nd international conference on humanoid robots (humanoids)*. IEEE; 2023, p. 1–8.
- [28] Cai W, Guo X, Wang J, Qin S, Zhao J, Tan Y. An improved advantage actor-critic algorithm for disassembly line balancing problems considering tools deterioration. In: *2022 IEEE international conference on systems, man, and cybernetics. SMC, 2022*, p. 3336–41. <http://dx.doi.org/10.1109/SMC53654.2022.9945173>.
- [29] Liu Y, Zhou M, Guo X. An improved Q-learning algorithm for human-robot collaboration two-sided disassembly line balancing problems. In: *2022 IEEE international conference on systems, man, and cybernetics. SMC, 2022*, p. 568–73. <http://dx.doi.org/10.1109/SMC53654.2022.9945263>.
- [30] Zhao X, Li C, Tang Y, Cui J. Reinforcement learning-based selective disassembly sequence planning for the end-of-life products with structure uncertainty. *IEEE Robot Autom Lett* 2021;6(4):7807–14. <http://dx.doi.org/10.1109/LRA.2021.3098248>.
- [31] Mao H, Liu Z, Qiu C. Adaptive disassembly sequence planning for VR maintenance training via deep reinforcement learning. *Int J Adv Manuf Technol* 2021;1–10.
- [32] Zakka K, Zeng A, Lee J, Song S. Form2fit: Learning shape priors for generalizable assembly from disassembly. In: *2020 IEEE international conference on robotics and automation. ICRA, IEEE; 2020*, p. 9404–10.
- [33] Liu Q, Liu Z, Xu W, Tang Q, Zhou Z, Pham DT. Human-robot collaboration in disassembly for sustainable manufacturing. *Int J Prod Res* 2019;57(12):4027–44.
- [34] Kristensen CB, Sørensen FA, Nielsen HB, Andersen MS, Bendtsen SP, Bøgh S. Towards a robot simulation framework for e-waste disassembly using reinforcement learning. *Procedia Manuf* 2019;38:225–32.
- [35] Chen S, Fang S, Tang R. A reinforcement learning based approach for multi-projects scheduling in cloud manufacturing. *Int J Prod Res* 2019;57(10):3080–98.
- [36] Tuncel E, Zeid A, Kamarthi S. Solving large scale disassembly line balancing problem with uncertainty using reinforcement learning. *J Intell Manuf* 2014;25:647–59.
- [37] Biehler M, Sun Y, Kode S, Li J, Shi J. PLURAL: 3D point cloud transfer learning via contrastive learning with augmentations. *IEEE Trans Autom Sci Eng* 2023.
- [38] Chen J, Bao E, Pan J. Classification and positioning of circuit board components based on improved YOLOv5. *Procedia Comput Sci* 2022;208:613–26.
- [39] Zheng S, Lan F, Baronti L, Pham DT, Castellani M. Automatic identification of mechanical parts for robotic disassembly using the PointNet deep neural network. *Int J Manuf Res* 2022;17(1):1–21.
- [40] Foo G, Kara S, Pagnucco M. Artificial learning for part identification in robotic disassembly through automatic rule generation in an ontology. *IEEE Trans Autom Sci Eng* 2022;20(1):296–309.
- [41] Brådlund H, Choux M, Cenkermaddi LR. Point cloud instance segmentation for automatic electric vehicle battery disassembly. In: *International conference on intelligent technologies and applications*. Springer; 2021, p. 247–58.
- [42] Su C, Hu J, Lou P. Mask R-CNN based laptop parts detector for disassembly. In: *Proceedings of the 2021 5th international conference on machine learning and soft computing*. 2021, p. 162–7.
- [43] Bai T, Yang J, Xu G, Yao D. An optimized railway fastener detection method based on modified faster R-CNN. *Measurement* 2021;182:109742.
- [44] Li X, Li M, Wu Y, Zhou D, Liu T, Hao F, Yue J, Ma Q. Accurate screw detection method based on faster R-CNN and rotation edge similarity for automatic screw disassembly. *Int J Comput Integr Manuf* 2021;34(11):1177–95.
- [45] Foo G, Kara S, Pagnucco M. Screw detection for disassembly of electronic waste using reasoning and re-training of a deep learning model. *Procedia CIRP* 2021;98:666–71.
- [46] Brogan DP, DiFilippo NM, Jouaneh MK. Deep learning computer vision for robotic disassembly and servicing applications. *Array* 2021;12:100094.
- [47] He C, Jin Z, Gu R, Qu H. Automatic disassembly and recovery device for mobile phone circuit board CPU based on machine vision. In: *Journal of physics: conference series*. 1684, IOP Publishing; 2020, 012137.
- [48] Zhang Y, Sun X, Loh KJ, Su W, Xue Z, Zhao X. Autonomous bolt loosening detection using deep learning. *Struct Health Monit* 2020;19(1):105–22.
- [49] DiFilippo NM, Jouaneh MK. Using the soar cognitive architecture to remove screws from different laptop models. *IEEE Trans Autom Sci Eng* 2018;16(2):767–80.
- [50] Vongbunyoung S, Vongseela P, Sreerattana-aporn J. A process demonstration platform for product disassembly skills transfer. *Procedia CIRP* 2017;61:281–6.
- [51] Friedrich C, Lechler A, Verl A. The control architecture RoViDiAsS—A robotic visual disassembly and assembly system. In: *2016 IEEE international conference on industrial technology. ICIT, IEEE; 2016*, p. 113–8.
- [52] Wegener K, Chen WH, Dietrich F, Dröder K, Kara S. Robot assisted disassembly for the recycling of electric vehicle batteries. *Procedia CIRP* 2015;29:716–21, URL <https://api.semanticscholar.org/CorpusID:108000987>.
- [53] Chu M, Chen W. Human-robot collaboration disassembly planning for end-of-life power batteries. *J Manuf Syst* 2023;69:271–91.
- [54] Yuan G, Liu X, Zhang C, Pham DT, Li Z. A new heuristic algorithm based on multi-criteria resilience assessment of human-robot collaboration disassembly for supporting spent lithium-ion battery recycling. *Eng Appl Artif Intell* 2023;126:106878.
- [55] Kay I, Farhad S, Mahajan A, Esmaeili R, Hashemi SR. Robotic disassembly of electric vehicles' battery modules for recycling. *Energies* 2022;15(13):4856.
- [56] Zhou Y, Peng Y, Li W, Pham DT. Stackelberg model-based human-robot collaboration in removing screws for product remanufacturing. *Robot Comput-Integr Manuf* 2022;77:102370.
- [57] Wu T, Zhang Z, Yin T, Zhang Y. Multi-objective optimisation for cell-level disassembly of waste power battery modules in human-machine hybrid mode. *Waste Manage* 2022;144:513–26.
- [58] Huang J, Pham DT, Li R, Qu M, Wang Y, Kerin M, Su S, Ji C, Mahomed O, Khalil R, et al. An experimental human-robot collaborative disassembly cell. *Comput Ind Eng* 2021;155:107189.
- [59] Wu C, Bi X, Pfrommer J, Cebulla A, Mangold S, Beyerer J. Sim2real transfer learning for point cloud segmentation: An industrial application case on autonomous disassembly. In: *Proceedings of the IEEE/CVF winter conference on applications of computer vision*. 2023, p. 4531–40.
- [60] Rosenberg S, Huster S, Baazouzi S, Glöser-Chahoud S, Al Assadi A, Schultmann F. Field study and multimethod analysis of an EV battery system disassembly. *Energies* 2022;15(15):5324.
- [61] Xu W, Tang Q, Liu J, Liu Z, Zhou Z, Pham DT. Disassembly sequence planning using discrete bees algorithm for human-robot collaboration in remanufacturing. *Robot Comput-Integr Manuf* 2020;62:101860.
- [62] Ramírez FJ, Aledo JA, Gamez JA, Pham DT. Economic modelling of robotic disassembly in end-of-life product recovery for remanufacturing. *Comput Ind Eng* 2020;142:106339.
- [63] Zhang Y, Lu H, Pham DT, Wang Y, Qu M, Lim J, Su S. Peg-hole disassembly using active compliance. *Royal Soc Open Sci* 2019;6(8):190476.
- [64] Laili Y, Tao F, Pham DT, Wang Y, Zhang L. Robotic disassembly re-planning using a two-pointer detection strategy and a super-fast bees algorithm. *Robot Comput-Integr Manuf* 2019;59:130–42.

- [65] Tao Y, Meng K, Lou P, Peng X, Qian X. Joint decision-making on automated disassembly system scheme selection and recovery route assignment using multi-objective meta-heuristic algorithm. *Int J Prod Res* 2019;57(1):124–42.
- [66] Ren Y, Zhang C, Zhao F, Xiao H, Tian G. An asynchronous parallel disassembly planning based on genetic algorithm. *European J Oper Res* 2018;269(2):647–60.
- [67] Feng Y, Gao Y, Tian G, Li Z, Hu H, Zheng H. Flexible process planning and end-of-life decision-making for product recovery optimization based on hybrid disassembly. *IEEE Trans Autom Sci Eng* 2018;16(1):311–26.
- [68] Liu J, Zhou Z, Pham DT, Xu W, Ji C, Liu Q. Robotic disassembly sequence planning using enhanced discrete bees algorithm in remanufacturing. *Int J Prod Res* 2018;56(9):3134–51.
- [69] Cheng H, Xu W, Ai Q, Liu Q, Zhou Z, Pham DT. Manufacturing capability assessment for human-robot collaborative disassembly based on multi-data fusion. *Procedia Manuf* 2017;10:26–36.
- [70] Xia K, Gao L, Wang L, Li W, Li X, Ijomah W. Service-oriented disassembly sequence planning for electrical and electronic equipment waste. *Electron Commer Res Appl* 2016;20:59–68.
- [71] Djogdom GVT, Meziane R, Otis MJ-D. Robust dynamic robot scheduling for collaborating with humans in manufacturing operations. *Robot Comput-Integr Manuf* 2024;88:102734.
- [72] Ghzouli R, Berger T, Johnsen EB, Wasowski A, Dragule S. Behavior trees and state machines in robotics applications. *IEEE Trans Softw Eng* 2023.
- [73] Pezzato C, Corbato CH, Bonhof S, Wisse M. Active inference and behavior trees for reactive action planning and execution in robotics. *IEEE Trans Robot* 2023;39(2):1050–69.
- [74] Rastegarpanah A, Gonzalez HC, Stolkin R. Semi-autonomous behaviour tree-based framework for sorting electric vehicle batteries components. *Robotics* 2021;10(2):82.
- [75] Ding Y, Xu W, Liu Z, Zhou Z, Pham DT. Robotic task oriented knowledge graph for human-robot collaboration in disassembly. *Procedia CIRP* 2019;83:105–10.
- [76] Ebinger T, Kaden S, Thomas S, Andre R, Amato NM, Thomas U. A general and flexible search framework for disassembly planning. In: 2018 IEEE international conference on robotics and automation. ICRA, IEEE; 2018, p. 3548–55.
- [77] Costa CM, Veiga G, Sousa A, Rocha L, Oliveira E, Cardoso HL, Thomas U. Automatic generation of disassembly sequences and exploded views from solid-works symbolic geometric relationships. In: 2018 IEEE international conference on autonomous robot systems and competitions. ICARSC, IEEE; 2018, p. 211–8.
- [78] Paxton C, Hundt A, Jonathan F, Guerin K, Hager GD. CoSTAR: Instructing collaborative robots with behavior trees and vision. In: 2017 IEEE international conference on robotics and automation. ICRA, IEEE; 2017, p. 564–71.
- [79] Gammell JD, Srinivasa SS, Barfoot TD. Informed RRT*: Optimal sampling-based path planning focused via direct sampling of an admissible ellipsoidal heuristic. In: 2014 IEEE/RSJ international conference on intelligent robots and systems. 2014, p. 2997–3004. <http://dx.doi.org/10.1109/IROS.2014.6942976>.
- [80] Hathaway J, Shaarawy A, Akeniz C, Afkian A, Stolkin R, Rastegarpanah A. Towards reuse and recycling of lithium-ion batteries: Tele-robotics for disassembly of electric vehicle batteries. *Front Robot AI* 2023;10.
- [81] Petereit J, Beyerer J, Asfour T, Gentes S, Hein B, Hanebeck UD, Kirchner F, Dillmann R, Götting HH, Weiser M, et al. ROBDEKON: Robotic systems for decontamination in hazardous environments. In: 2019 IEEE international symposium on safety, security, and rescue robotics. SSR, IEEE; 2019, p. 249–55.
- [82] Marturi N, Rastegarpanah A, Takahashi C, Adjigle M, Stolkin R, Zurek S, Kopicki M, Talha M, Kuo JA, Bekiroglu Y. Towards advanced robotic manipulation for nuclear decommissioning: A pilot study on tele-operation and autonomy. In: 2016 international conference on robotics and automation for humanitarian applications. RAHA, IEEE; 2016, p. 1–8.
- [83] Gungor A, Gupta SM, Pochampally K, Kamarthi SV. Complications in disassembly line balancing. In: Environmentally conscious manufacturing. 4193, SPIE; 2001, p. 289–98.
- [84] Mei K, Fang Y. Multi-robotic disassembly line balancing using deep reinforcement learning. In: International manufacturing science and engineering conference. 85079, American Society of Mechanical Engineers; 2021, V002T07A005.
- [85] Pornsing C, Watanasungsit A. Discrete particle swarm optimization for disassembly sequence planning. In: 2014 IEEE international conference on management of innovation and technology. IEEE; 2014, p. 480–5.
- [86] Farhan AS, Bassiouny AM, Afif YT, Gamil AA, Alsheikh M, Kamal AM, Elenay KS, Bahour KA, Awad M, Maged S. Autonomous non-destructive assembly/disassembly of electronic components using a robotic arm. In: 2021 16th international conference on computer engineering and systems (ICCES). 2021, p. 1–7. <http://dx.doi.org/10.1109/ICCES54031.2021.9686083>.
- [87] Williams JAS. A review of research towards computer integrated demanufacturing for materials recovery. *Int J Comput Integr Manuf* 2007;20(8):773–80.
- [88] Vongbunyoung S, Kara S, Pagnucco M. Learning and revision in cognitive robotics disassembly automation. *Robot Comput-Integr Manuf* 2015;34:79–94.
- [89] Chang M, Ong S, Nee A. Approaches and challenges in product disassembly planning for sustainability. *Procedia Cirp* 2017;60:506–11.
- [90] Lee M-L, Behdad S, Liang X, Zheng M. Disassembly sequence planning considering human-robot collaboration. In: 2020 American control conference. ACC, IEEE; 2020, p. 2438–43.
- [91] Garey MR, Johnson DS, Stockmeyer L. Some simplified NP-complete problems. In: Proceedings of the sixth annual ACM symposium on theory of computing. 1974, p. 47–63.
- [92] Hartono N, Javier Ramírez F, Pham D. Optimisation of robotic disassembly sequence plans for sustainability using the multi-objective bees algorithm. In: Intelligent production and manufacturing optimisation—the bees algorithm approach. Springer; 2022, p. 337–63.
- [93] Lan F, Wang Y, Pham DT, Liu J, Huang J, Ji C, Su S, Xu W, Liu Q, Zhou Z. Interlocking problem in automatic disassembly planning and two solutions. In: Informatics in control, automation and robotics: 15th international conference, ICINCO 2018, porto, Portugal, July 29–31, 2018, revised selected papers 15. Springer; 2020, p. 193–213.
- [94] Wang Y, Lan F, Liu J, Huang J, Su S, Ji C, Pham DT, Xu W, Liu Q, Zhou Z. Interlocking problems in disassembly sequence planning. *Int J Prod Res* 2021;59(15):4723–35.
- [95] Chang P, Padif T. Sim2real2sim: Bridging the gap between simulation and real-world in flexible object manipulation. In: 2020 fourth IEEE international conference on robotic computing. IRC, IEEE; 2020, p. 56–62.
- [96] Zhao W, Queralta JP, Westerlund T. Sim-to-real transfer in deep reinforcement learning for robotics: a survey. In: 2020 IEEE symposium series on computational intelligence. SSCI, IEEE; 2020, p. 737–44.
- [97] Muratore F, Gienger M, Peters J. Assessing transferability from simulation to reality for reinforcement learning. *IEEE Trans Pattern Anal Mach Intell* 2019;43(4):1172–83.
- [98] Alves SF, Uribe-Quevedo A, Chen D, Morris J, Radmard S. Developing a VR simulator for robotics navigation and human robot interactions employing digital twins. In: 2022 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops. VRW, IEEE; 2022, p. 121–5.
- [99] Kadian A, Truong J, Gokaslan A, Clegg A, Wijmans E, Lee S, Savva M, Chernova S, Batra D. Sim2real predictivity: Does evaluation in simulation predict real-world performance? *IEEE Robot Autom Lett* 2020;5(4):6670–7.
- [100] He W, Dong Y. Adaptive fuzzy neural network control for a constrained robot using impedance learning. *IEEE Trans Neural Netw Learn Syst* 2017;29(4):1174–86.
- [101] Tang Y, Turowski M. Adaptive fuzzy system for disassembly process planning with uncertainty. *J Chinese Instit Industr Eng* 2007;24(1):20–9.
- [102] Moore KE, Gungor A, Gupta SM. A Petri net approach to disassembly process planning. *Comput Ind Eng* 1998;35(1–2):165–8.
- [103] Chen Y-L, Lin F. Modeling of discrete event systems using finite state machines with parameters. In: Proceedings of the 2000. IEEE international conference on control applications conference proceedings (cat. no. 00CH37162). IEEE; 2000, p. 941–6.
- [104] Endsley E, Almeida E, Tilbury D. Modular finite state machines: Development and application to reconfigurable manufacturing cell controller generation. *Control Eng Pract* 2006;14(10):1127–42.
- [105] König L, Mostaghim S, Schmeck H. Decentralized evolution of robotic behavior using finite state machines. *Int J Intell Comput Cybern* 2009;2(4):695–723.
- [106] Kim N, Shin D, Wysk R, Rothrock L. Using finite state automata (FSA) for formal modelling of affordances in human-machine cooperative manufacturing systems. *Int J Prod Res* 2010;48(5):1303–20.
- [107] Xia K, Gao L, Wang L, Li W, Chao K-M. A simplified teaching-learning-based optimization algorithm for disassembly sequence planning. In: 2013 IEEE 10th international conference on e-business engineering. IEEE; 2013, p. 393–8.
- [108] Aguinaga I, Borro D, Matey L. Parallel RRT-based path planning for selective disassembly planning. *Int J Adv Manuf Technol* 2008;36:1221–33.
- [109] Colledanchise M, Ögren P. Behavior trees in robotics and AI: An introduction. CRC Press; 2018.
- [110] Cortés J, Jaillat L, Siméon T. Disassembly path planning for complex articulated objects. *IEEE Trans Robot* 2008;24(2):475–81.
- [111] Sales DO, Correa DO, Fernandes LC, Wolf DF, Osório FS. Adaptive finite state machine based visual autonomous navigation system. *Eng Appl Artif Intell* 2014;29:152–62.
- [112] Cui J, Yang C, Zhang J, Tian S, Liu J, Xu W. Robotic disassembly sequence planning considering parts failure features. *IET Collaborat Intell Manuf* 2023;5(1):e12074.
- [113] Parsa S, Saadat M. Human-robot collaboration disassembly planning for end-of-life product disassembly process. *Robot Comput-Integr Manuf* 2021;71:102170.
- [114] Min S-s, Zhu X-j, Zhu X. Mechanical product disassembly and/or graph construction. In: 2010 international conference on measuring technology and mechatronics automation. 2, IEEE; 2010, p. 627–31.
- [115] Ye F, Perrett J, Zhang L, Laili Y, Wang Y. A self-evolving system for robotic disassembly sequence planning under uncertain interference conditions. *Robot Comput-Integr Manuf* 2022;78:102392.
- [116] Xu W, Cui J, Liu B, Liu J, Yao B, Zhou Z. Human-robot collaborative disassembly line balancing considering the safe strategy in remanufacturing. *J Clean Prod* 2021;324:129158.
- [117] Yu J, Zhang H, Jiang Z, Yan W, Wang Y, Zhou Q. Disassembly task planning for end-of-life automotive traction batteries based on ontology and partial destructive rules. *J Manuf Syst* 2022;62:347–66.

- [118] Zhang HC, Kuo T. A graph-based disassembly sequence planning for EOL product recycling. In: Twenty first IEEE/cPMT international electronics manufacturing technology symposium proceedings 1997 IEMT symposium. IEEE; 1997, p. 140–51.
- [119] Moore KE, Güngör A, Gupta SM. Petri net approach to disassembly process planning for products with complex AND/OR precedence relationships. *European J Oper Res* 2001;135(2):428–49.
- [120] Fernandez R, Zerhouni N. Modeling and analysis of disassembly systems using continuous Petri nets. In: Proceedings of the 2001 IEEE international symposium on assembly and task planning (ISATP2001). assembly and disassembly in the twenty-first century. (cat. no. 01TH8560). IEEE; 2001, p. 232–7.
- [121] Tian G, Zhou M, Li P. Disassembly sequence planning considering fuzzy component quality and varying operational cost. *IEEE Trans Autom Sci Eng* 2017;15(2):748–60.
- [122] Chen JC, Chen Y-Y, Chen T-L, Yang Y-C. An adaptive genetic algorithm-based and AND/OR graph approach for the disassembly line balancing problem. *Eng Optim* 2022;54(9):1583–99.
- [123] Bentaha ML, Dolgui A, Battaia O, Riggs RJ, Hu J. Profit-oriented partial disassembly line design: dealing with hazardous parts and task processing times uncertainty. *Int J Prod Res* 2018;56(24):7220–42.
- [124] Ren Y, Meng L, Zhang C, Zhao F, Saif U, Huang A, Mendis GP, Sutherland JW. An efficient metaheuristics for a sequence-dependent disassembly planning. *J Cleaner Product* 2020;245:118644.
- [125] Xu C, Wei H, Guo X, Liu S, Qi L, Zhao Z. Human-robot collaboration multi-objective disassembly line balancing subject to task failure via multi-objective artificial bee colony algorithm. *IFAC-PapersOnLine* 2020;53(5):1–6.
- [126] Liu Y, Zhou M, Guo X. An improved Q-learning algorithm for human-robot collaboration two-sided disassembly line balancing problems. In: 2022 IEEE international conference on systems, man, and cybernetics. SMC, IEEE; 2022, p. 568–73.
- [127] Hellmuth JF, DiFilippo NM, Jouaneh MK. Assessment of the automation potential of electric vehicle battery disassembly. *J Manuf Syst* 2021;59:398–412.
- [128] Torres F, Puente S, Díaz C. Automatic cooperative disassembly robotic system: Task planner to distribute tasks among robots. *Control Eng Pract* 2009;17(1):112–21.
- [129] Lykov A, Dronova M, Naglov N, Litvinov M, Satsevich S, Bazhenov A, Berman V, Shcherbak A, Tsetserukou D. LLM-MARS: Large language model for behavior tree generation and NLP-enhanced dialogue in multi-agent robot systems. 2023, arXiv preprint arXiv:2312.09348.
- [130] Alzubaidi L, Zhang J, Humaidi AJ, Al-Dujaili A, Duan Y, Al-Shamma O, Santamaria J, Fadhel MA, Al-Amidie M, Farhan L. Review of deep learning: concepts, CNN architectures, challenges, applications, future directions. *J Big Data* 2021;8:1–74.
- [131] Redmon J, Divvala S, Girshick R, Farhadi A. You only look once: Unified, real-time object detection. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2016, p. 779–88.
- [132] Lee M-L, Liu W, Behdad S, Liang X, Zheng M. Robot-assisted disassembly sequence planning with real-time human motion prediction. *IEEE Trans Syst, Man, Cybernet: Syst* 2022;53(1):438–50.
- [133] Choux M, Marti Bigorra E, Tyapin I. Task planner for robotic disassembly of electric vehicle battery pack. *Metals* 2021;11(3):387.
- [134] Wong C-C, Yeh L-Y, Liu C-C, Tsai C-Y, Aoyama H. Manipulation planning for object re-orientation based on semantic segmentation keypoint detection. *Sensors* 2021;21(7):2280.
- [135] Yildirim S, Rana ZA, Tang G. Development of vision guided real-time trajectory planning system for autonomous ground refuelling operations using hybrid dataset. In: AIAA SCITECH 2023 forum. 2023, p. 1148.
- [136] Bello SA, Yu S, Wang C, Adam JM, Li J. Deep learning on 3D point clouds. *Remote Sens* 2020;12(11):1729.
- [137] Park J, Xu C, Zhou Y, Tomizuka M, Zhan W. Detmatch: Two teachers are better than one for joint 2d and 3d semi-supervised object detection. In: European conference on computer vision. Springer; 2022, p. 370–89.
- [138] Pang G, Neumann U. 3D point cloud object detection with multi-view convolutional neural network. In: 2016 23rd international conference on pattern recognition. ICPR, IEEE; 2016, p. 585–90.
- [139] Jaritz M, Gu J, Su H. Multi-view pointnet for 3d scene understanding. In: Proceedings of the IEEE/CVF international conference on computer vision workshops. 2019.
- [140] Bilal MT, Tyapin I, Choux MMH. Fusion of multiview images for EV battery disassembly. In: 2023 11th international conference on control, mechatronics and automation. ICCMA, IEEE; 2023, p. 19–24.
- [141] Brunton SL, Kutz JN. Data-driven science and engineering: Machine learning, dynamical systems, and control. Cambridge University Press; 2022.
- [142] Clifton J, Laber E. Q-learning: Theory and applications. *Annu Rev Stat Appl* 2020;7:279–301.
- [143] Dankwa S, Zheng W. Twin-delayed ddpg: A deep reinforcement learning technique to model a continuous movement of an intelligent robot agent. In: Proceedings of the 3rd international conference on vision, image and signal processing. 2019, p. 1–5.
- [144] Deisenroth MP, Neumann G, Peters J, et al. A survey on policy search for robotics. *Found Trends Robot* 2013;2(1–2):1–142.
- [145] Gu S, Holly E, Lillicrap T, Levine S. Deep reinforcement learning for robotic manipulation with asynchronous off-policy updates. In: 2017 IEEE international conference on robotics and automation. ICRA, IEEE; 2017, p. 3389–96.
- [146] Rastegarpanah A, Afkarian A, Hathaway J, Stolkin R. An online hyper-volume action bounding approach for accelerating the process of deep reinforcement learning from multiple controllers. *Authorea*; 2023.
- [147] Lian B, Xue W, Xie Y, Lewis FL, Davoudi A. Off-policy inverse Q-learning for discrete-time antagonistic unknown systems. *Automatica* 2023;155:111171.
- [148] Nguyen H, Dang HB, Dao PN. On-policy and off-policy Q-learning strategies for spacecraft systems: An approach for time-varying discrete-time without controllability assumption of augmented system. *Aerosp Sci Technol* 2024;108972.
- [149] Chen L, Dong C, He S, Dai S-L. Adaptive optimal formation control for unmanned surface vehicles with guaranteed performance using actor-critic learning architecture. *Internat J Robust Nonlinear Control* 2023;33(8):4504–22.
- [150] Wen G, Chen CL, Li W. Simplified optimized control using reinforcement learning algorithm for a class of stochastic nonlinear systems. *Inform Sci* 2020;517:230–43. <http://dx.doi.org/10.1016/j.ins.2019.12.039>.
- [151] Han M, Zhang L, Wang J, Pan W. Actor-critic reinforcement learning for control with stability guarantee. *IEEE Robot Autom Lett* 2020. <http://dx.doi.org/10.1109/LRA.2020.3011351>.
- [152] Ouyang Y, Sun C, Dong L. Actor-critic learning based coordinated control for a dual-arm robot with prescribed performance and unknown backlash-like hysteresis. *ISA Trans* 2021. <http://dx.doi.org/10.1016/j.isatra.2021.08.005>.
- [153] Gong L, Sun T, Li X, Lin K, Diaz-Rodríguez N, Filliat D, Zhang Z, Zhang J. Demonstration guided actor-critic deep reinforcement learning for fast teaching of robots in dynamic environments. *IFAC-PapersOnLine* 2020. <http://dx.doi.org/10.1016/j.ifacol.2021.04.227>.
- [154] Newaz AAR, Alam T. Hierarchical task and motion planning through deep reinforcement learning. In: 2021 fifth IEEE international conference on robotic computing. IRC, IEEE; 2021, p. 100–5.
- [155] Rajendran G, Uma V, O'Brien B. Unified robot task and motion planning with extended planner using ROS simulator. *J King Saud Univer-Comput Inf Sci* 2022;34(9):7468–81.
- [156] Bustamante S, Quere G, Leidner D, Vogel J, Stulp F. CATs: Task planning for shared control of assistive robots with variable autonomy. In: 2022 international conference on robotics and automation. ICRA, IEEE; 2022, p. 3775–82.
- [157] Kebria PM, Khosravi A, Nahavandi S, Homaifar A, Saif M. Experimental comparison study on joint and cartesian space control schemes for a teleoperation system under time-varying delay. In: 2019 IEEE international conference on industrial technology. ICIT, IEEE; 2019, p. 108–13.
- [158] Singh J, Srinivasan AR, Neumann G, Kucukylmaz A. Haptic-guided teleoperation of a 7-dof collaborative robot arm with an identical twin master. *IEEE Trans Hapt* 2020;13(1):246–52.
- [159] Xu G, Zhang Z, Li Z, Guo X, Qi L, Liu X. Multi-objective discrete brainstorming optimizer to solve the stochastic multiple-product robotic disassembly line balancing problem subject to disassembly failures. *Mathematics* 2023;11(6). <http://dx.doi.org/10.3390/math11061557>, URL <https://www.mdpi.com/2227-7390/11/6/1557>.
- [160] Ming H, Liu Q, Pham DT. Multi-robotic disassembly line balancing with uncertain processing time. *Procedia CIRP* 2019;83:71–6.
- [161] Laili Y, Wang Y, Fang Y, Pham DT, Laili Y, Wang Y, Fang Y, Pham DT. Modelling of robotic disassembly line balancing. *Optim Robot Disassembly Remanuf* 2022;71–83.
- [162] ElSayed A, Kongar E, Gupta SM, Sobh T. A robotic-driven disassembly sequence generator for end-of-life electronic products. *J Intell Robot Syst* 2012;68:43–52.
- [163] Malekhouyan S, Aghsami A, Rabbani M. An integrated multi-stage vehicle routing and mixed-model job-shop-type robotic disassembly sequence scheduling problem for e-waste management system. *Int J Comput Integr Manuf* 2021;34(11):1237–62.
- [164] Kheder M, Trigui M, Aifaoui N. Disassembly sequence planning based on a genetic algorithm. *Proc Inst Mech Eng C* 2015;229(12):2281–90.
- [165] Hu Y, Liu C, Zhang M, Jia Y, Xu Y. A novel simulated annealing-based hyper-heuristic algorithm for stochastic parallel disassembly line balancing in smart remanufacturing. *Sensors* 2023;23(3):1652.
- [166] Alshibli M, El Sayed A, Kongar E, Sobh TM, Gupta SM. Disassembly sequencing using tabu search. *J Intell Robot Syst* 2016;82:69–79.
- [167] Wang K, Gao L, Li X, Li P. Energy-efficient robotic parallel disassembly sequence planning for end-of-life products. *IEEE Trans Autom Sci Eng* 2022;19(2):1277–85. <http://dx.doi.org/10.1109/TASE.2021.3061506>.
- [168] Zhong L, Youchao S, Ekene Gabriel O, Haiqiao W. Disassembly sequence planning for maintenance based on metaheuristic method. *Airc Eng Aerosp Technol* 2011;83(3):138–45.
- [169] Zhang S, Liu P, Guo X, Wang J, Qin S, Tang Y. An improved tabu search algorithm for multi-robot hybrid disassembly line balancing problems. In: 2022 international conference on cyber-physical social intelligence. ICCSI, IEEE; 2022, p. 315–20.

- [170] Kim H-W, Lee D-H. An optimal algorithm for selective disassembly sequencing with sequence-dependent set-ups in parallel disassembly environment. *Int J Prod Res* 2017;55(24):7317–33.
- [171] Liao H-y, Chen Y, Hu B, Behdad S. Optimization-based disassembly sequence planning under uncertainty for human–robot collaboration. *J Mech Des* 2023;145(2):022001.
- [172] Son C. Intelligent rule-based sequence planning algorithm with fuzzy optimization for robot manipulation tasks in partially dynamic environments. *Inform Sci* 2016;342:209–21.
- [173] Gullapalli V, Franklin JA, Benbrahim H. Acquiring robot skills via reinforcement learning. *IEEE Control Syst Mag* 1994;14(1):13–24.
- [174] Apolinariska AA, Pacher M, Li H, Cote N, Pastrana R, Gramazio F, Kohler M. Robotic assembly of timber joints using reinforcement learning. *Autom Constr* 2021;125:103569.
- [175] Nguyen K, Dang VT, Pham DD, Dao PN. Formation control scheme with reinforcement learning strategy for a group of multiple surface vehicles. *Internat J Robust Nonlinear Control* 2024;34(3):2252–79.
- [176] Neu G, Jonsson A, Gómez V. A unified view of entropy-regularized markov decision processes. *Universitat Pompeu Fabra* 2017.
- [177] Kamanichi C, Diddigi RB, Bhatnagar S. Successive over-relaxation Q-learning. *IEEE Control Syst Lett* 2019;4(1):55–60.
- [178] Yershov DS, Frazzoli E. Asymptotically optimal feedback planning using a numerical hamilton-jacobi-bellman solver and an adaptive mesh refinement. *The Int J Robot Res* 2016;35(5):565–84.
- [179] Luo B, Wu H-N, Huang T, Liu D. Reinforcement learning solution for HJB equation arising in constrained optimal control problem. *Neural Netw* 2015;71:150–8.
- [180] Mukherjee A, Liu J. Bridging physics-informed neural networks with reinforcement learning: Hamilton-Jacobi-Bellman proximal policy optimization (HJBPO). *New front Learn, Contr Dyn Syst* 2023.
- [181] Della Penna G, Magazzeni D, Mercurio F, Intrigila B. Upmuphi: A tool for universal planning on PDDL+ problems. In: *Proceedings of the international conference on automated planning and scheduling*. 19, 2009, p. 106–13.
- [182] Mesesan G, Roa MA, Icer E, Althoff M. Hierarchical path planner using workspace decomposition and parallel task-space rrt. In: *2018 IEEE/RSJ international conference on intelligent robots and systems. IROS, IEEE*; 2018, p. 1–9.
- [183] Erol K, Hendler JA, Nau DS. UMCP: A sound and complete procedure for hierarchical task-network planning.. In: *Aips*. 94, 1994, p. 249–54.
- [184] Kaelbling LP, Lozano-Pérez T. Hierarchical task and motion planning in the now. In: *2011 IEEE international conference on robotics and automation. IEEE*; 2011, p. 1470–7.
- [185] Alford R, Shivashankar V, Roberts M, Frank J, Aha DW. Hierarchical planning: Relating task and goal decomposition with task sharing.. In: *IJCAI*. 2016, p. 3022–9.
- [186] Suárez-Hernández A, Alenyà G, Torras C. Interleaving hierarchical task planning and motion constraint testing for dual-arm manipulation. In: *2018 IEEE/RSJ international conference on intelligent robots and systems. IROS, IEEE*; 2018, p. 4061–6.
- [187] Luo J, Xu C, Geng X, Feng G, Fang K, Tan L, Schaal S, Levine S. Multi-stage cable routing through hierarchical imitation learning. *IEEE Trans Robot* 2024.
- [188] Gregorić J, Seder M, Petrović I. Autonomous hierarchy creation for computationally feasible near-optimal path planning in large environments. *Robot Autonom Syst* 2024;172:104584.
- [189] Miao R, Jia Q, Sun F, Chen G, Huang H. Hierarchical understanding in robotic manipulation: A knowledge-based framework. In: *Actuators*. 13, MDPI; 2024, p. 28.
- [190] Shah N, Srivastava S. Hierarchical planning and learning for robots in stochastic settings using zero-shot option invention. In: *Proc. AAAI*. 2023, p. 10.
- [191] Banfi J, Messing A, Kroninger C, Stump E, Hutchinson S, Roy N. Hierarchical planning for heterogeneous multi-robot routing problems via learned subteam performance. *IEEE Robot Autom Lett* 2022;7(2):4464–71.
- [192] Rico FM, Morelli M, Espinoza H, Lera FJR, Olivera VM. Optimized execution of PDDL plans using behavior trees. In: *AAMAS*. 2021, p. 1596–8.
- [193] Lee J, Nam C, Park J, Kim C. Tree search-based task and motion planning with prehensile and non-prehensile manipulation for obstacle rearrangement in clutter. In: *2021 IEEE international conference on robotics and automation. ICRA, IEEE*; 2021, p. 8516–22.
- [194] Kast B, Schmitt PS, Albrecht S, Feiten W, Zhang J. Hierarchical planner with composable action models for asynchronous parallelization of tasks and motions. In: *2020 fourth IEEE international conference on robotic computing. IRC, IEEE*; 2020, p. 143–50.
- [195] Görner M, Haschke R, Ritter H, Zhang J. Moveit! task constructor for task-level motion planning. In: *2019 international conference on robotics and automation. ICRA, IEEE*; 2019, p. 190–6.
- [196] Munoz-Morera J, Alarcon F, Maza I, Ollero A. Combining a hierarchical task network planner with a constraint satisfaction solver for assembly operations involving routing problems in a multi-robot context. *Int J Adv Robot Syst* 2018;15(3):1729881418782088.
- [197] Rovida F, Grossmann B, Krüger V. Extended behavior trees for quick definition of flexible robotic tasks. In: *2017 IEEE/RSJ international conference on intelligent robots and systems. IROS, IEEE*; 2017, p. 6793–800.
- [198] Johannsmeier L, Haddadin S. A hierarchical human-robot interaction-planning framework for task allocation in collaborative industrial assembly processes. *IEEE Robot Autom Lett* 2016;2(1):41–8.
- [199] Wang Y, Sun H, Chen G, Jia Q, Yu B, et al. Hierarchical task planning for multiarm robot with multiconstraint. *Math Probl Eng* 2016;2016.
- [200] Hayes B, Scassellati B. Autonomously constructing hierarchical task networks for planning and human-robot collaboration. In: *2016 IEEE international conference on robotics and automation. ICRA, IEEE*; 2016, p. 5469–76.
- [201] Stock S, Mansouri M, Pecora F, Hertzberg J. Online task merging with a hierarchical hybrid task planner for mobile service robots. In: *2015 IEEE/RSJ international conference on intelligent robots and systems. IROS, IEEE*; 2015, p. 6459–64.
- [202] Kast B, Dietrich V, Albrecht S, Feiten W, Zhang J. A hierarchical planner based on set-theoretic models: Towards automating the automation for autonomous systems. In: *ICINCO* (1). 2019, p. 249–60.
- [203] Castaman N, Pagello E, Menegatti E, Pretto A. Receding horizon task and motion planning in changing environments. *Robot Auton Syst* 2021;145:103863.
- [204] Rahman SR, Whitesides S. Sliding local planners for sampling-based path planning. In: *2017 first IEEE international conference on robotic computing. IRC, IEEE*; 2017, p. 271–6.
- [205] Hadfield-Menell D. Execution cost optimization for hierarchical planning in the now (Ph.D. thesis), Massachusetts Institute of Technology; 2013.
- [206] Şucan IA, Kavrakı LE. Accounting for uncertainty in simultaneous task and motion planning using task motion multigraphs. In: *2012 IEEE international conference on robotics and automation. IEEE*; 2012, p. 4822–8.
- [207] Shah N, Vasudevan DK, Kumar K, Kamojihala P, Srivastava S. Anytime integrated task and motion policies for stochastic environments. In: *2020 IEEE international conference on robotics and automation. ICRA, IEEE*; 2020, p. 9285–91.
- [208] Guo M, Zavlanos MM. Probabilistic motion planning under temporal tasks and soft constraints. *IEEE Trans Autom Control* 2018;63(12):4051–66.
- [209] Yang S, Mao X, Liu W. Towards an extended pomdp planning approach with adjoint action model for robotic task. In: *2020 IEEE international conference on systems, man, and cybernetics. SMC, IEEE*; 2020, p. 1412–9.
- [210] Zhao Z, Lee WS, Hsu D. Large language models as commonsense knowledge for large-scale task planning. *Adv Neural Inf Process Syst* 2024;36.
- [211] Safronov E, Colledanchise M, Natale L. Task planning with belief behavior trees. In: *2020 IEEE/RSJ international conference on intelligent robots and systems. IROS, IEEE*; 2020, p. 6870–7.
- [212] Safronov E, Vilzmann M, Tsetserukou D, Kondak K. Asynchronous behavior trees with memory aimed at aerial vehicles with redundancy in flight controller. In: *2019 IEEE/RSJ international conference on intelligent robots and systems. IROS, IEEE*; 2019, p. 3113–8.
- [213] Dantam NT, Kingston ZK, Chaudhuri S, Kavrakı LE. An incremental constraint-based framework for task and motion planning. *The Int J Robot Res* 2018;37(10):1134–51.
- [214] Garrett CR, Lozano-Pérez T, Kaelbling LP. Pddlstream: Integrating symbolic planners and blackbox samplers via optimistic adaptive planning. In: *Proceedings of the international conference on automated planning and scheduling*. 30, 2020, p. 440–8.
- [215] Curtis A, Kaelbling L, Jain S. Task-directed exploration in continuous pomdps for robotic manipulation of articulated objects. In: *2023 IEEE international conference on robotics and automation. ICRA, IEEE*; 2023, p. 3721–8.
- [216] Thomas A, Mastrogianni F, Baglietto M. Mtp: Motion-planning-aware task planning for navigation in belief space. *Robot Auton Syst* 2021;141:103786.
- [217] Phiquepal C, Toussaint M. Combined task and motion planning under partial observability: An optimization-based approach. In: *2019 international conference on robotics and automation. ICRA, IEEE*; 2019, p. 9000–6.
- [218] Zhang C, Shah JA. Co-optimizing task and motion planning. In: *2016 IEEE/RSJ international conference on intelligent robots and systems. IROS, IEEE*; 2016, p. 4750–6.
- [219] Saha S, Julius AA. Task and motion planning for manipulator arms with metric temporal logic specifications. *IEEE Robot Automat Lett* 2017;3(1):379–86.
- [220] Aliakbar A, Lagriffoul F, Rosell J. Combined heuristic task and motion planning for bi-manual robots. *Auton Robots* 2018;43:1575–90. <http://dx.doi.org/10.1007/S10514-018-9817-3>.
- [221] Zimmermann S, Hakimifard G, Zamora M, Poranne R, Coros S. A multi-level optimization framework for simultaneous grasping and motion planning. *IEEE Robot Autom Lett* 2020;5:2966–72. <http://dx.doi.org/10.1109/LRA.2020.2974684>.
- [222] Leu J, Zhang G, Sun L, Tomizuka M. Efficient robot motion planning via sampling and optimization. In: *2021 American Control Conference (ACC)*. 2021, p. 4196–202. <http://dx.doi.org/10.23919/ACC50511.2021.9483146>.
- [223] Mukherjee S, Aine S, Likhachev M. Mplp: Massively parallelized lazy planning. *IEEE Robot Autom Lett* 2022;7(3):6067–74.
- [224] Cheng Y, Sun L, Tomizuka M. Human-aware robot task planning based on a hierarchical task model. *IEEE Robot Autom Lett* 2021;6(2):1136–43.

- [225] Faroni M, Beschi M, Ghidini S, Pedrocchi N, Umbrico A, Orlandini A, Cesta A. A layered control approach to human-aware task and motion planning for human-robot collaboration. In: 2020 29th IEEE international conference on robot and human interactive communication (RO-MAN). IEEE; 2020, p. 1204–10.
- [226] Umbrico A, Cesta A, Cialdea Mayer M, Orlandini A, Platinu m: A new framework for planning and acting. In: AI* IA 2017 advances in artificial intelligence: xVith international conference of the Italian association for artificial intelligence, bari, Italy, November 14–17, 2017, proceedings 16. Springer; 2017, p. 498–512.
- [227] Gombolay MC, Wilcox RJ, Shah JA. Fast scheduling of robot teams performing tasks with temporospatial constraints. *IEEE Trans Robot* 2018;34(1):220–39.
- [228] Mandalika A, Choudhury S, Salzman O, Srinivasa S. Generalized lazy search for robot motion planning: Interleaving search and edge evaluation via event-based toggles. In: Proceedings of the international conference on automated planning and scheduling. 29, 2019, p. 745–53.
- [229] Ratliff N, Zucker M, Bagnell JA, Srinivasa S. CHOMP: Gradient optimization techniques for efficient motion planning. In: 2009 IEEE international conference on robotics and automation. IEEE; 2009, p. 489–94.
- [230] Kalakrishnan M, Chitta S, Theodorou E, Pastor P, Schaal S. STOMP: Stochastic trajectory optimization for motion planning. In: 2011 IEEE international conference on robotics and automation. IEEE; 2011, p. 4569–74.
- [231] Park C, Pan J, Manocha D. ITOMP: Incremental trajectory optimization for real-time replanning in dynamic environments. In: Proceedings of the international conference on automated planning and scheduling. 22, 2012, p. 207–15.
- [232] Lagriffoul F, Andres B. Combining task and motion planning: A culprit detection problem. *The Int J Robot Res* 2016;35(8):890–927.
- [233] Mirabel J, Lamiraux F. Constraint graphs: Unifying task and motion planning for navigation and manipulation among movable obstacles. 2016, URL <https://hal.science/hal-01281348>; working paper or preprint.
- [234] Lagriffoul F, Dimitrov D, Bidot J, Saffiotti A, Karlsson L. Efficiently combining task and motion planning using geometric constraints. *The Int J Robot Res* 2014;33(14):1726–47.
- [235] Srivastava S, Fang E, Riano L, Chitnis R, Russell S, Abbeel P. Combined task and motion planning through an extensible planner-independent interface layer. In: 2014 IEEE international conference on robotics and automation. ICRA, IEEE; 2014, p. 639–46.
- [236] Mansouri M, Pecora F. More knowledge on the table: Planning with space, time and resources for robots. In: 2014 IEEE international conference on robotics and automation. ICRA, IEEE; 2014, p. 647–54.
- [237] Edelkamp S, Lahijanian M, Magazzeni D, Plaku E. Integrating temporal reasoning and sampling-based motion planning for multigoal problems with dynamics and time windows. *IEEE Robot Autom Lett* 2018;3(4):3473–80.
- [238] Basile F, Caccavale F, Chiacchio P, Coppola J, Curatella C. Task-oriented motion planning for multi-arm robotic systems. *Robot Comput-Integr Manuf* 2012;28(5):569–82.
- [239] LaValle SM. Planning algorithms. Cambridge University Press; 2006.
- [240] Fang X, Garrett CR, Eppner C, Lozano-Pérez T, Kaelbling LP, Fox D. DiMSam: Diffusion models as samplers for task and motion planning under partial observability. 2023, arXiv preprint arXiv:2306.13196.
- [241] Wang J, Meng MQ-H. Optimal path planning using generalized voronoi graph and multiple potential functions. *IEEE Trans Industr Electr* 2020;67(12):10621–30.
- [242] Migimatsu T, Bohg J. Object-centric task and motion planning in dynamic environments. *IEEE Robot Autom Lett* 2020;5(2):844–51.
- [243] Thomason W, Knepper RA. A unified sampling-based approach to integrated task and motion planning. In: The international symposium of robotics research. Springer; 2019, p. 773–88.
- [244] Garrett CR, Lozano-Perez T, Kaelbling LP. Ffrob: Leveraging symbolic planning for efficient task and motion planning. *The Int J Robot Res* 2018;37(1):104–36.
- [245] Huh J, Lee B, Lee DD. Adaptive motion planning with high-dimensional mixture models. In: 2017 IEEE international conference on robotics and automation. ICRA, 2017, p. 3740–7. <http://dx.doi.org/10.1109/ICRA.2017.7989431>.
- [246] Salzman O, Halperin D. Asymptotically near-optimal RRT for fast, high-quality motion planning. *IEEE Trans Robot* 2016;32(3):473–83.
- [247] Zito C, Stolkin R, Kopicki M, Wyatt JL. Two-level RRT planning for robotic push manipulation. In: 2012 IEEE/RSJ international conference on intelligent robots and systems. IEEE; 2012, p. 678–85.
- [248] Rios-Martinez J, Spalanzani A, Laugier C. Understanding human interaction for probabilistic autonomous navigation using risk-RRT approach. In: 2011 IEEE/RSJ international conference on intelligent robots and systems. IEEE; 2011, p. 2014–9.
- [249] Karaman S, Frazzoli E. Sampling-based algorithms for optimal motion planning. *The Int J Robot Res* 2011;30(7):846–94.
- [250] Kuffner JJ, LaValle SM. RRT-connect: An efficient approach to single-query path planning. In: Proceedings 2000 iCRA. millennium conference. IEEE international conference on robotics and automation. symposia proceedings (cat. no. 00CH37065). 2, IEEE; 2000, p. 995–1001.
- [251] Melchior NA, Simmons R. Particle RRT for path planning with uncertainty. In: Proceedings 2007 IEEE international conference on robotics and automation. IEEE; 2007, p. 1617–24.
- [252] Mendez-Mendez J, Kaelbling LP, Lozano-Pérez T. Embodied lifelong learning for task and motion planning. In: Conference on robot learning. PMLR; 2023, p. 2134–50.
- [253] Wang X, He Z, Peng X. Artificial-intelligence-generated content with diffusion models: A literature review. *Mathematics* 2024;12(7):977.
- [254] Bäckström C, Nebel B. Complexity results for SAS+ planning. *Comput Intell* 1995;11(4):625–55.
- [255] Economist I. A circular approach to tackling e-waste. 2023, [Online]. Available: <https://impact.economist.com/sustainability/circular-economies/a-circular-approach-to-tackling-e-waste>; impact.economist.com.
- [256] Álvarez-de-los Mozos E, Rentería-Bilbao A, Díaz-Martín F. WEEE recycling and circular economy assisted by collaborative robots. *Appl Sci* 2020;10(14):4800.
- [257] Wagner F, Peeters J, De Keyser J, Dufloy J, Dewulf W. Evaluation of the quality of postconsumer plastics obtained from disassembly-based recycling strategies. *Polym Eng Sci* 2018;58(4):485–92.
- [258] IEA. Prospects for electric vehicle deployment – global EV outlook 2023 – analysis. 2023, [Online]. Available: <https://www.iea.org/reports/global-ev-outlook-2023/prospects-for-electric-vehicle-deployment>; IEA.
- [259] Rebellion. About. 2024, [Online]. Available: <https://rebellion-project.eu/about/>; Rebellion.
- [260] Zhao Y, Pohl O, Bhatt AI, Collis GE, Mahon PJ, Rüter T, Hollenkamp AF. A review on battery market trends, second-life reuse, and recycling. *Sust Chem* 2021;2(1):167–205.
- [261] Rastegarparnah A, Asif ME, Stolkin R. Hybrid neural networks for enhanced predictions of remaining useful life in lithium-ion batteries. *Batteries* 2024;10(3):106.
- [262] Zorn M, Ionescu C, Klohs D, Zühl K, Kisseler N, Daldrup A, Hams S, Zheng Y, Offermanns C, Flamme S, et al. An approach for automated disassembly of lithium-ion battery packs and high-quality recycling using computer vision, labeling, and material characterization. *Recycling* 2022;7(4):48.
- [263] Tan WJ, Chin CMM, Garg A, Gao L. A hybrid disassembly framework for disassembly of electric vehicle batteries. *Int J Energy Res* 2021;45(5):8073–82.
- [264] Aguinaga I, Borro D, Matey L. Path-planning techniques for the simulation of disassembly tasks. *Assem Autom* 2007;27(3):207–14.
- [265] Fortune. Commercial aircraft disassembly, dismantling & recycling market growth, 2027. 2024, [Online]. Available: <https://www.fortunebusinessinsights.com/commercial-aircraft-disassembly-dismantling-and-recycling-market-103584>.
- [266] Najmon JC, Raeisi S, Tovar A. Review of additive manufacturing technologies and applications in the aerospace industry. *Add Manuf Aerospace Indust* 2019;7–31.
- [267] Katnam KB, Da Silva L, Young T. Bonded repair of composite aircraft structures: A review of scientific challenges and opportunities. *Prog Aerosp Sci* 2013;61:26–42.
- [268] Liang X, Xiao R, Zhang J. A review on digital twin for robotics in smart manufacturing. In: 2022 IEEE 17th conference on industrial electronics and applications (ICIEA). 2022, p. 1510–5. <http://dx.doi.org/10.1109/ICIEA54703.2022.10006119>.
- [269] Liu Y, Wang X, Yang K, Pan Y, Wang Q. Architecture and implementation of high-fidelity digital twins for industrial robots. In: 2023 International conference on frontiers of robotics and software engineering (FRSE). 2023, p. 207–14. <http://dx.doi.org/10.1109/FRSE58934.2023.00036>.
- [270] Choi SH, Park K-B, Roh DH, Lee JY, Mohammed M, Ghasemi Y, Jeong H. An integrated mixed reality system for safety-aware human-robot collaboration using deep learning and digital twin generation. *Robot Comput-Integr Manuf* 2022;73:102258.
- [271] Douthwaite JA, Lesage B, Gleirscher M, Calinescu R, Aitken J, Alexander R, Law J. A modular digital twinning framework for safety assurance of collaborative robotics. *Front Robot AI* 2021;8. <http://dx.doi.org/10.3389/frobot.2021.758099>.
- [272] Xu W, Cui J, Li L, Yao B, Tian S, Zhou Z. Digital twin-based industrial cloud robotics: Framework, control approach and implementation. *J Manuf Syst* 2021;58:196–209.
- [273] Liu A, Zhang Y, Yao Y. Digital twins for hand gesture-guided human-robot collaboration systems. *Proc Inst Mech Eng B* 2024;09544054231223783.
- [274] Qu W, Li J, Zhang R, Liu S, Bao J. Adaptive planning of human-robot collaborative disassembly for end-of-life lithium-ion batteries based on digital twin. *J Intell Manuf* 2023;1–23.
- [275] Liu J, Xu Z, Xiong H, Lin Q, Xu W, Zhou Z. Digital twin-driven robotic disassembly sequence dynamic planning under uncertain missing condition. *IEEE Trans Ind Inf* 2023.

- [276] Li C, Zheng P, Li S, Pang Y, Lee CK. AR-assisted digital twin-enabled robot collaborative manufacturing system with human-in-the-loop. *Robot Comput-Integr Manuf* 2022;76:102321.
- [277] Liu Y, Xu H, Liu D, Wang L. A digital twin-based sim-to-real transfer for deep reinforcement learning-enabled industrial robot grasping. *Robot Comput-Integr Manuf* 2022;78:102365.
- [278] Wang K, Li Y, Guo J, Gao L, Li X. Dynamic balancing of U-shaped robotic disassembly lines using an effective deep reinforcement learning approach. *IEEE Trans Ind Inf* 2024;1–11. <http://dx.doi.org/10.1109/TII.2023.3348811>.
- [279] Mineo C, Wong C, Vasilev M, Cowan B, MacLeod CN, Pierce SG, Yang E. Interfacing toolbox for robotic arms with real-time adaptive behaviour capabilities. 2019.
- [280] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I. Attention is all you need. *Adv Neural Inform Process Syst* 2017;30.
- [281] Xie B, Xi X, Zhao X, Wang Y, Song W, Gu J, Zhu S. ChatGPT for robotics: A new approach to human-robot interaction and task planning. In: *International conference on intelligent robotics and applications*. Springer; 2023, p. 365–76.
- [282] Khandelwal A, Weihs L, Mottaghi R, Kembhavi A. Simple but effective: Clip embeddings for embodied ai. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022, p. 14829–38.
- [283] Chen X, Wang X, Changpinyo S, Piergiovanni A, Padlewski P, Salz D, Goodman SA, Grycner A, Mustafa B, Beyer L, Kolesnikov A, Puigcerver J, Ding N, Rong K, Akbari H, Mishra G, Xue L, Thapliyal A, Bradbury J, Kuo W, Seyedhosseini M, Jia C, Ayan BK, Riquelme C, Steiner A, Angelova A, Zhai X, Housby N, Soricut R. Pali: A jointly-scaled multilingual language-image model. 2023, p. 33, [Online]. Available: <https://research.google/pubs/pali-a-jointly-scaled-multilingual-language-image-model/>.
- [284] Driess D, Xia F, Sajjadi MS, Lynch C, Chowdhery A, Ichter B, Wahid A, Tompson J, Vuong Q, Yu T, et al. Palm-e: An embodied multimodal language model. Google 2023.
- [285] Zitkovich B, Yu T, Xu S, Xu P, Xiao T, Xia F, Wu J, Wohlhart P, Welker S, Wahid A, Vuong Q, Vanhoucke V, Tran H, Soricut R, Singh A, Singh J, Sermanet P, Sanketi PR, Salazar G, Ryoo MS, Reymann K, Rao K, Pertsch K, Mordatch I, Michalewski H, Lu Y, Levine S, Lee L, Lee T-WE, Leal I, Kuang Y, Kalashnikov D, Julian R, Joshi NJ, Irpan A, Ichter B, Hsu J, Herzog A, Hausman K, Gopalakrishnan K, Fu C, Florence P, Finn C, Dubey KA, Driess D, Ding T, Choromanski KM, Chen X, Chebotar Y, Carbajal J, Brown N, Brohan A, Arenas MG, Han K. RT-2: Vision-language-action models transfer web knowledge to robotic control. In: *Tan J, Toussaint M, Darvish K, editors. Proceedings of the 7th conference on robot learning. Proceedings of machine learning research*, 229, PMLR; 2023, p. 2165–83, URL <https://proceedings.mlr.press/v229/zitkovich23a.html>.
- [286] Leal I, Choromanski K, Jain D, Dubey A, Varley J, Ryoo M, Lu Y, Liu F, Sindhwani V, Vuong Q, et al. Sara-rt: Scaling up robotics transformers with self-adaptive robust attention. Google 2023. URL <https://sites.google.com/view/rtsara/>.
- [287] Ahn M, Dwibedi D, Finn C, Arenas MG, Gopalakrishnan K, Hausman K, brianichter, Irpan A, Joshi NJ, Julian R, Kirmani S, Leal I, Lee T-WE, Levine S, Lu Y, Maddineni S, Rao K, Sadigh D, Sanketi PR, Sermanet P, Vuong Q, Welker S, Xia F, Xiao T, Xu P, Xu S, Xu Z. AutoRT: Embodied foundation models for large scale orchestration of robotic agents. 2024, URL <https://openreview.net/forum?id=xVlcbh0poD>.
- [288] Torreno A, Onaindia E, Komenda A, Štolba M. Cooperative multi-agent planning: A survey. *ACM Comput Surv* 2017;50(6):1–32.
- [289] Bechon P, Lesire C, Barbier M. Hybrid planning and distributed iterative repair for multi-robot missions with communication losses. *Auton Robots* 2020;44(3):505–31.
- [290] Chen Y, Rosolia U, Ames AD. Decentralized task and path planning for multi-robot systems. *IEEE Robot Autom Lett* 2021;6(3):4337–44.
- [291] Liu R, Nageotte F, Zanne P, de Mathelin M, Dresch-Langley B. Deep reinforcement learning for the control of robotic manipulation: a focussed mini-review. *Robotics* 2021;10(1):22.
- [292] Yu Y. Towards sample efficient reinforcement learning. In: *IJCAI*. 2018, p. 5739–43.
- [293] Levine S, Pastor P, Krizhevsky A, Ibarz J, Quillen D. Learning hand-eye coordination for robotic grasping with deep learning and large-scale data collection. *The Int J Robot Res* 2018;37(4–5):421–36.
- [294] Csurka G. Deep visual domain adaptation. In: *2020 22nd international symposium on symbolic and numeric algorithms for scientific computing. SYNASC, IEEE*; 2020, p. 1–8.
- [295] Jiang Y, Zhang T, Ho D, Bai Y, Liu CK, Levine S, Tan J. Simgan: Hybrid simulator identification for domain adaptation via adversarial reinforcement learning. In: *2021 IEEE international conference on robotics and automation. ICRA, IEEE*; 2021, p. 2884–90.
- [296] Chen C, Chen Z, Jiang B, Jin X. Joint domain alignment and discriminative feature learning for unsupervised deep domain adaptation. In: *Proceedings of the AAAI conference on artificial intelligence*. 33, 2019, p. 3296–303.
- [297] Loghmani MR, Robbiano L, Planamente M, Park K, Caputo B, Vincze M. Unsupervised domain adaptation through inter-modal rotation for rgb-d object recognition. *IEEE Robot Autom Lett* 2020;5(4):6631–8.
- [298] Li C, Du D, Zhang L, Wen L, Luo T, Wu Y, Zhu P. Spatial attention pyramid network for unsupervised domain adaptation. In: *Computer vision–ECCV 2020: 16th European conference, Glasgow, UK, August 23–28, 2020, proceedings, part XIII 16*. Springer; 2020, p. 481–97.
- [299] Pandey P, Tyagi AK, Ambekar S, Prathosh A. Unsupervised domain adaptation for semantic segmentation of NIR images through generative latent search. In: *Computer vision–ECCV 2020: 16th European conference, Glasgow, UK, August 23–28, 2020, proceedings, part VI 16*. Springer; 2020, p. 413–29.